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(12) **United States Patent**
Koito et al.

(10) **Patent No.:** **US 12,411,363 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **LIGHTING DEVICE AND OPTICAL ELEMENT**

(71) Applicant: **Japan Display Inc.**, Tokyo (JP)

(72) Inventors: **Takeo Koito**, Tokyo (JP); **Kojiro Ikeda**, Tokyo (JP); **Tae Kurokawa**, Tokyo (JP)

(73) Assignee: **Japan Display Inc.**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 135 days.

(21) Appl. No.: **18/492,067**

(22) Filed: **Oct. 23, 2023**

(65) **Prior Publication Data**

US 2024/0045237 A1 Feb. 8, 2024

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2022/019928, filed on May 11, 2022.

(30) **Foreign Application Priority Data**

May 21, 2021 (JP) 2021-086465

(51) **Int. Cl.**
G02F 1/00 (2006.01)
G02F 1/01 (2006.01)
G02F 1/03 (2006.01)

(52) **U.S. Cl.**
CPC **G02F 1/0045** (2013.01); **G02F 1/0105** (2013.01); **G02F 1/0316** (2013.01); **G02F 1/0327** (2013.01)

(58) **Field of Classification Search**
CPC G02F 1/0045
See application file for complete search history.

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Primary Examiner — Chanceity N Robinson

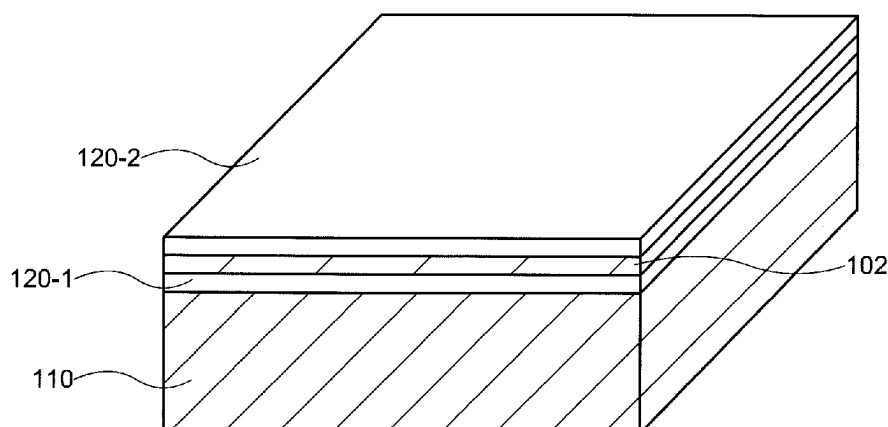
(74) *Attorney, Agent, or Firm* — K&L Gates LLP

(57) **ABSTRACT**

A lighting device includes a light source, a first liquid crystal cell over the light source, and a second liquid crystal cell over the first liquid crystal cell. The light source includes light-emitting elements arranged in a matrix shape with m rows and n columns. The first and second liquid crystal cells each include: a first substrate; first electrode groups located over the first substrate and arranged in a matrix shape with m rows and n columns; a liquid crystal layer over the first electrode groups; and a second substrate over the liquid crystal layer. In each of the first and the second liquid crystal cells, the first electrode groups each have first electrodes extending in a row direction, and the light-emitting elements in a jth row and a kth column overlaps the first electrode group located in the jth row and the kth column.

19 Claims, 44 Drawing Sheets

100



(56)

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FIG. 1

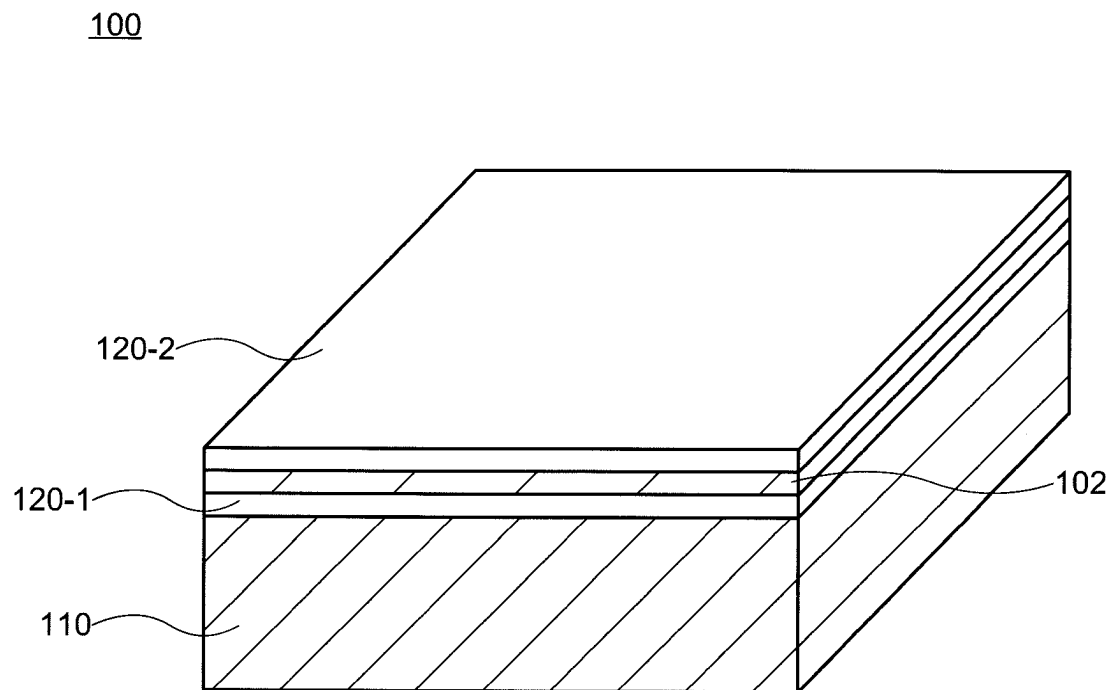


FIG. 2A

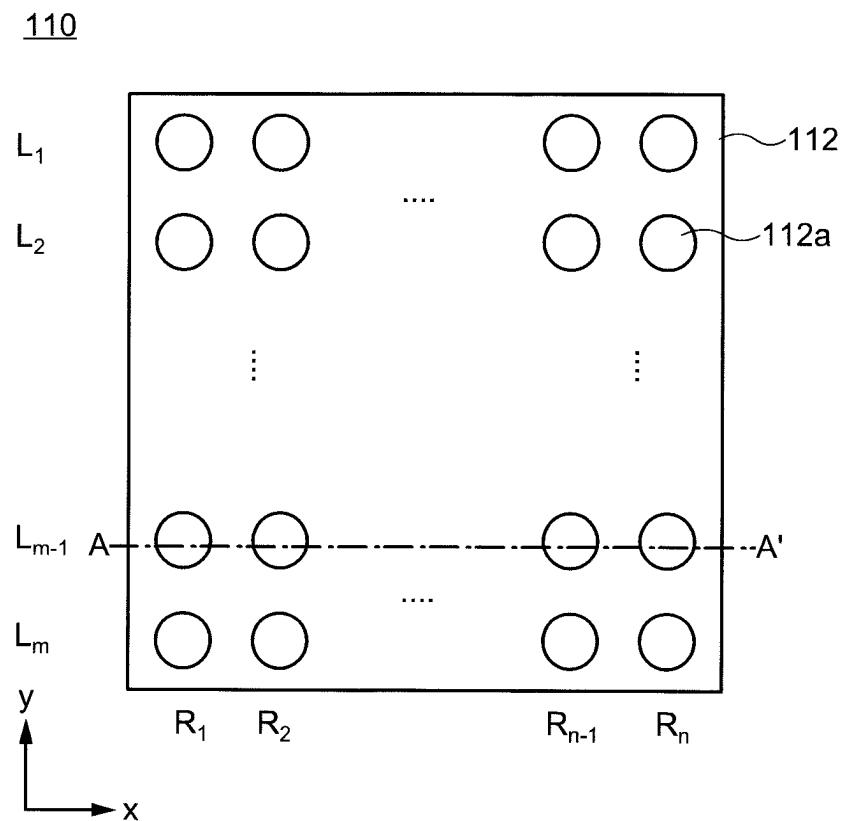


FIG. 2B

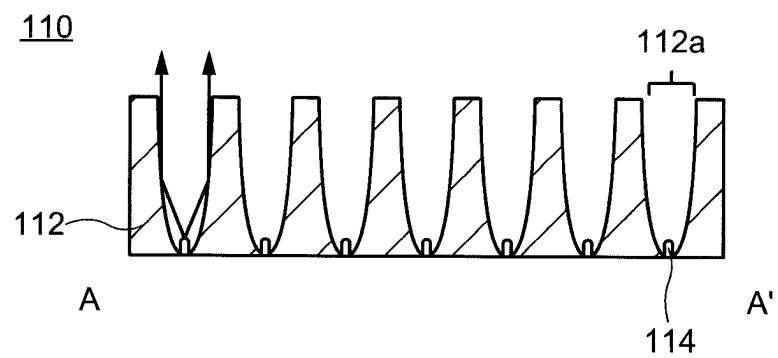


FIG. 3A

110

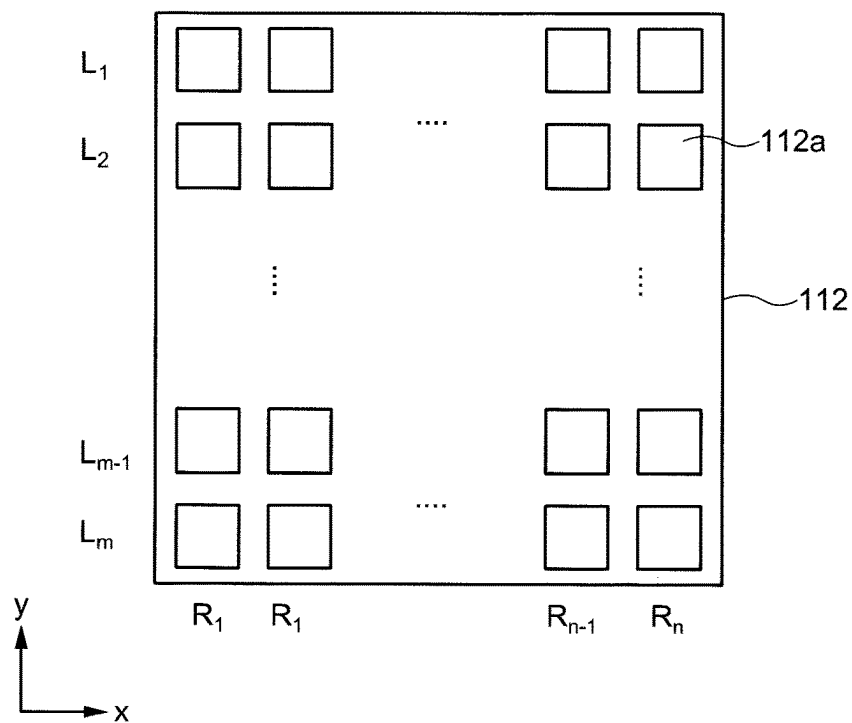


FIG. 3B

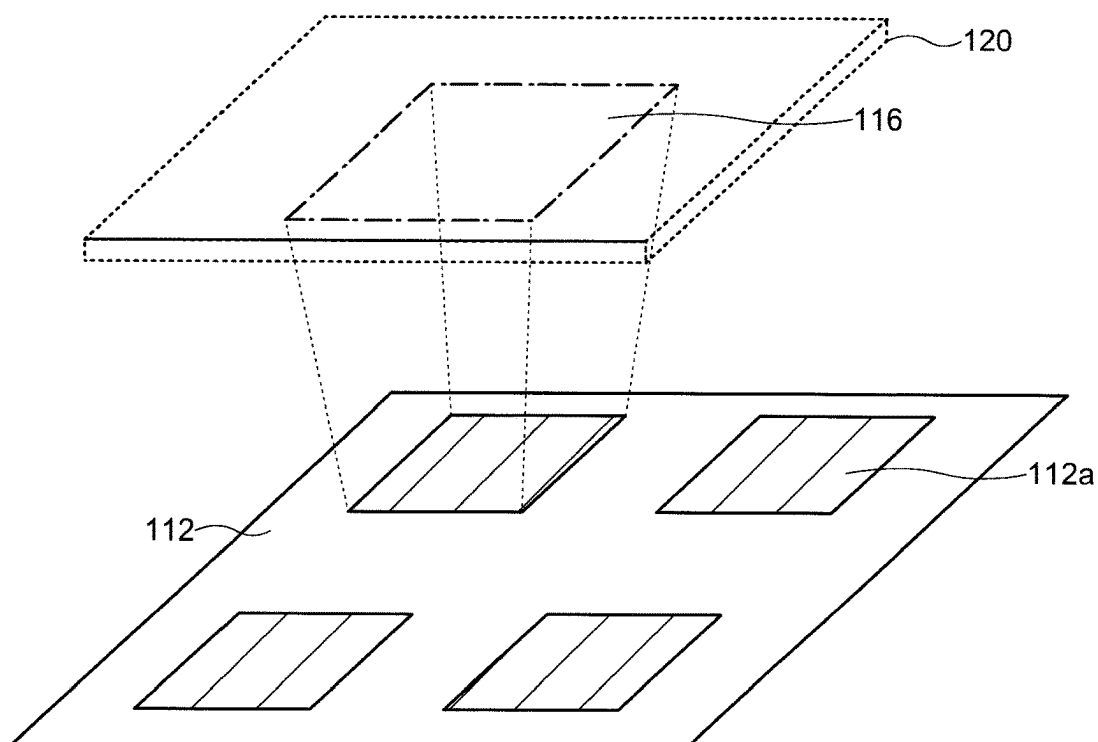


FIG. 4A

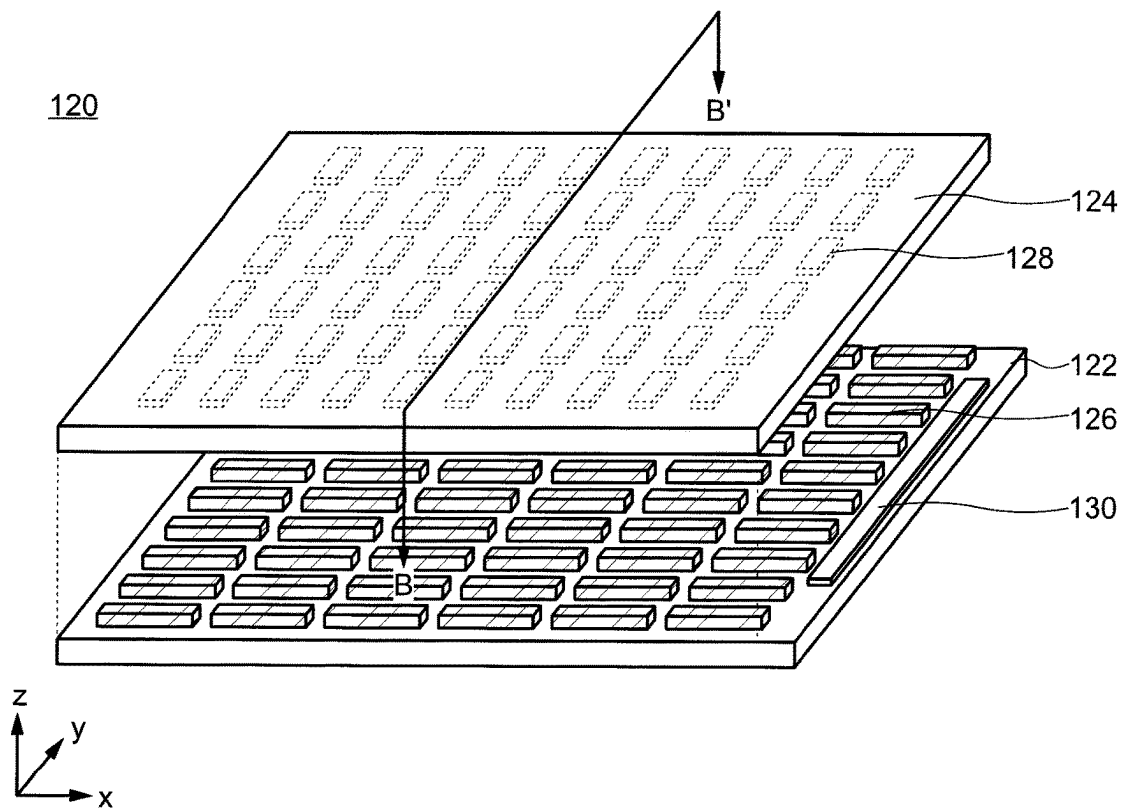


FIG. 4B

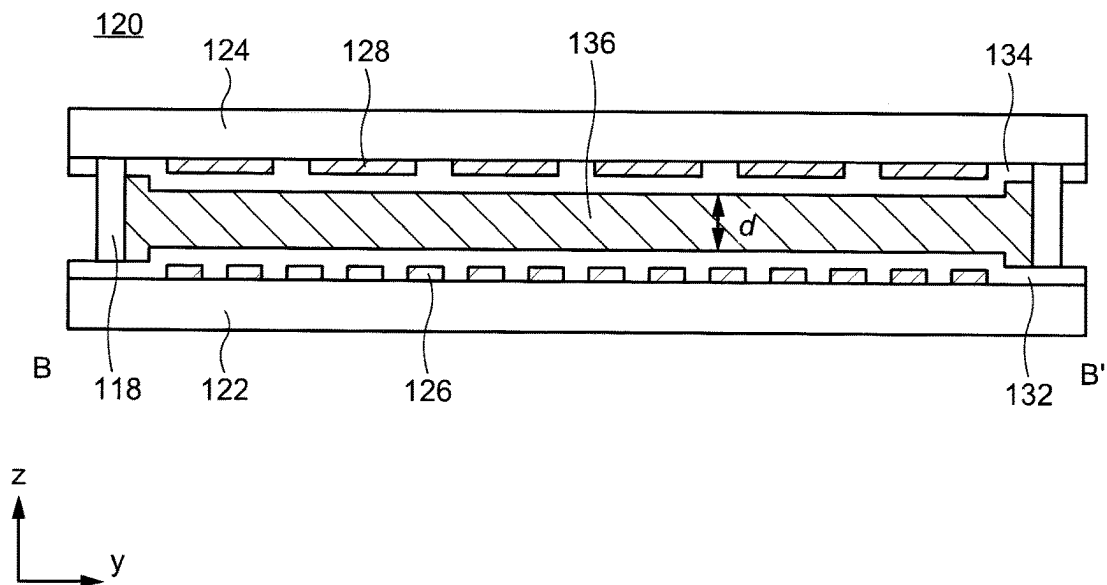


FIG. 5A

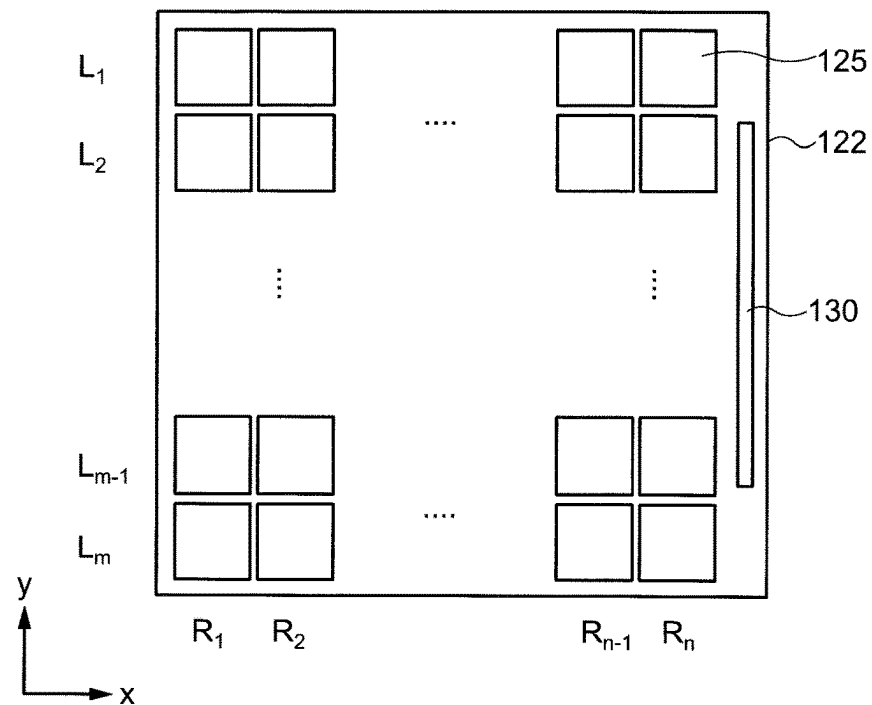


FIG. 5B

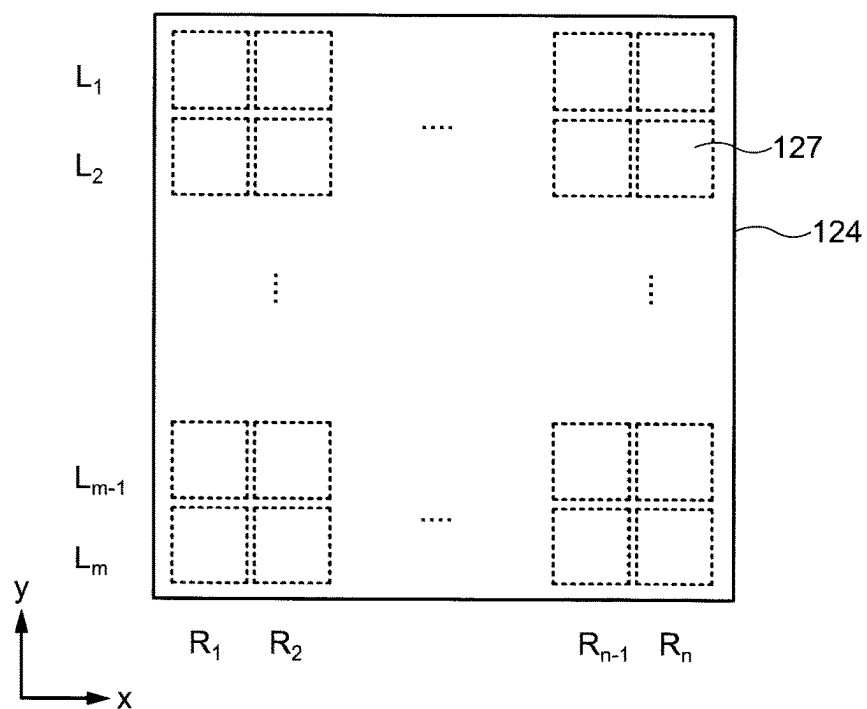


FIG. 6

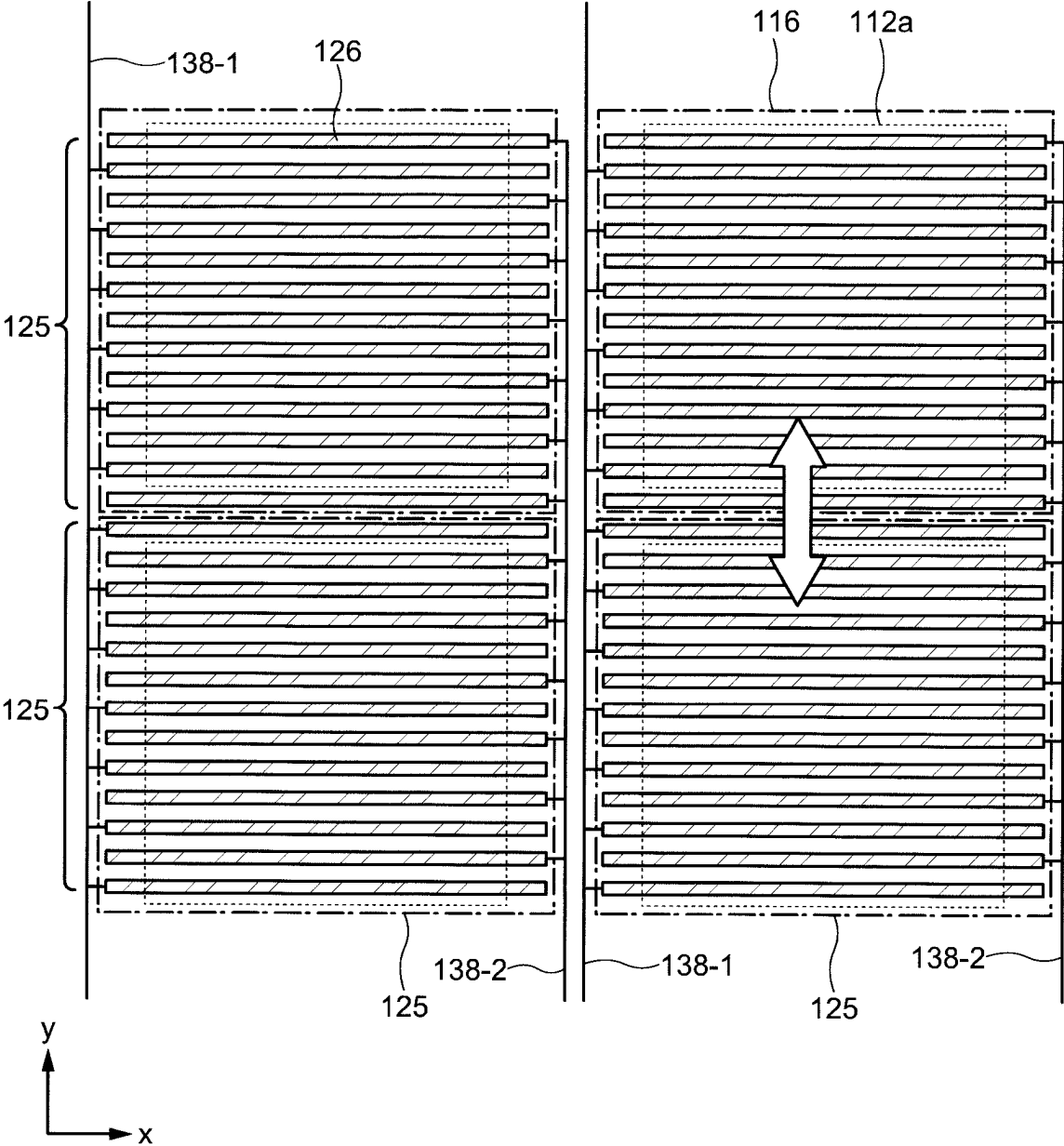


FIG. 7

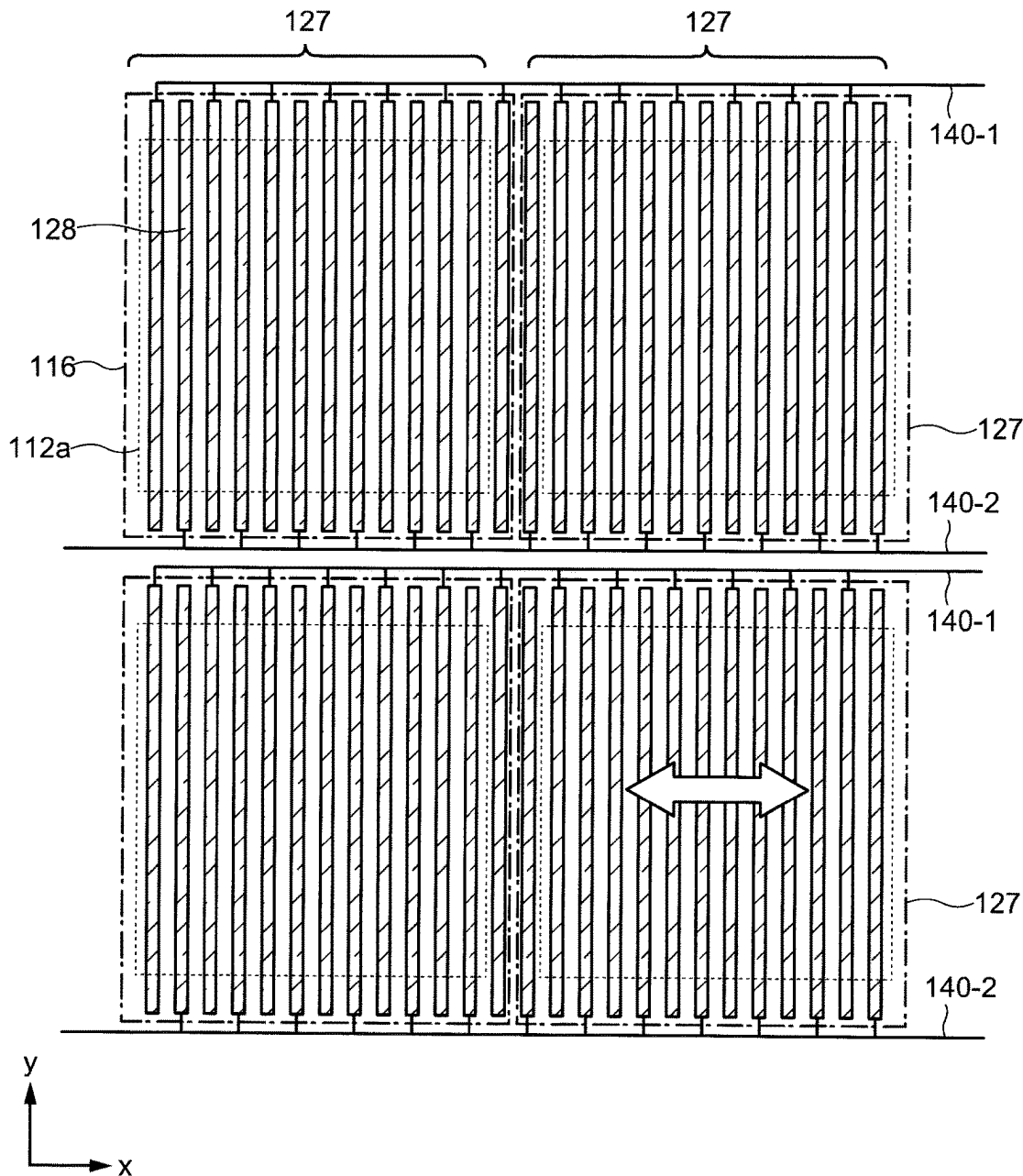


FIG. 8A

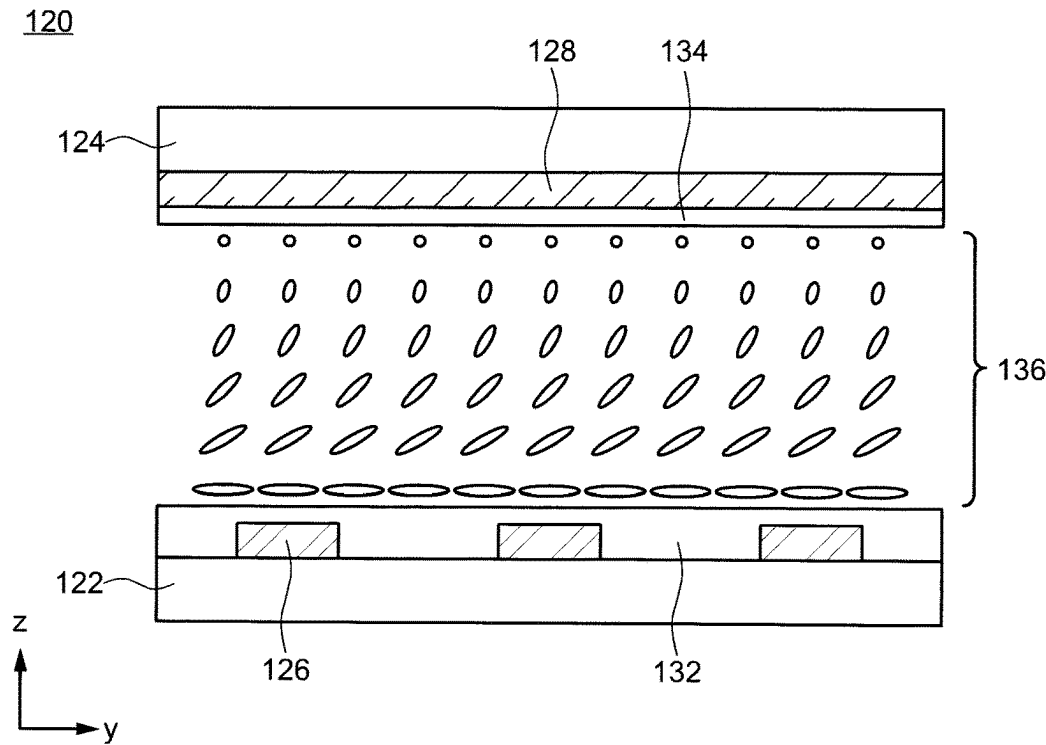


FIG. 8B

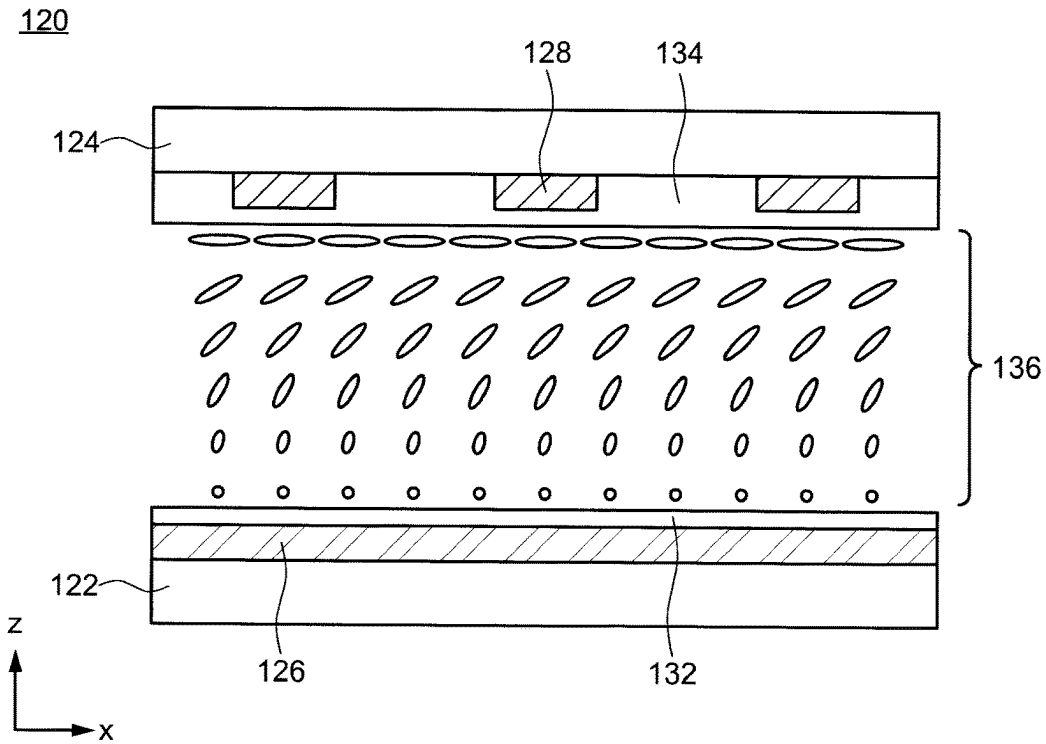


FIG. 9A

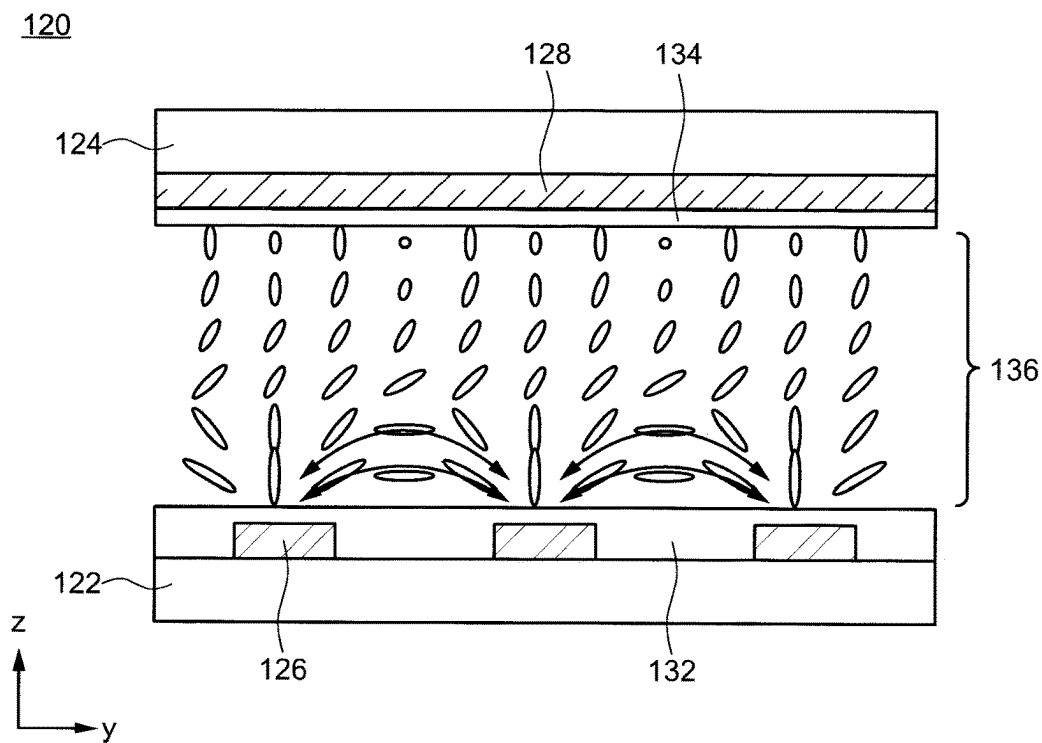


FIG. 9B

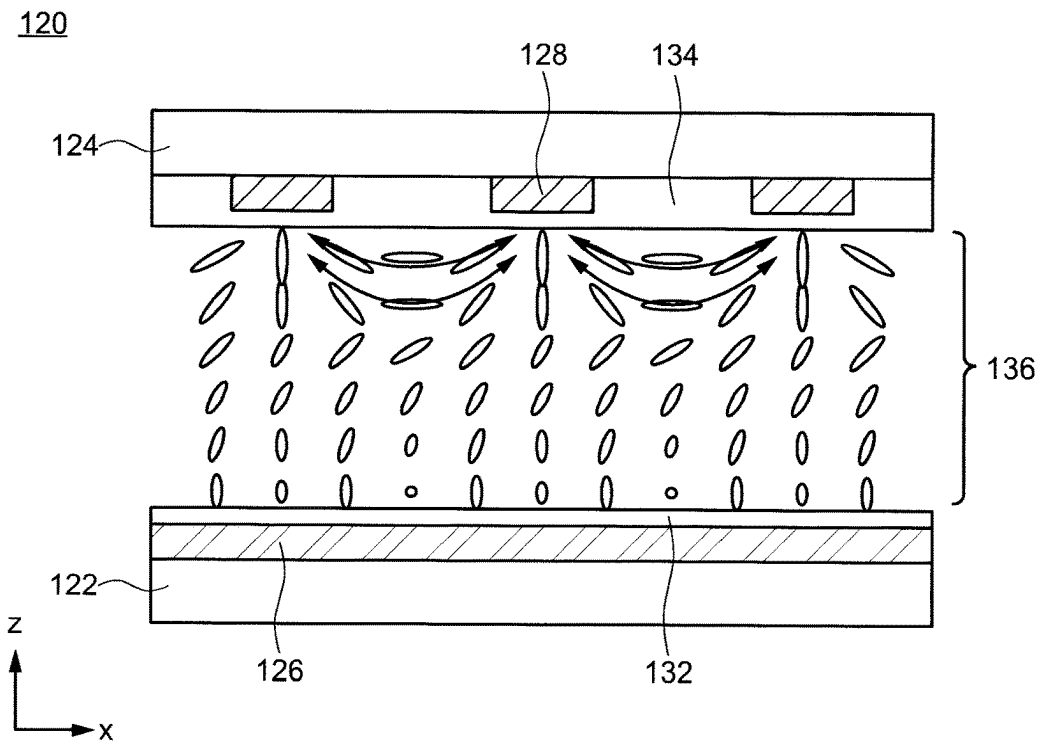


FIG. 10

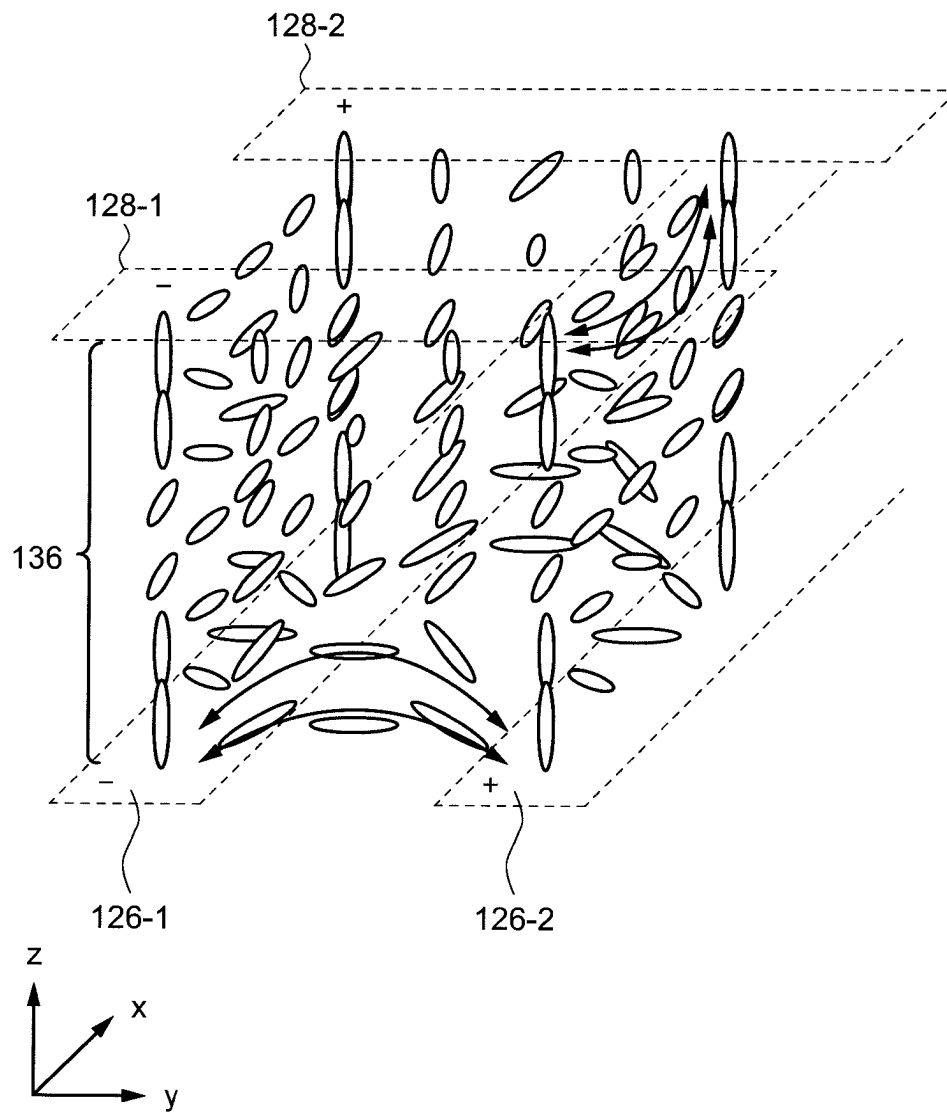


FIG. 11

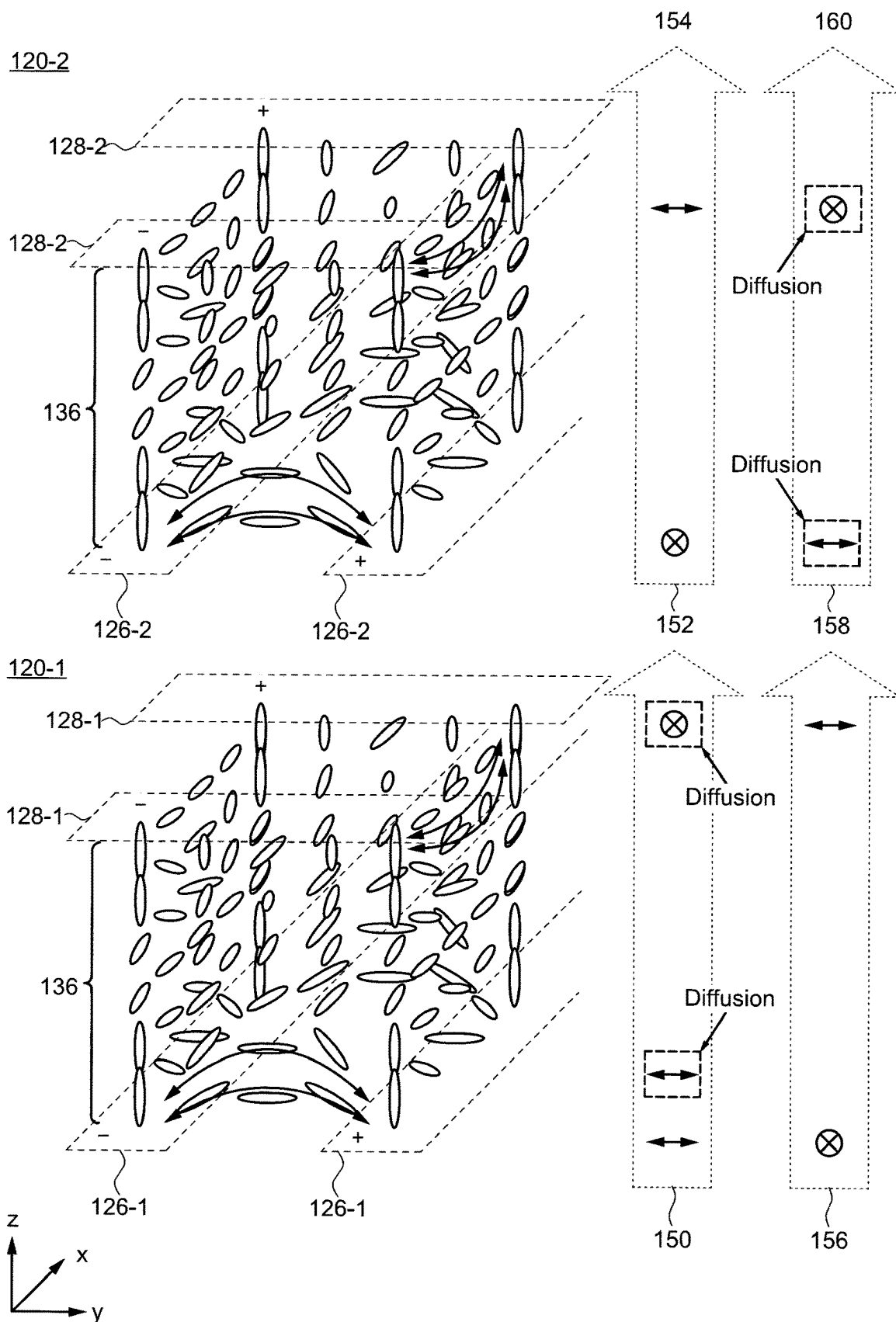
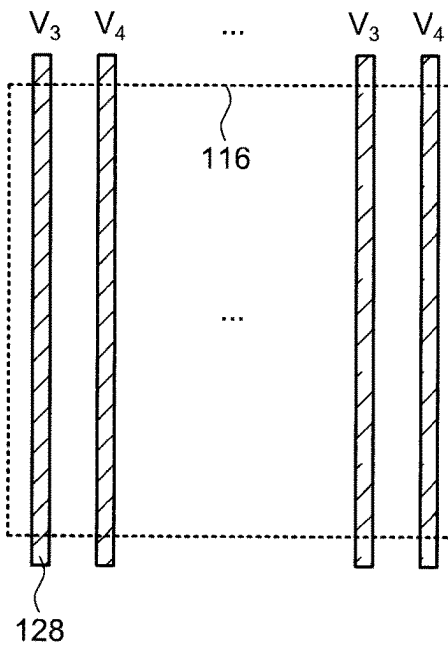
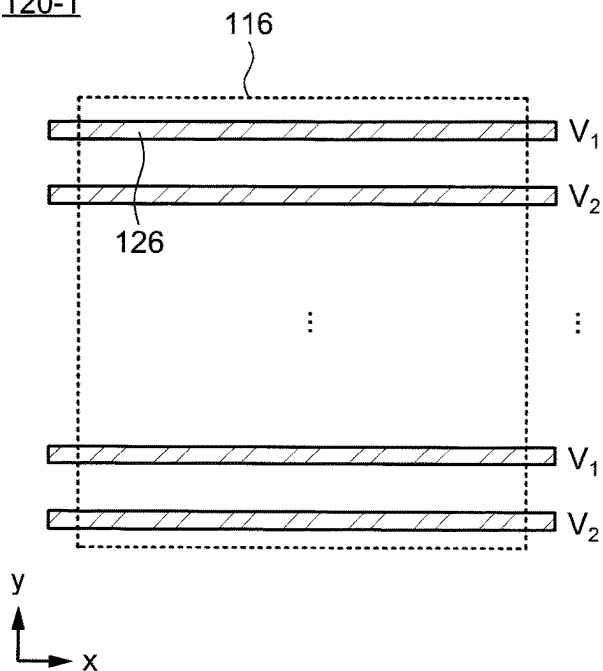


FIG. 12

120-1



120-2

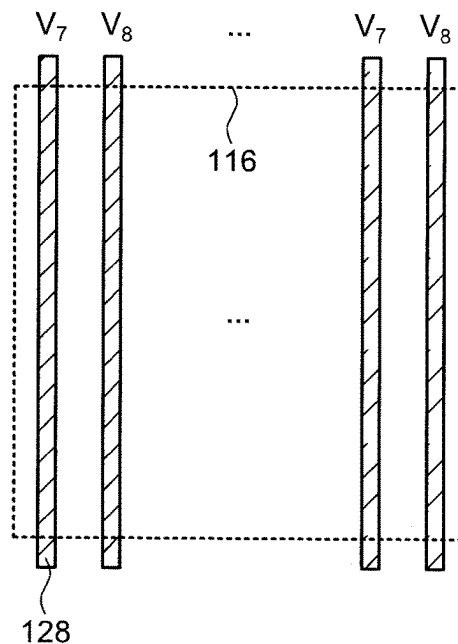
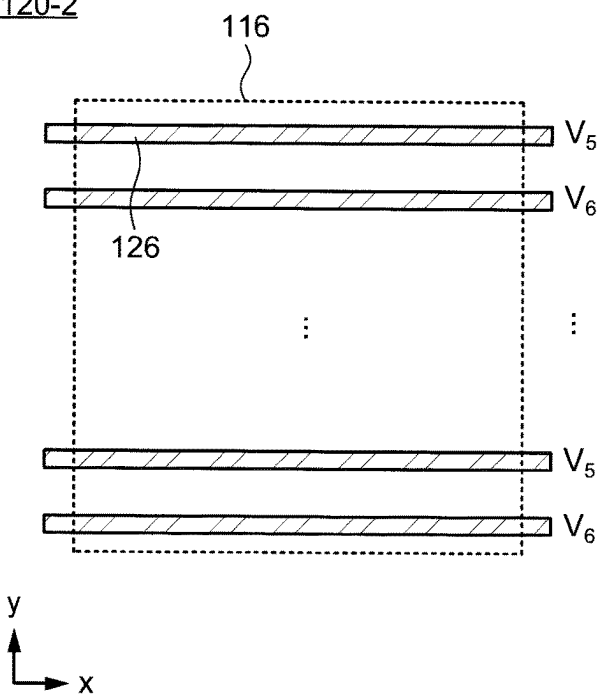


FIG. 13A

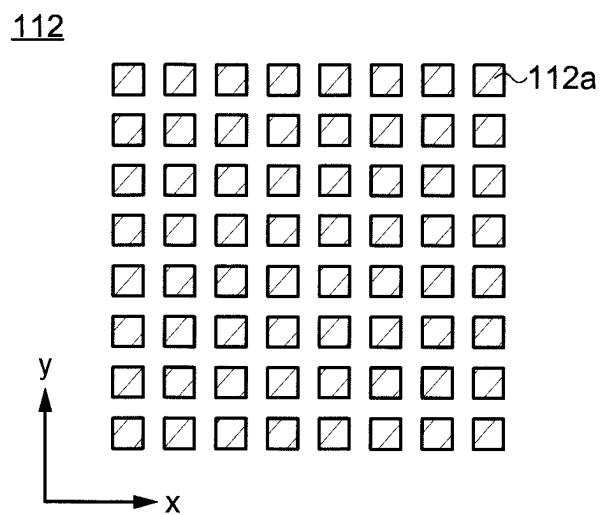


FIG. 13B

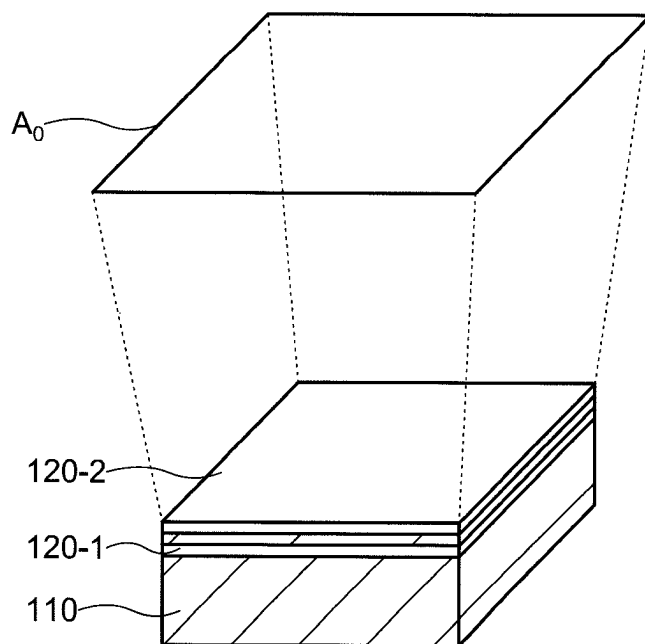


FIG. 14A

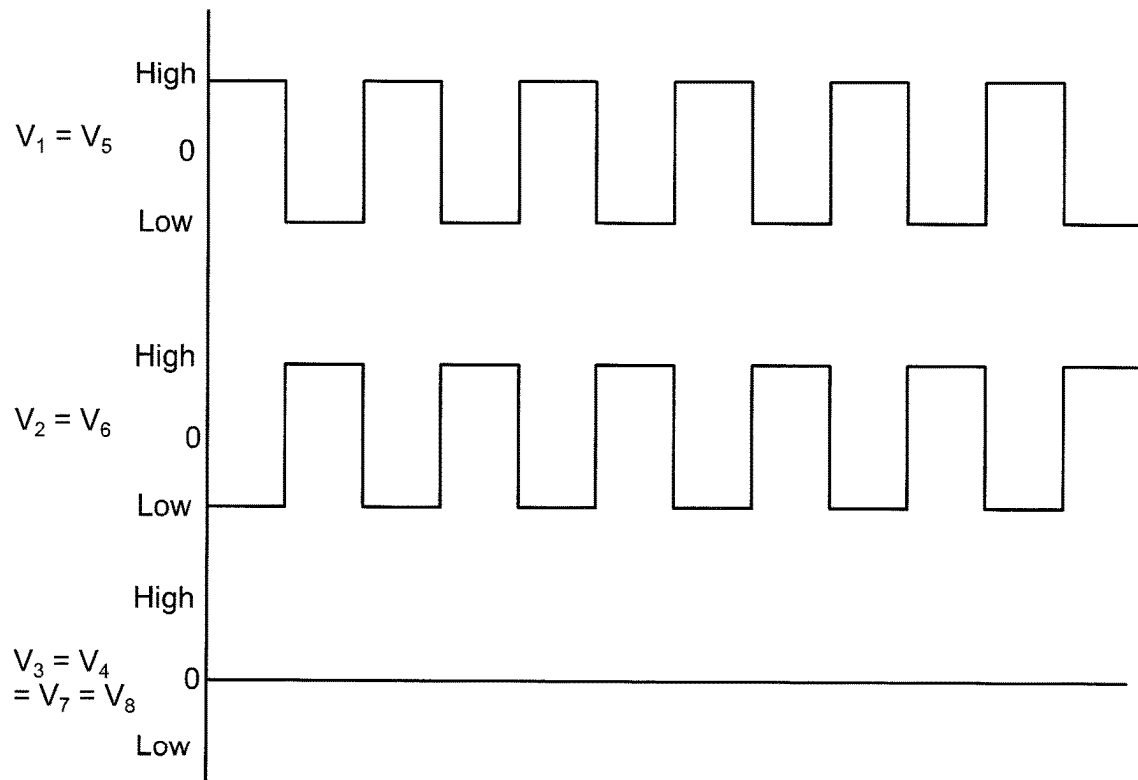


FIG. 14B

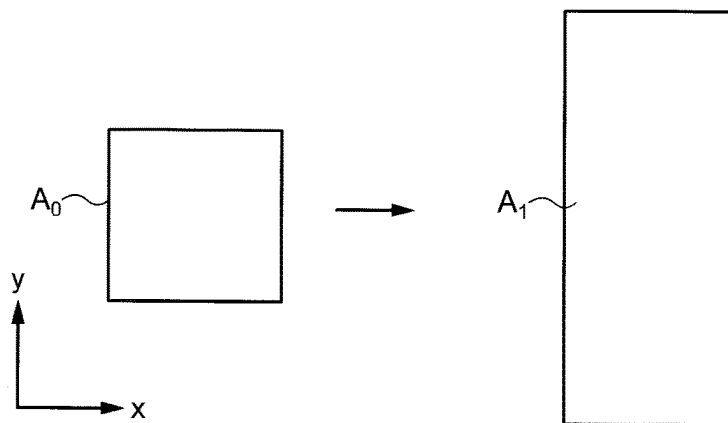


FIG. 15A

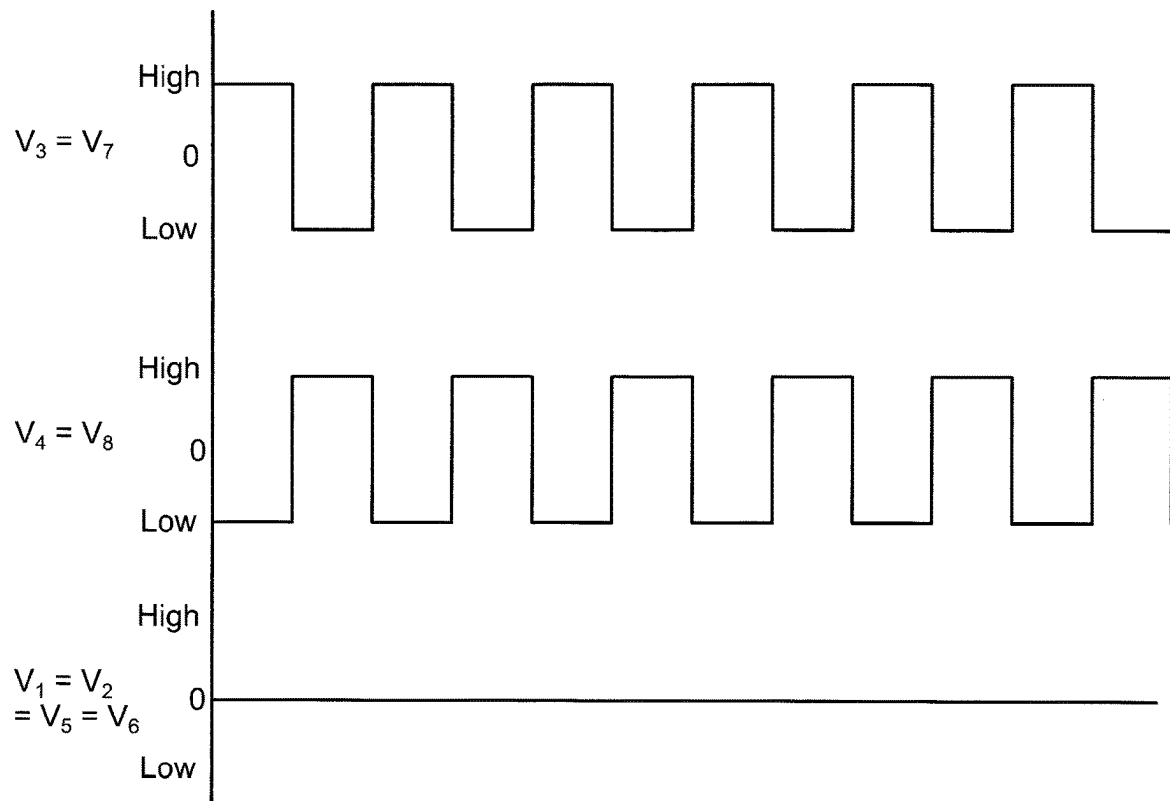


FIG. 15B

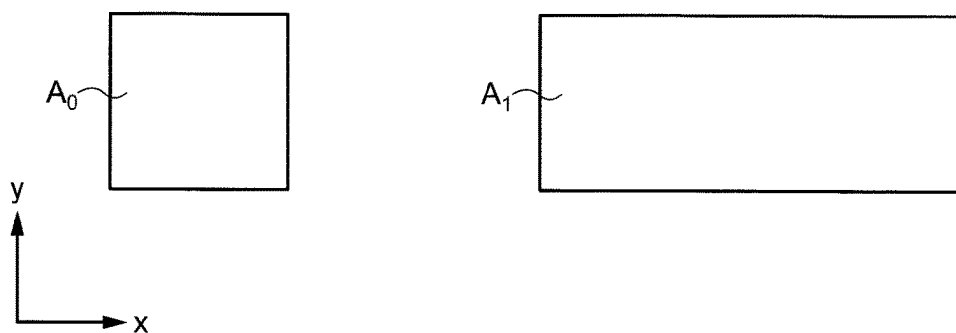


FIG. 16A

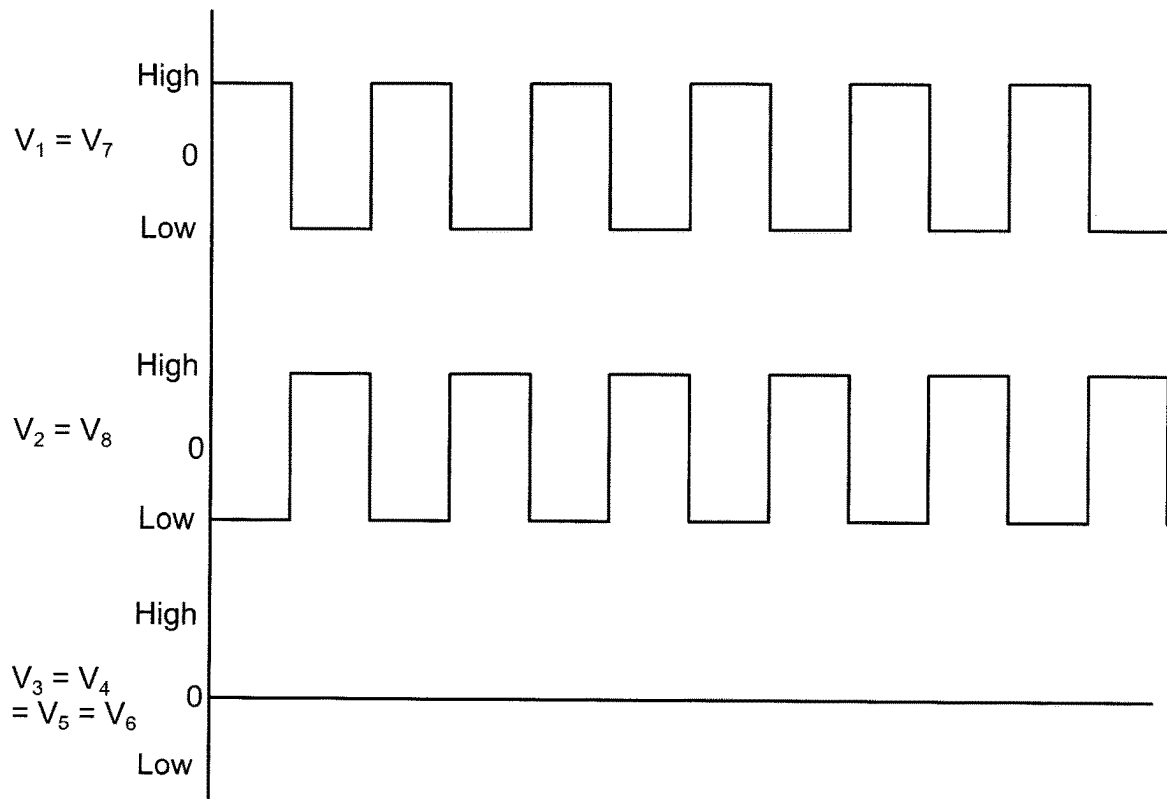


FIG. 16B

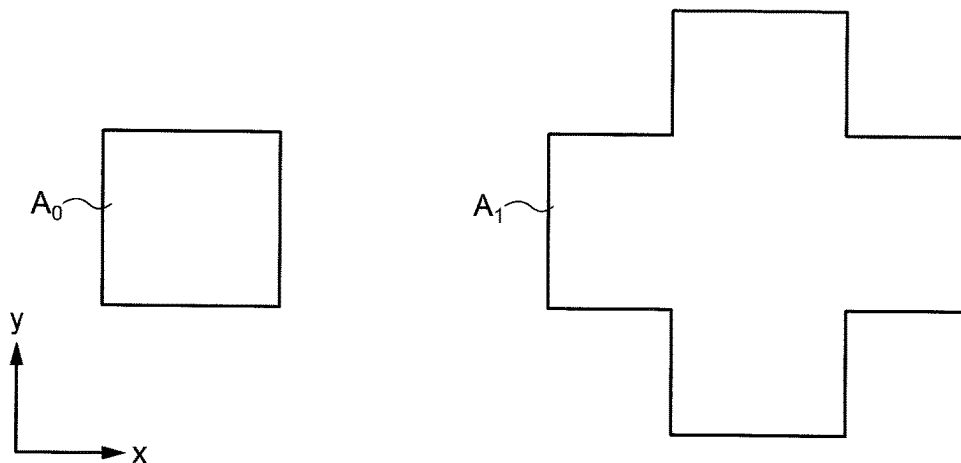


FIG. 17A

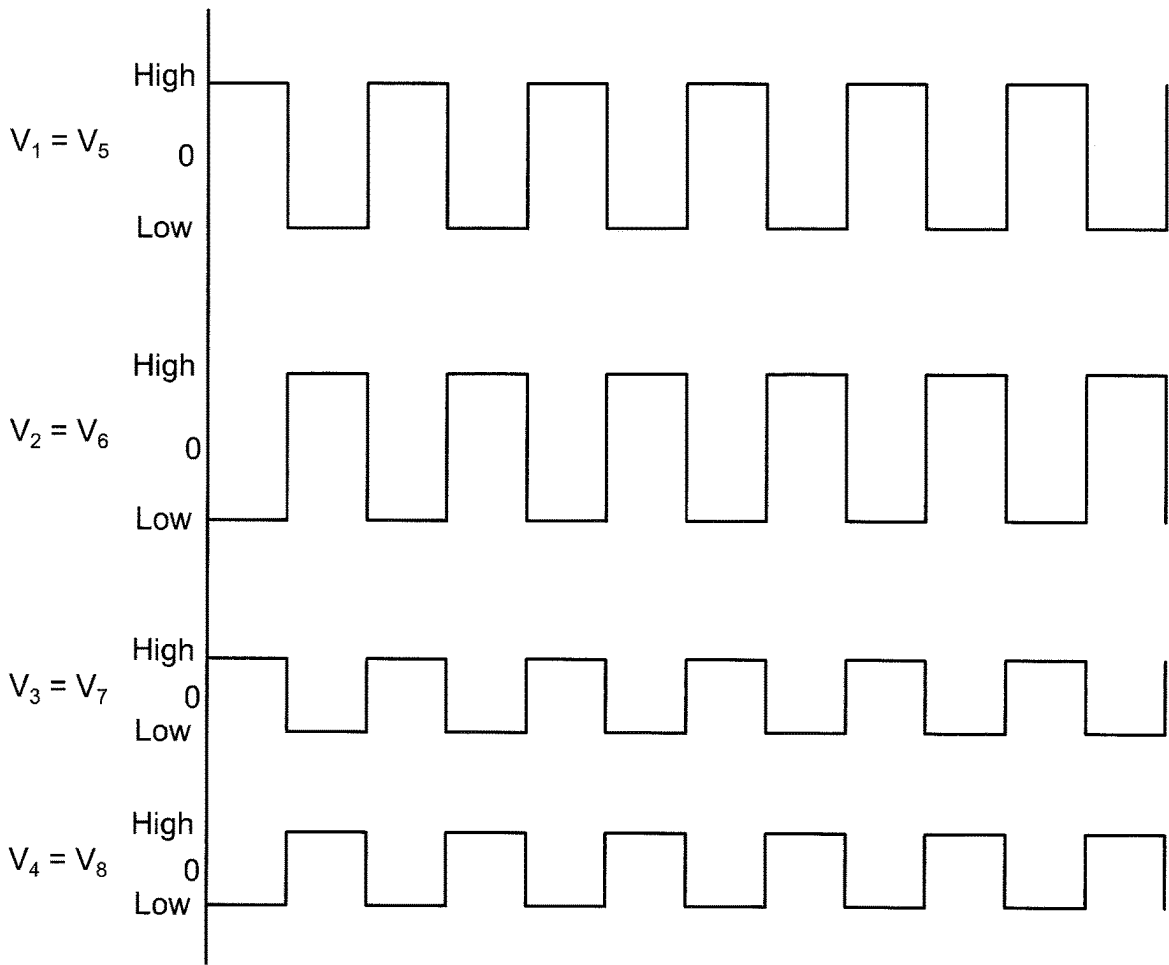


FIG. 17B

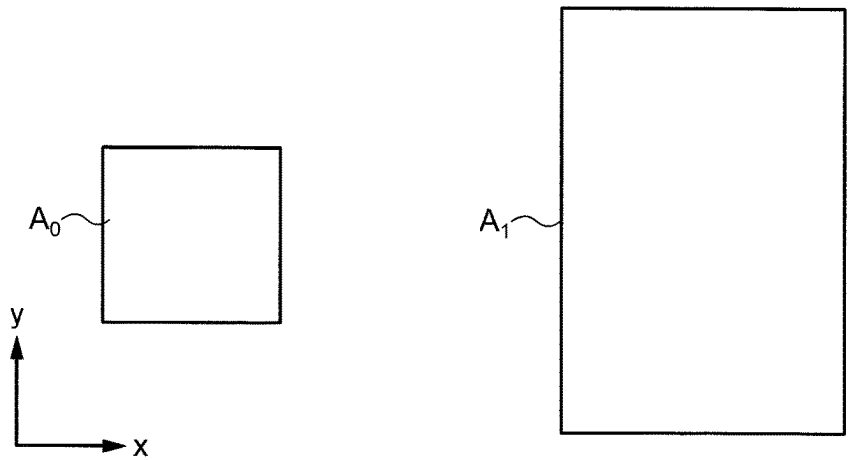


FIG. 18A

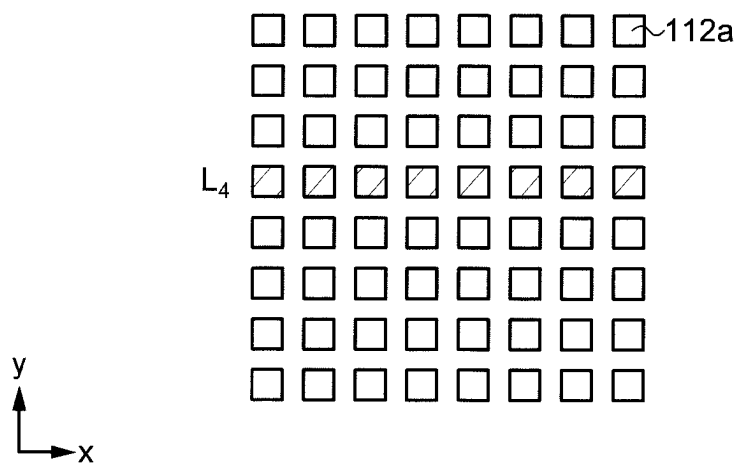
112

FIG. 18B

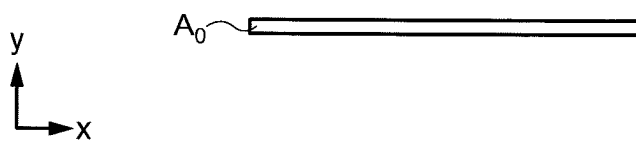


FIG. 18C

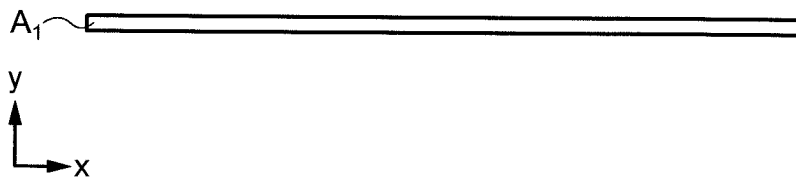


FIG. 19A

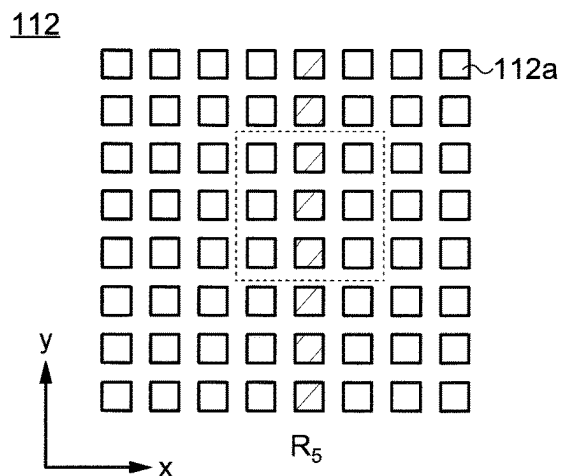


FIG. 19B

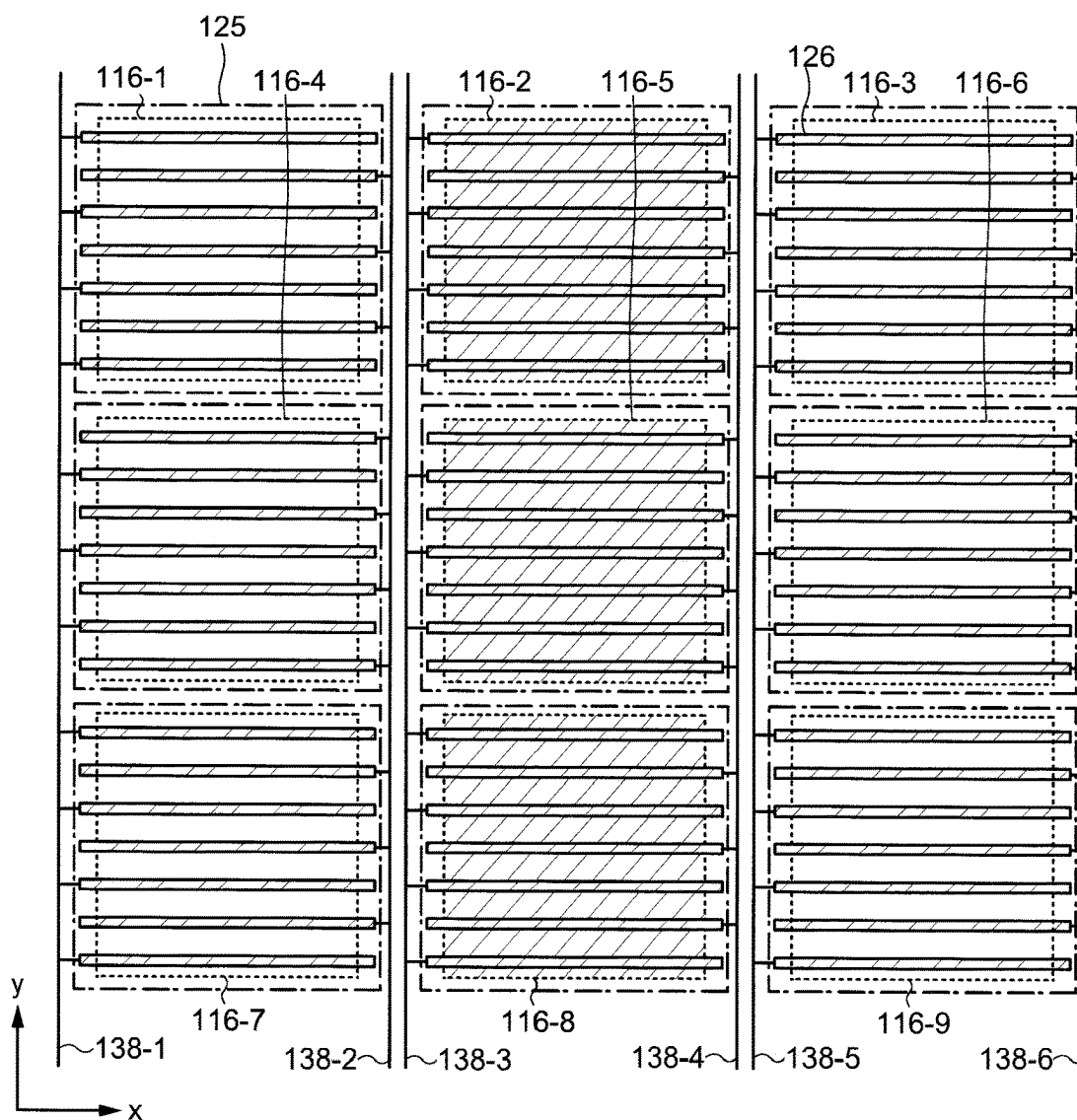


FIG. 20

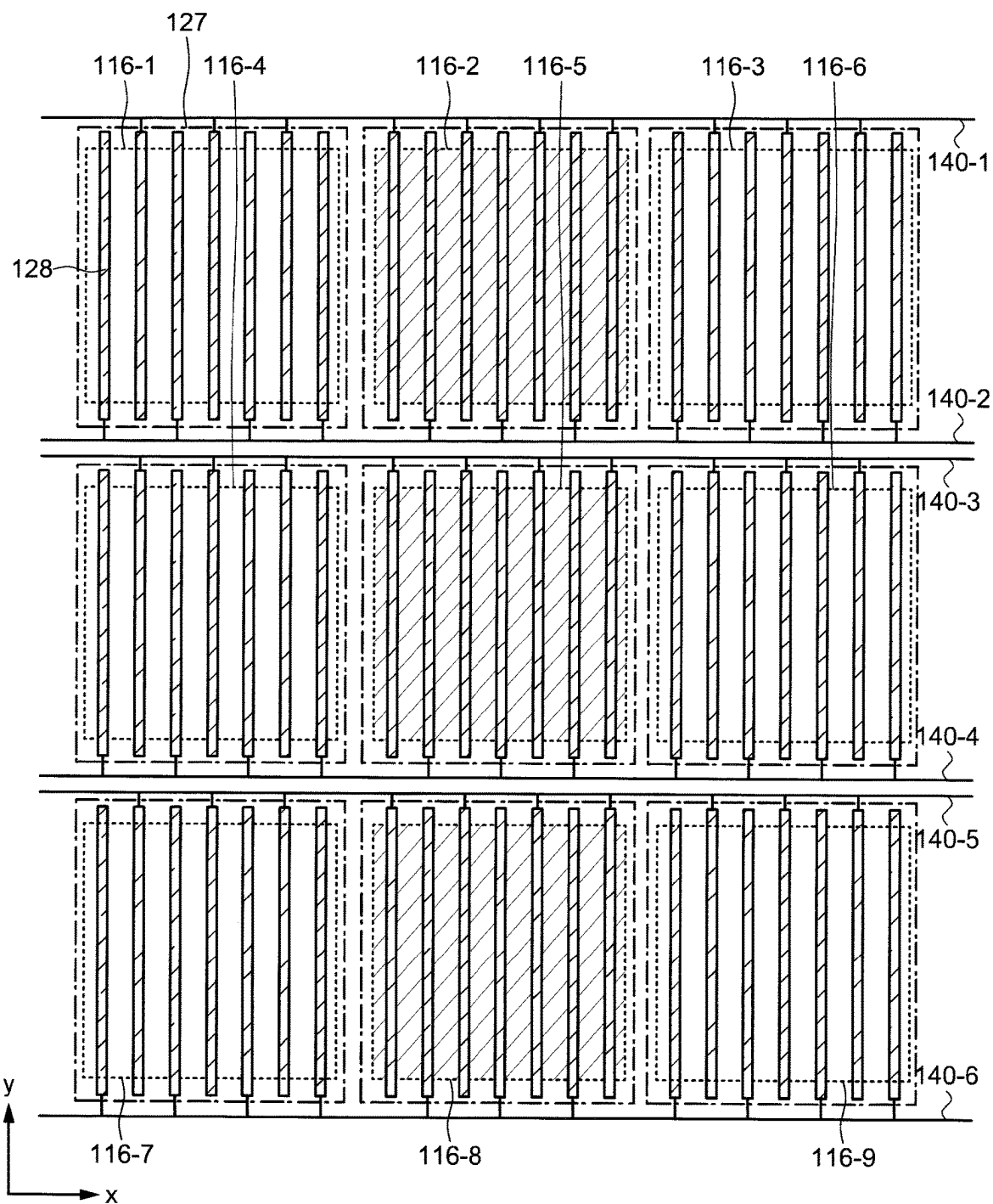


FIG. 21

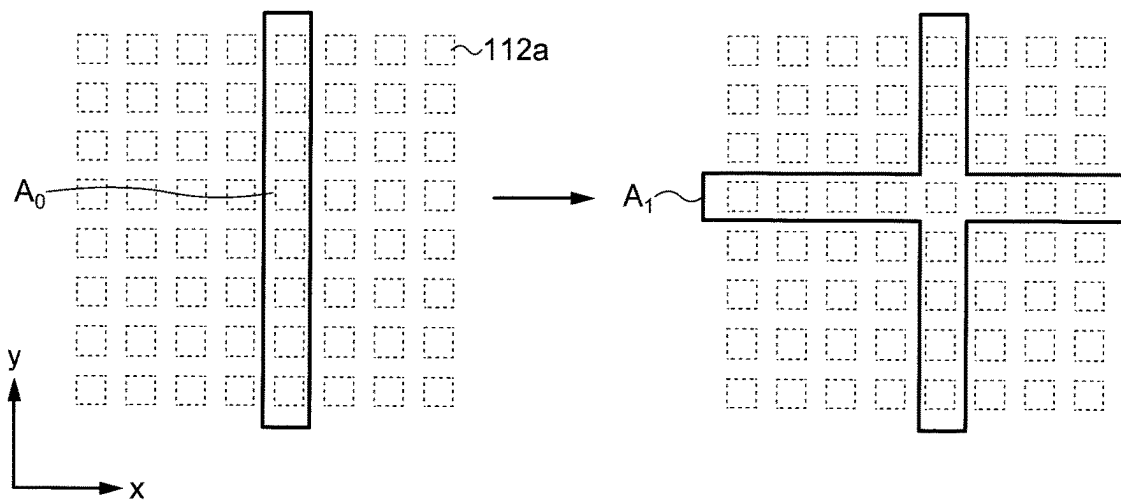


FIG. 22A

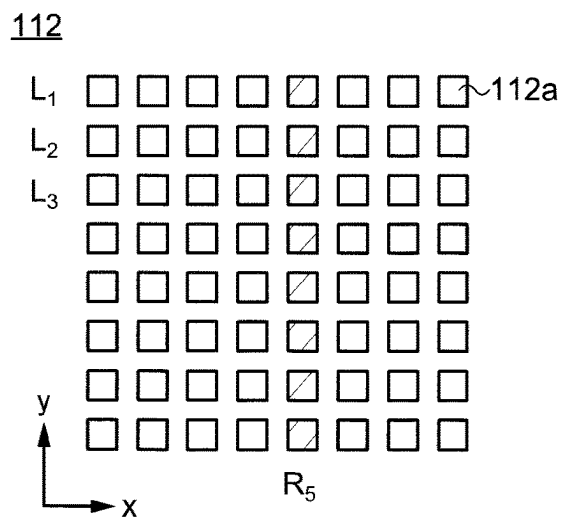


FIG. 22B

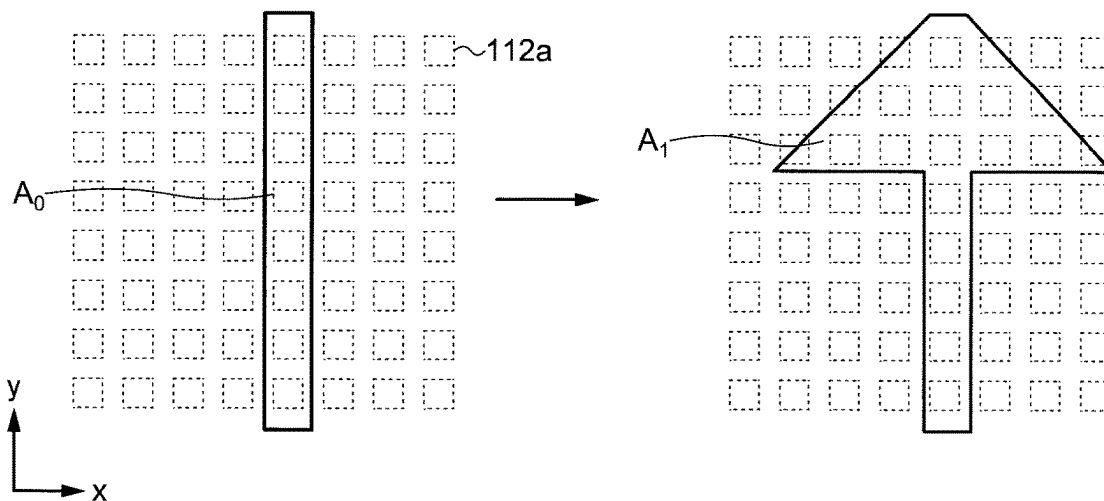


FIG. 23A

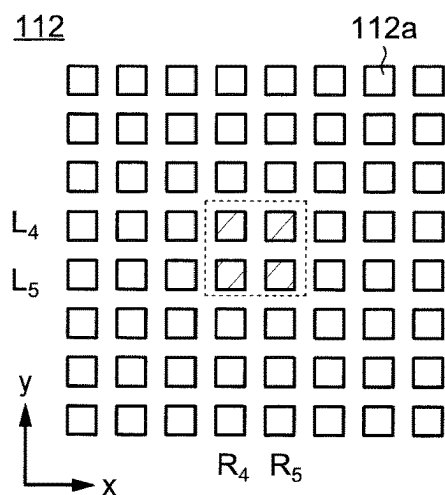


FIG. 23C

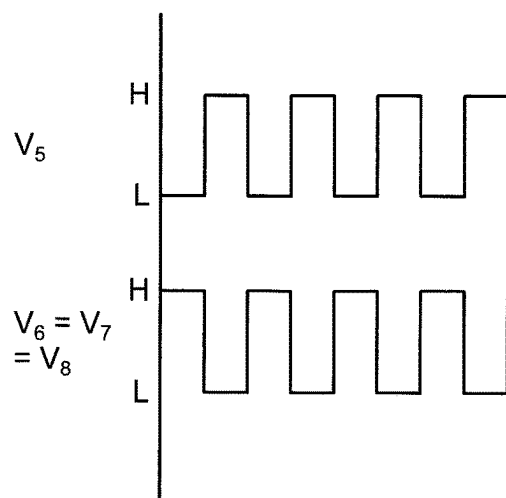


FIG. 23B

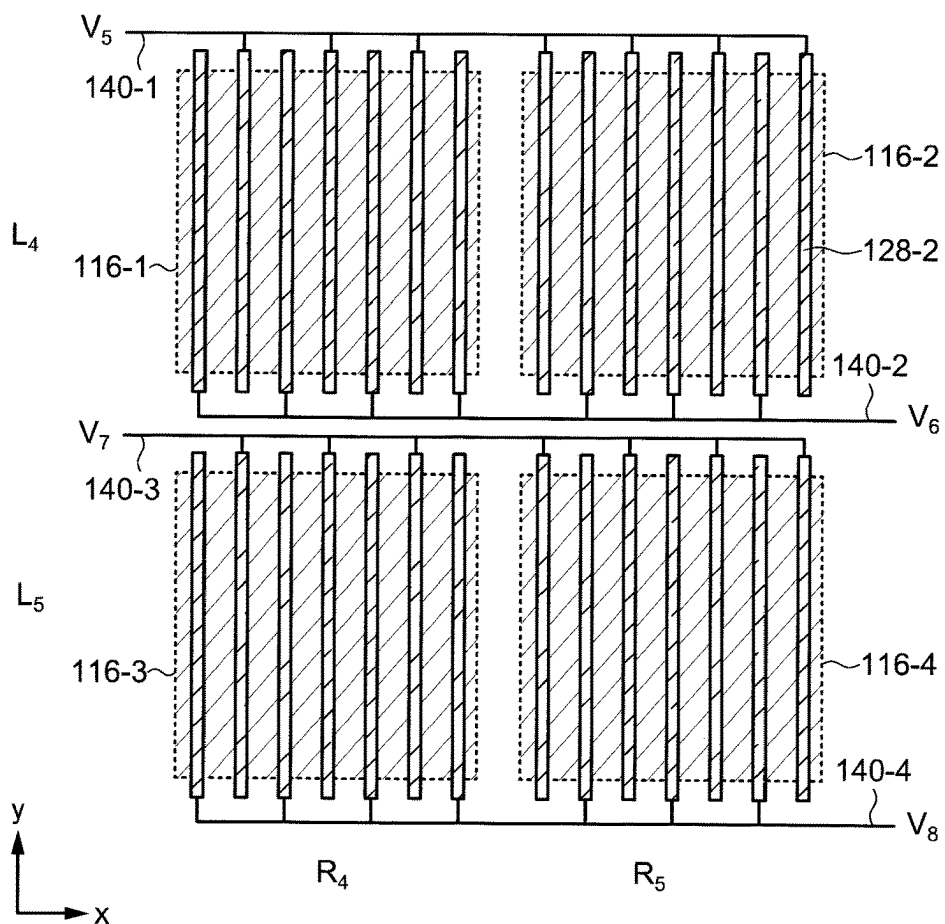


FIG. 24A

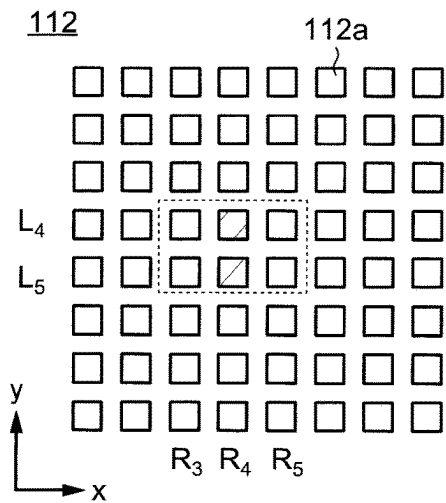


FIG. 24B

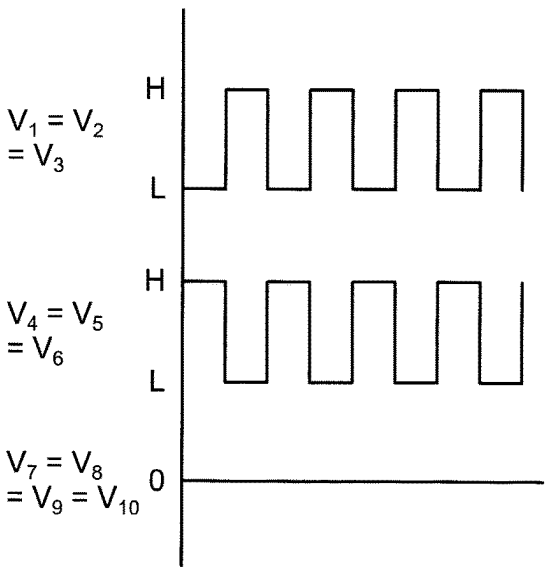


FIG. 24C

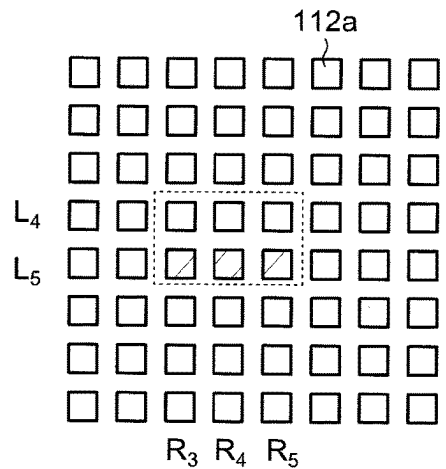


FIG. 24D

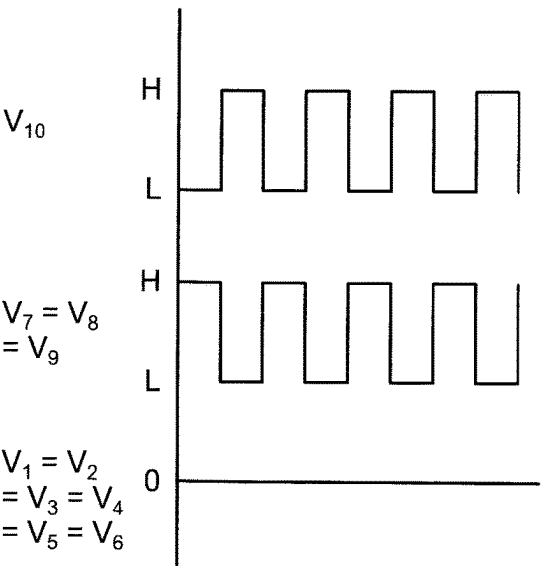


FIG. 25

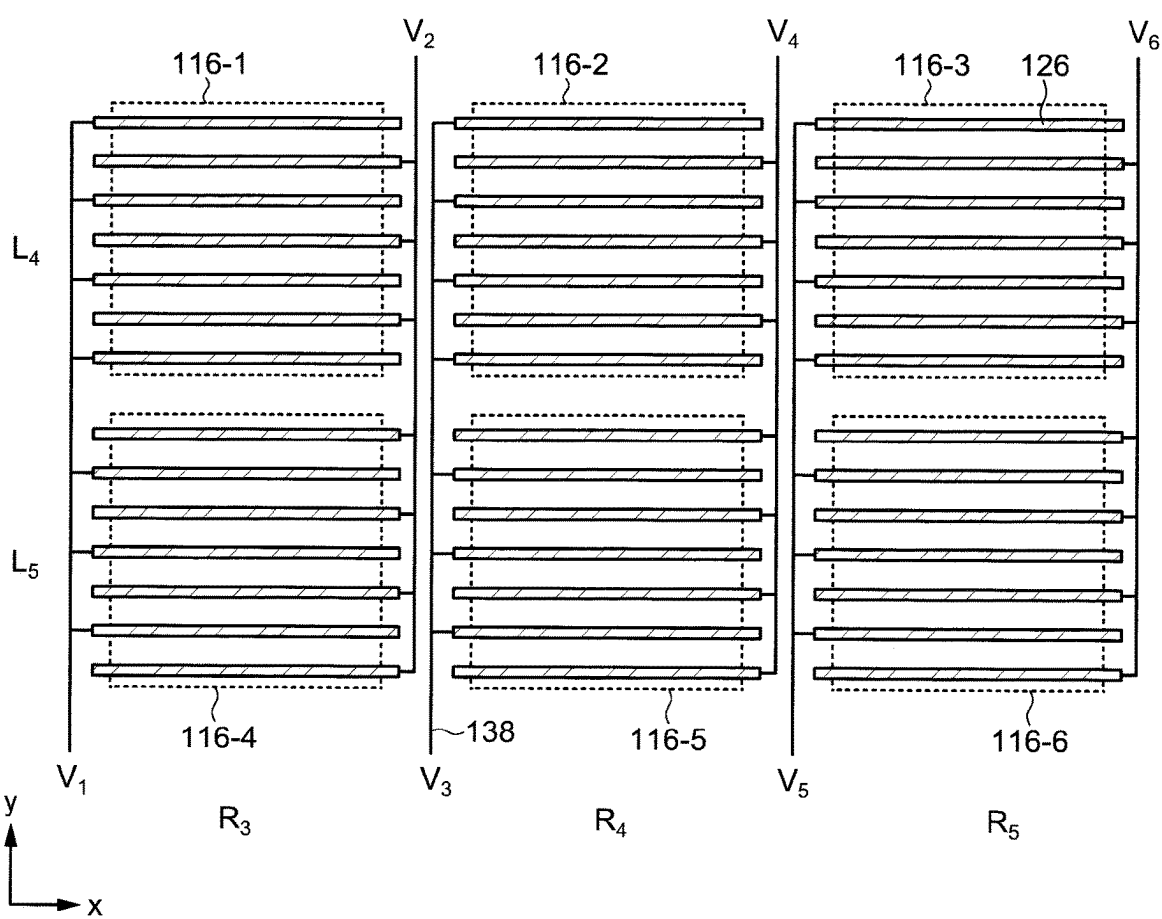


FIG. 26

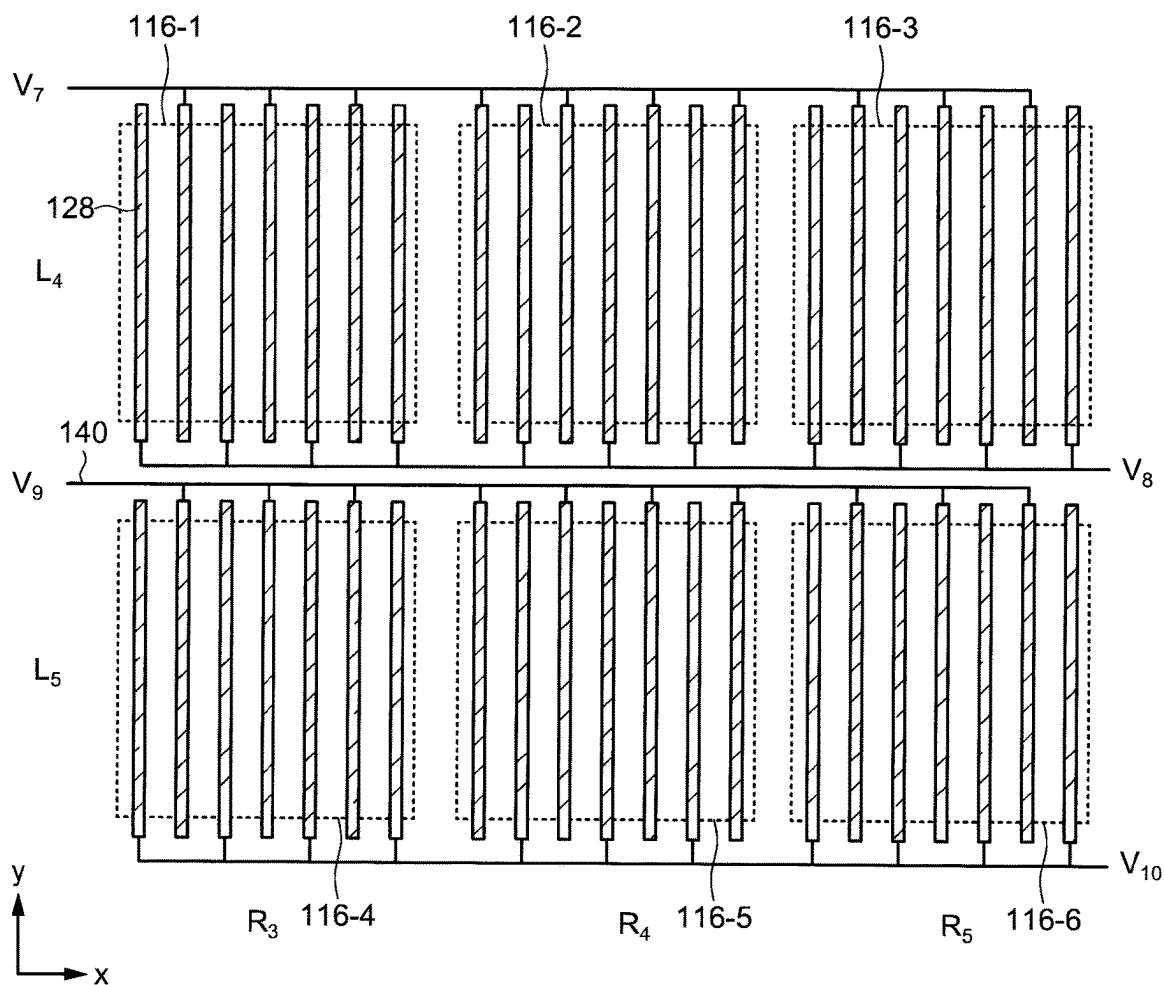


FIG. 27

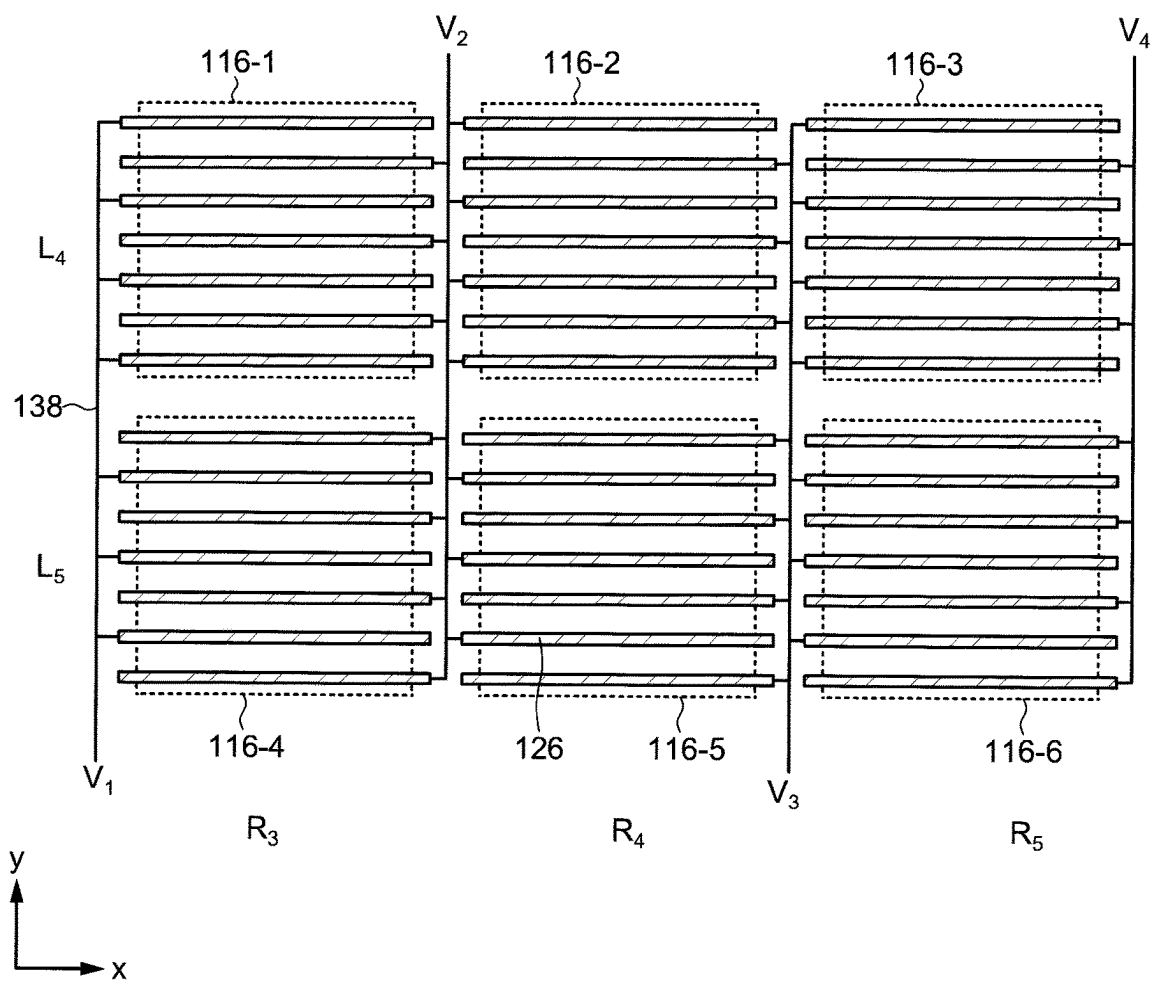


FIG. 28

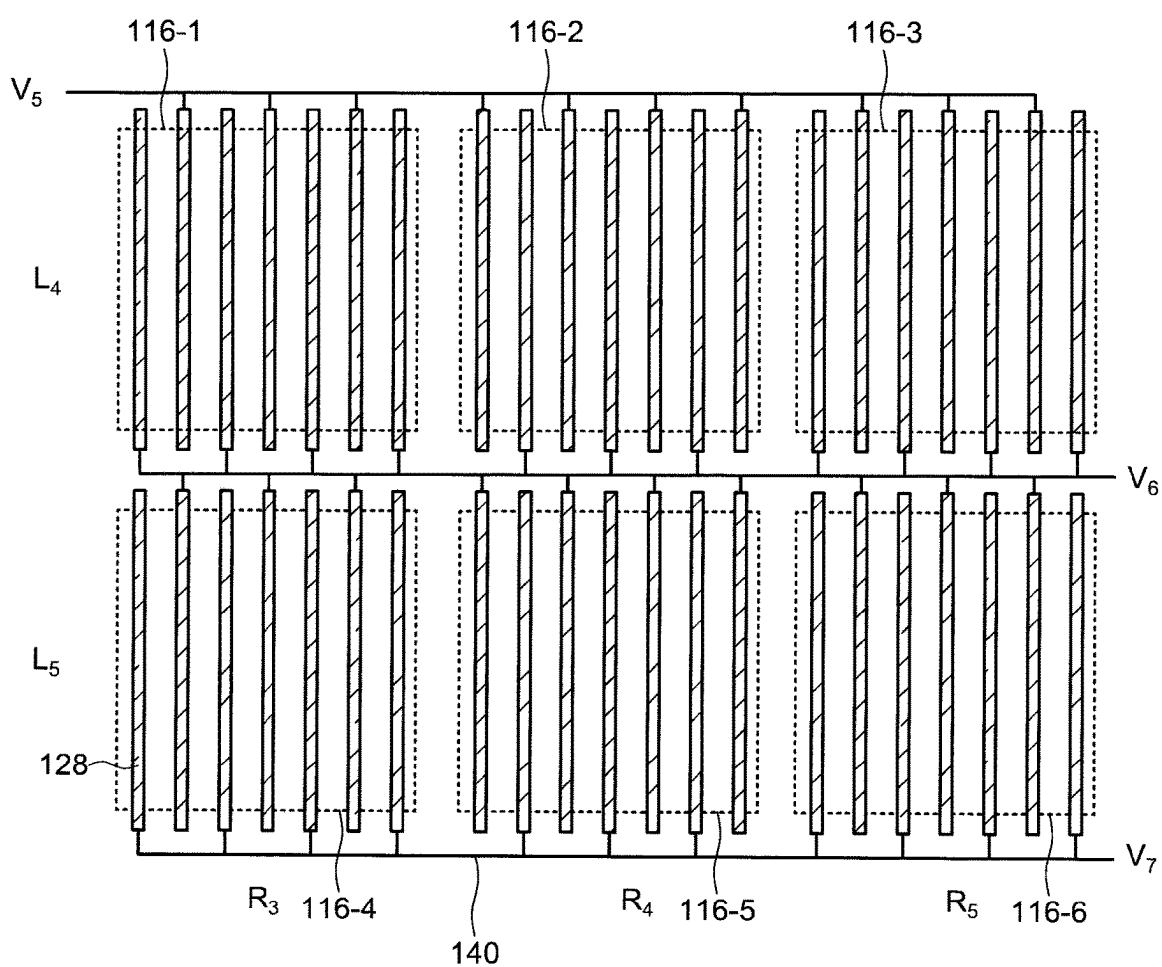


FIG. 29A

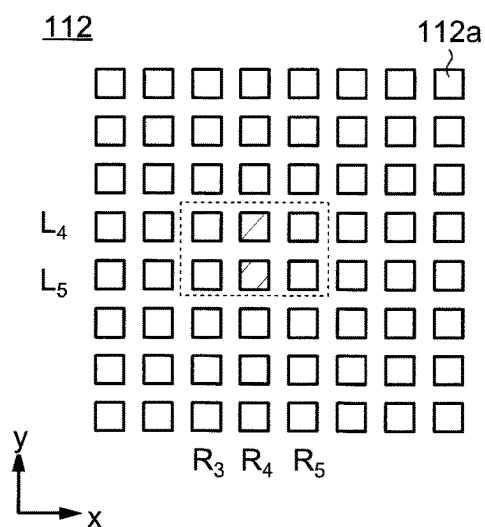


FIG. 29B

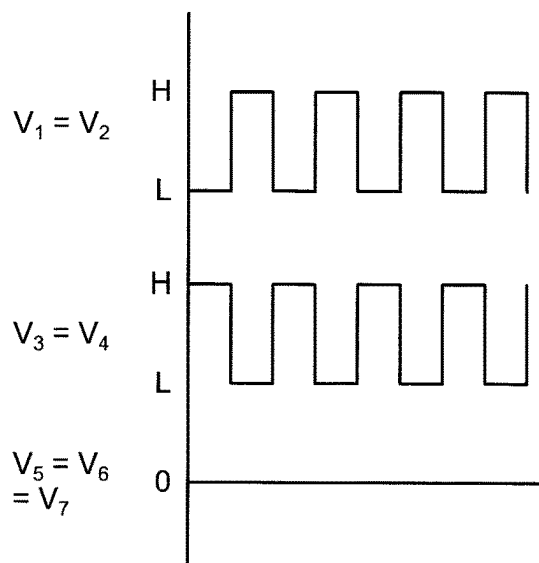


FIG. 29C

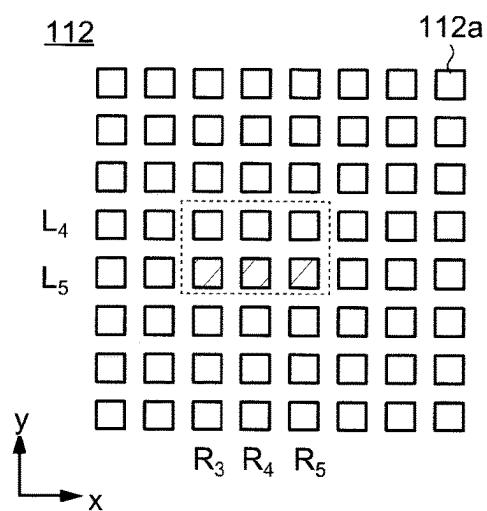


FIG. 29D

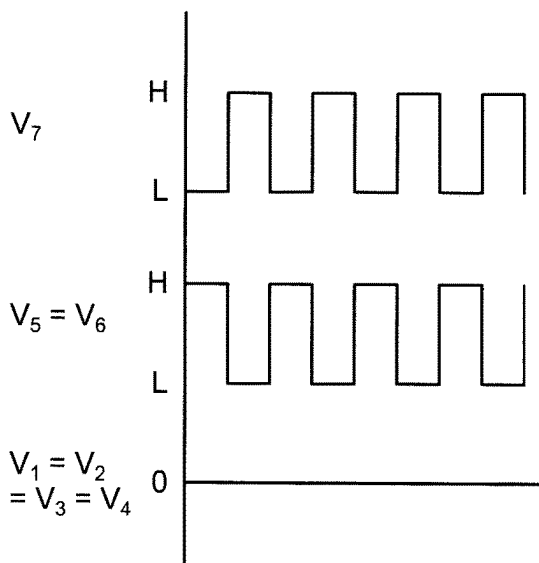


FIG. 30

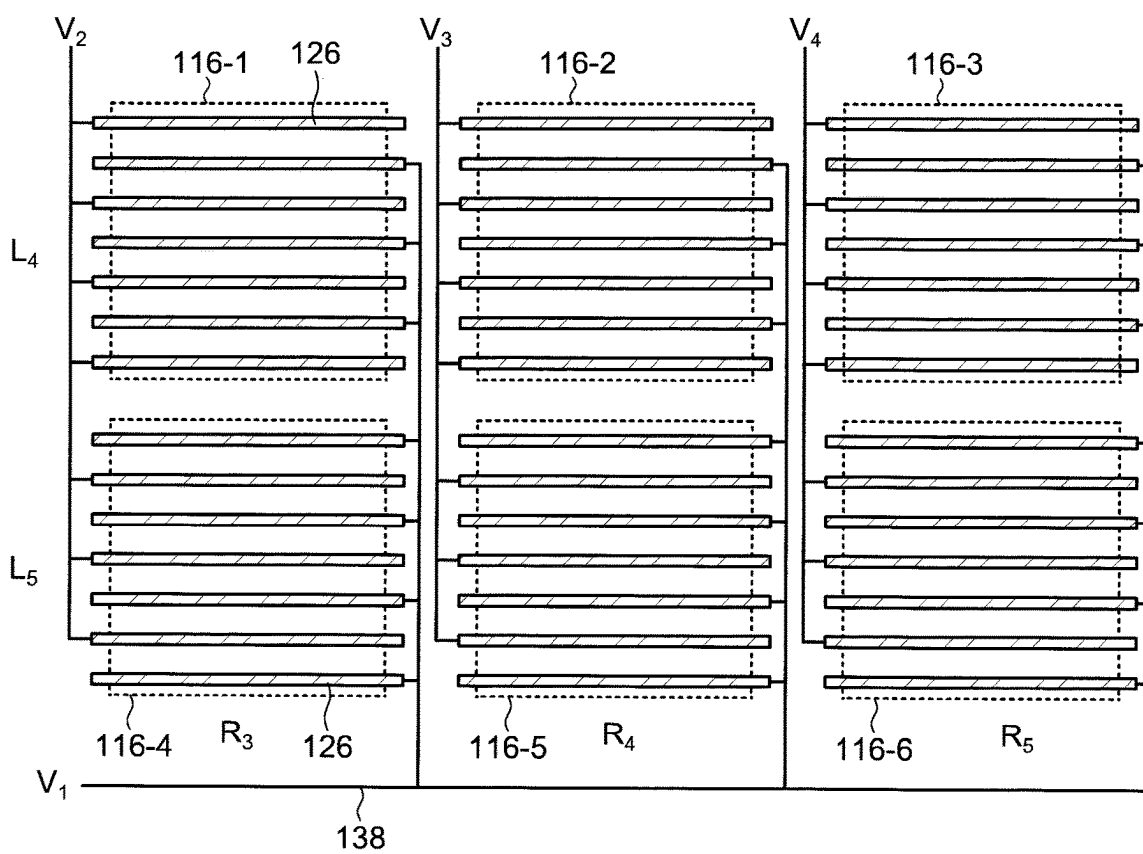


FIG. 31

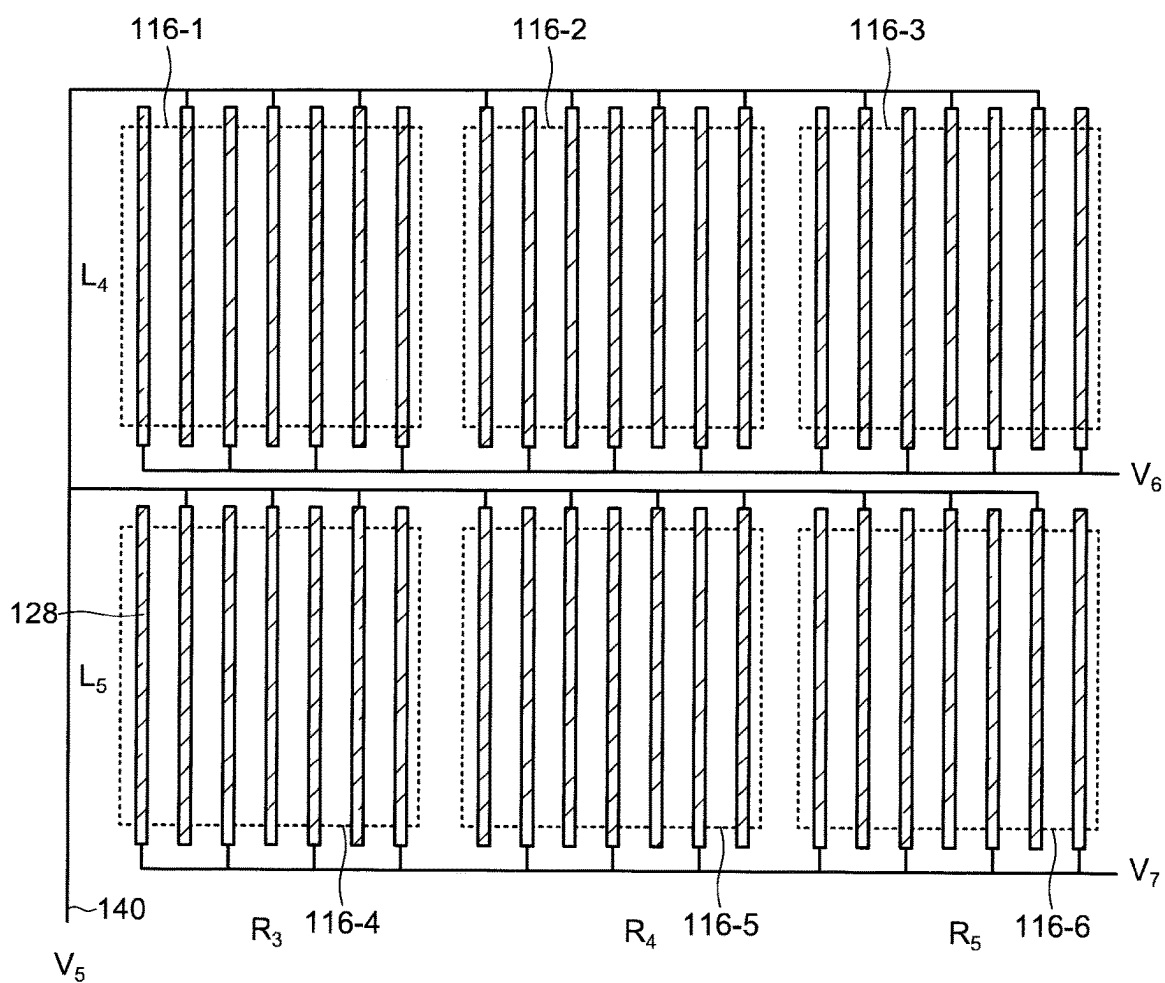


FIG. 32A

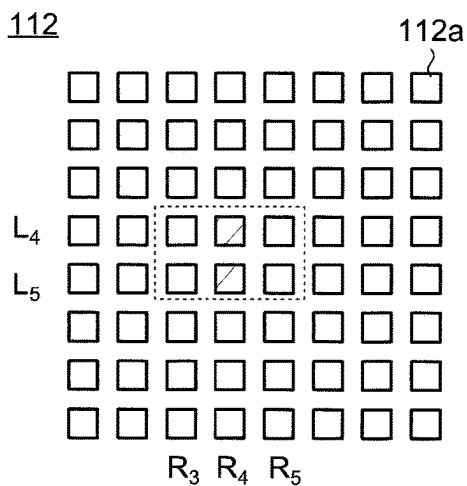


FIG. 32B

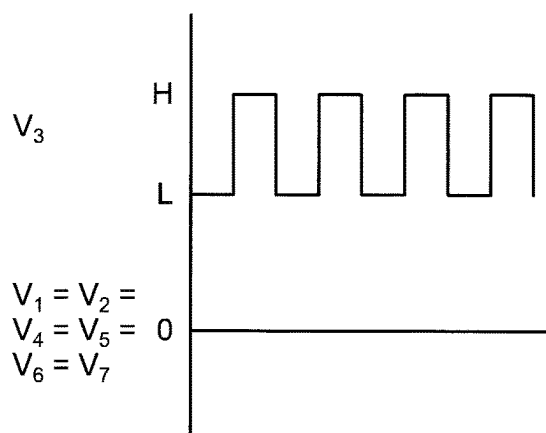


FIG. 32C

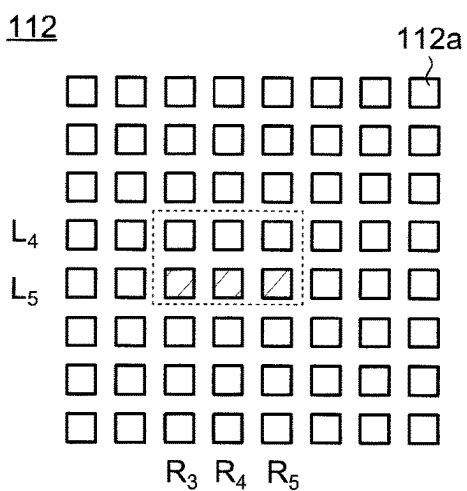


FIG. 32D

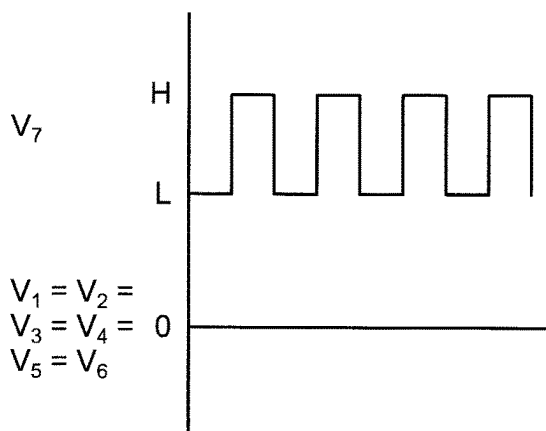


FIG. 33A

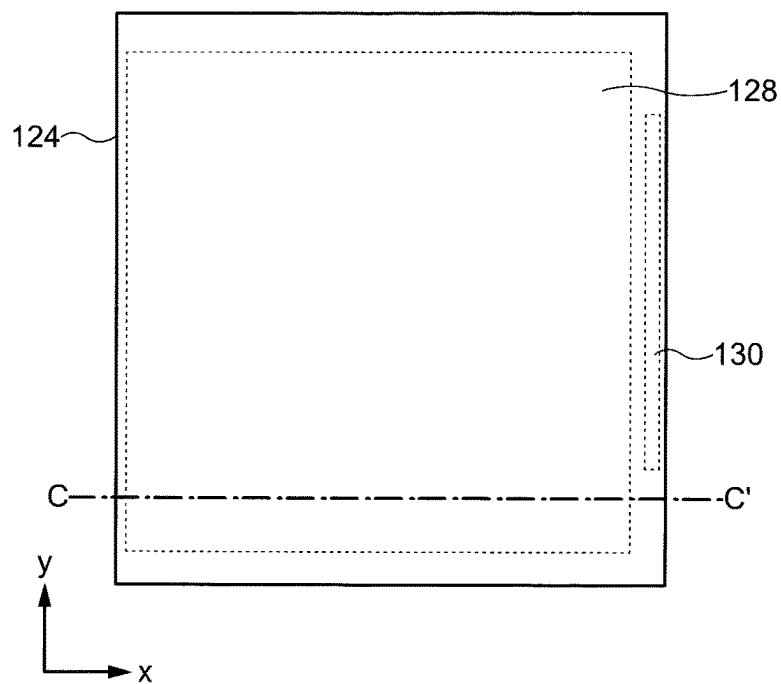


FIG. 33B

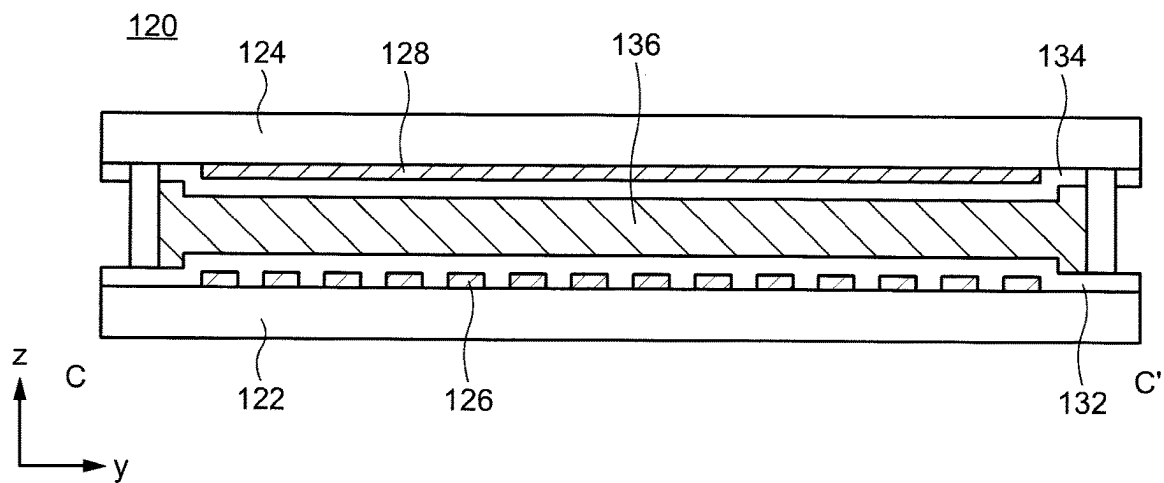


FIG. 34A

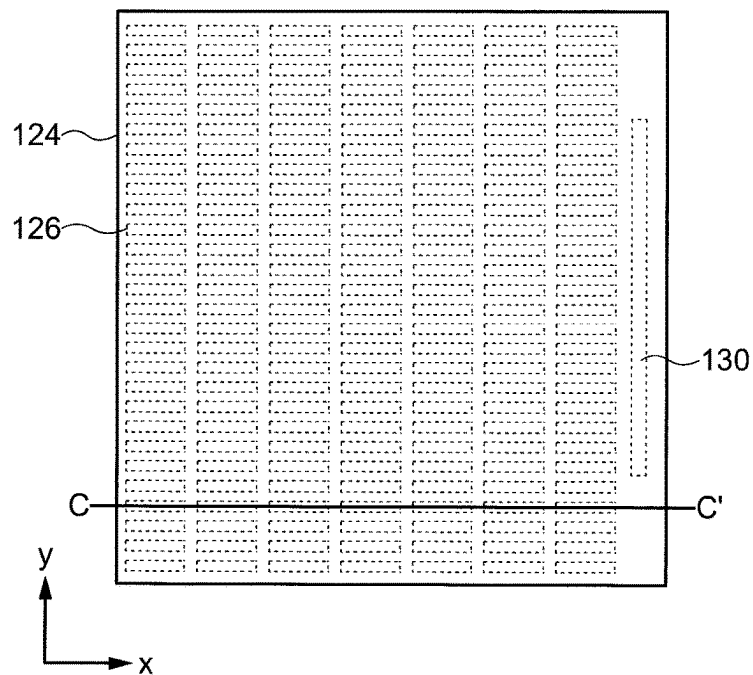


FIG. 34B

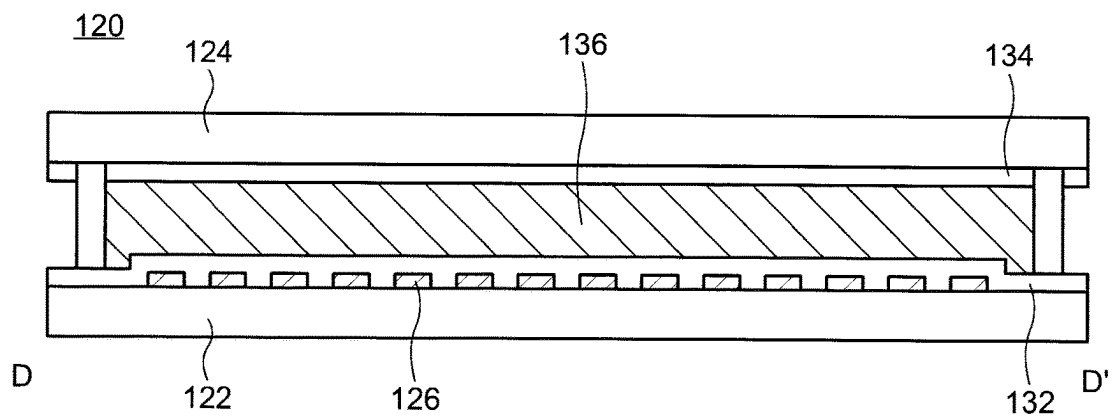


FIG. 35A

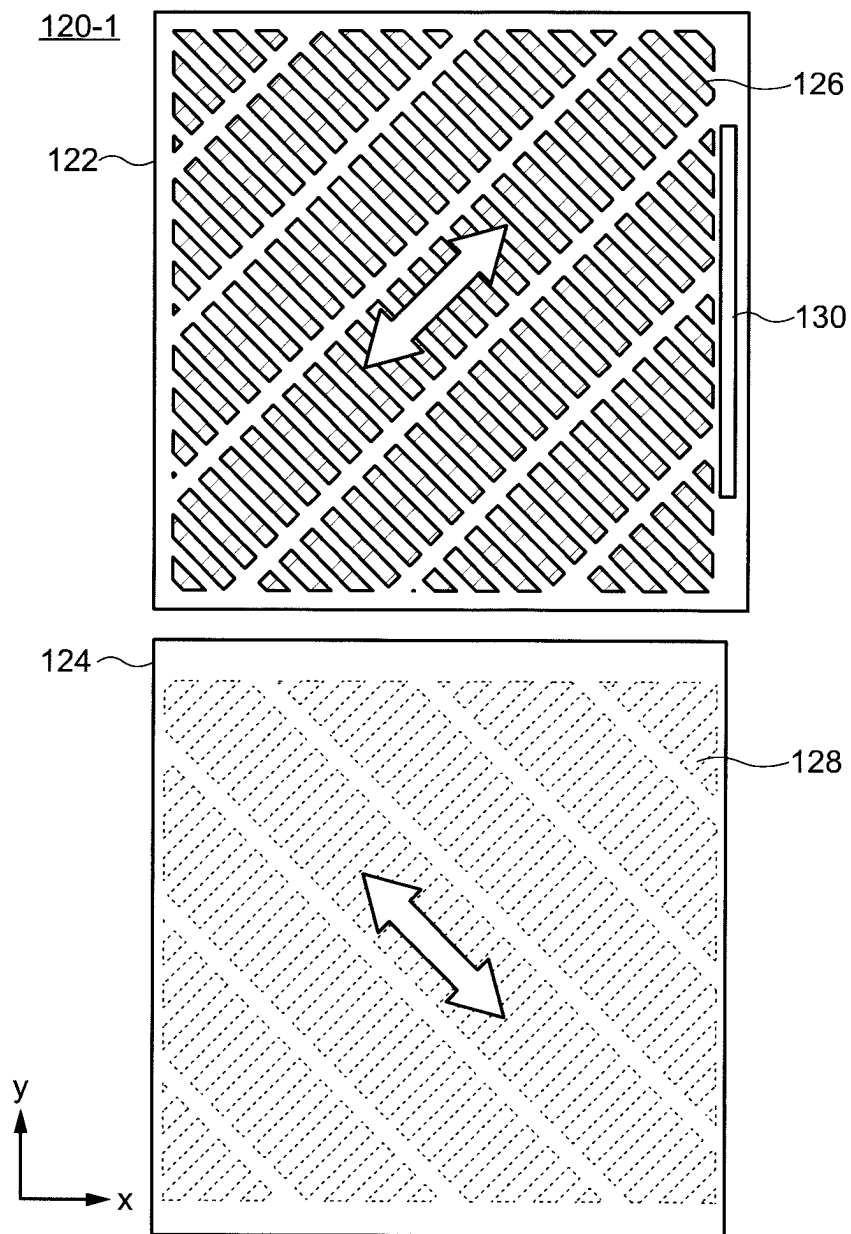


FIG. 35B

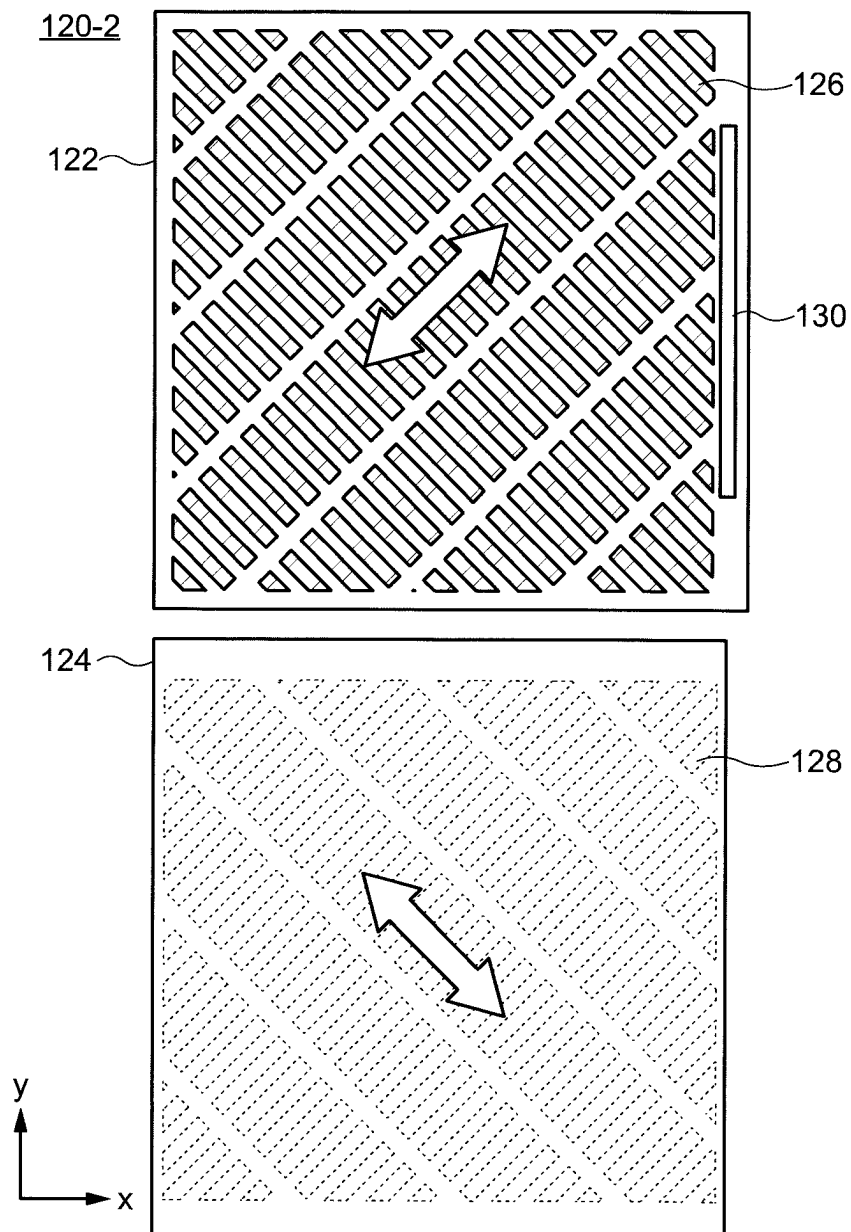


FIG. 36A

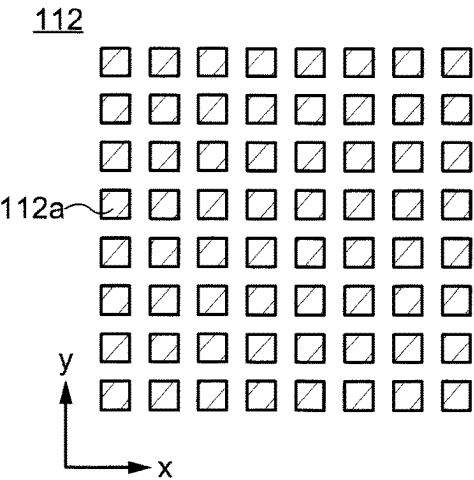


FIG. 36B

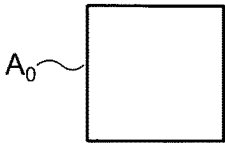


FIG. 36C

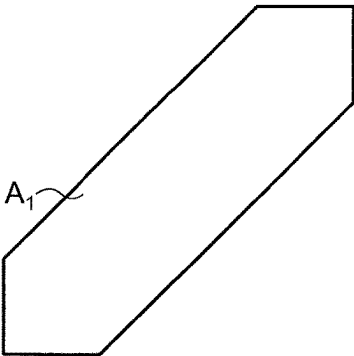


FIG. 36D

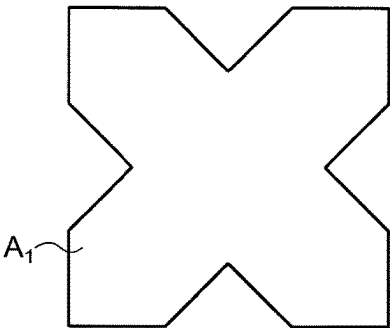


FIG. 36E

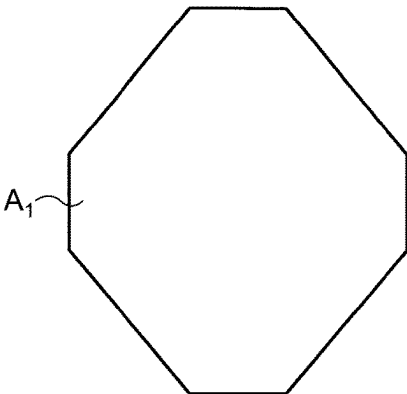


FIG. 36F

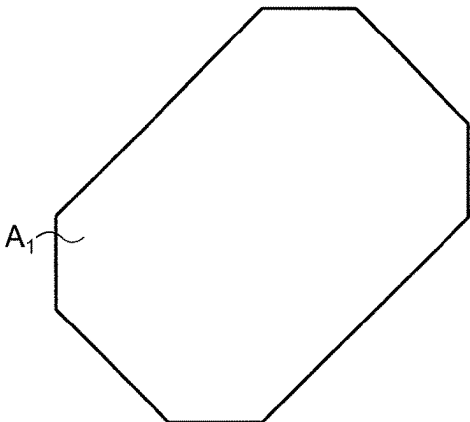


FIG. 37A

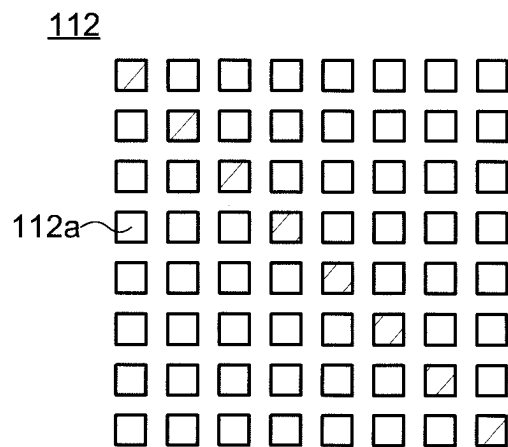


FIG. 37B

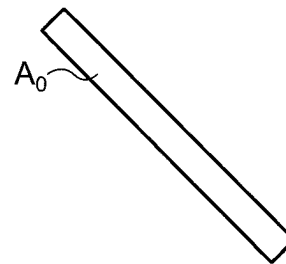


FIG. 37C

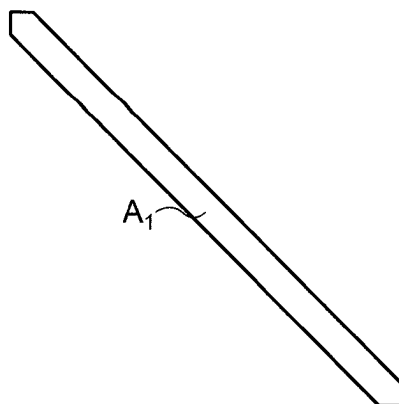


FIG. 38A

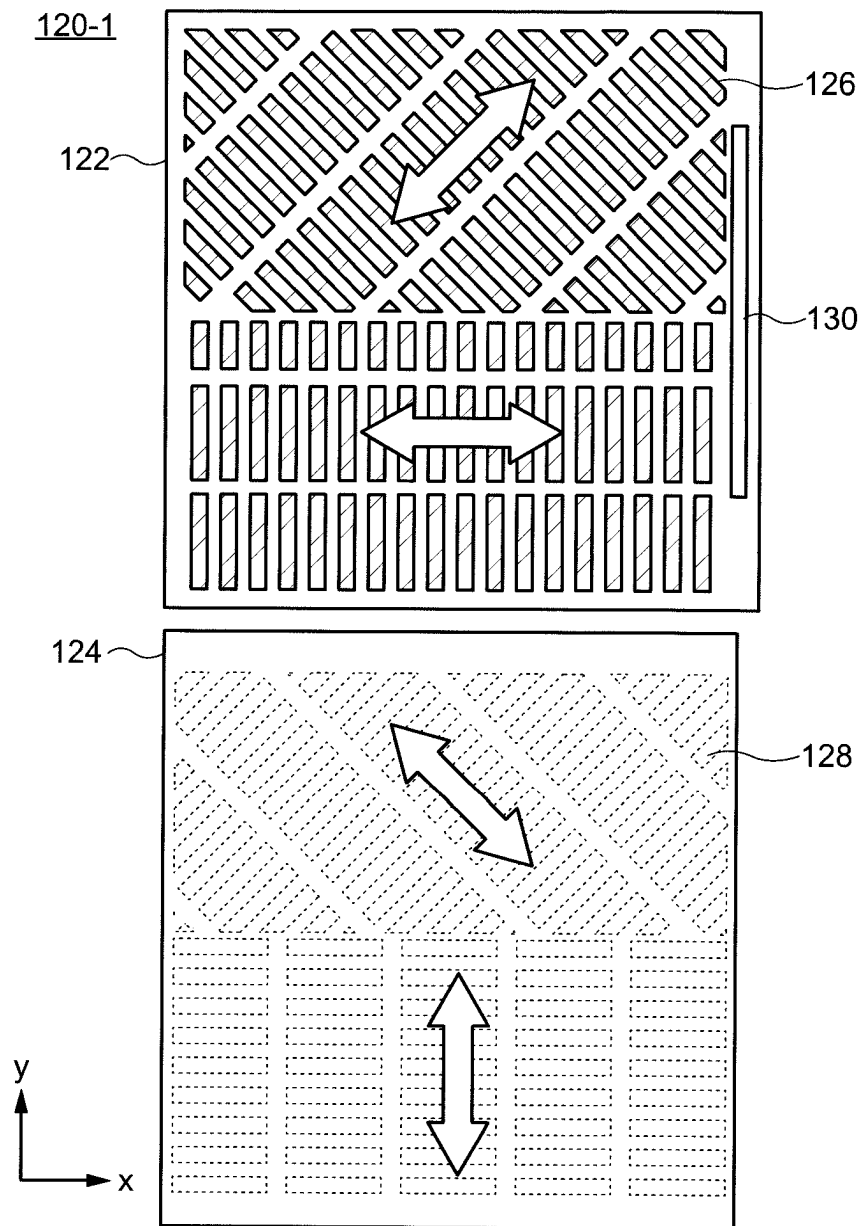


FIG. 38B

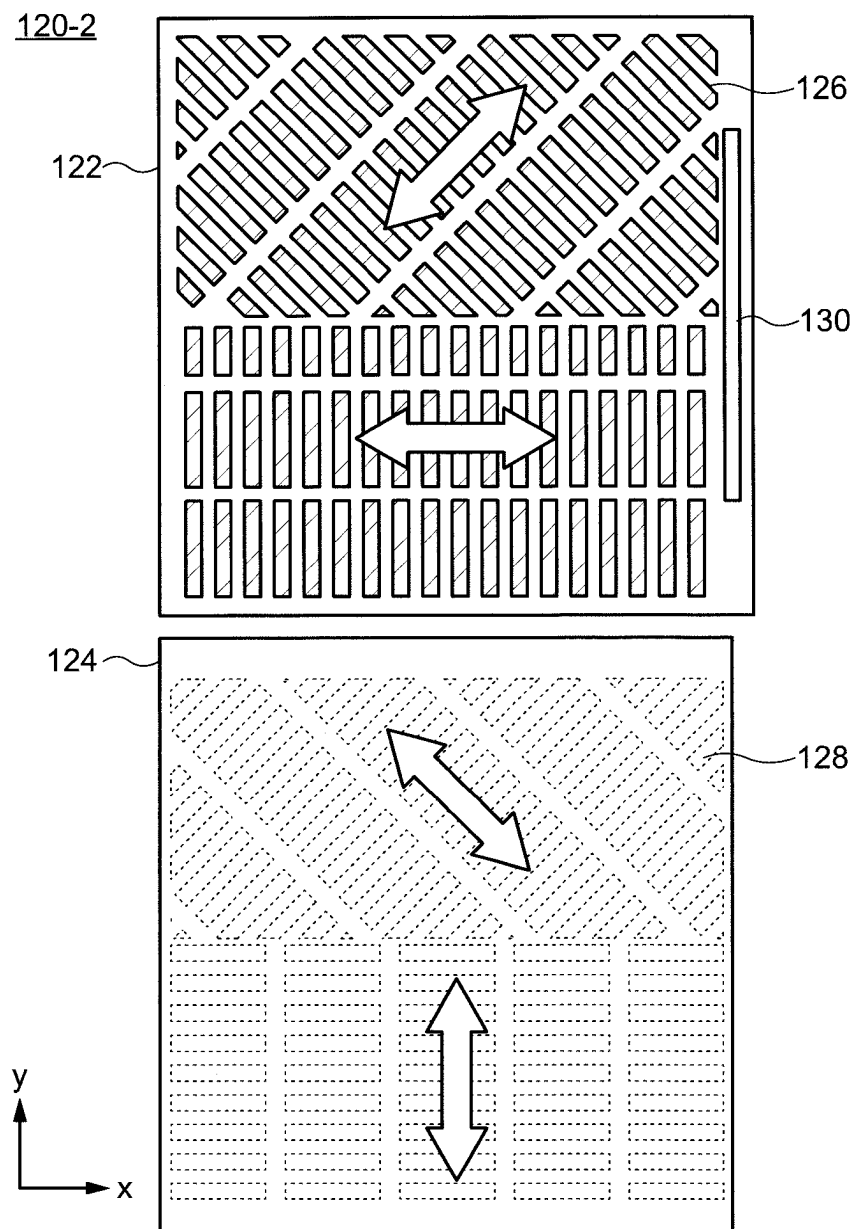


FIG. 39

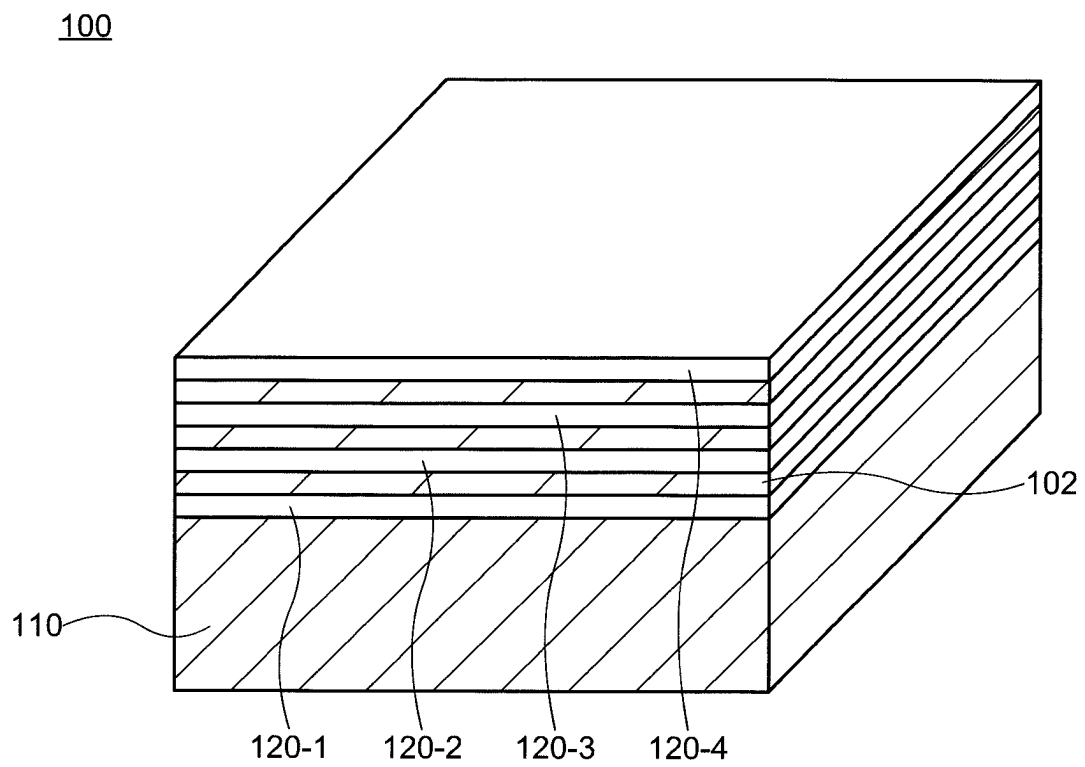


FIG. 40

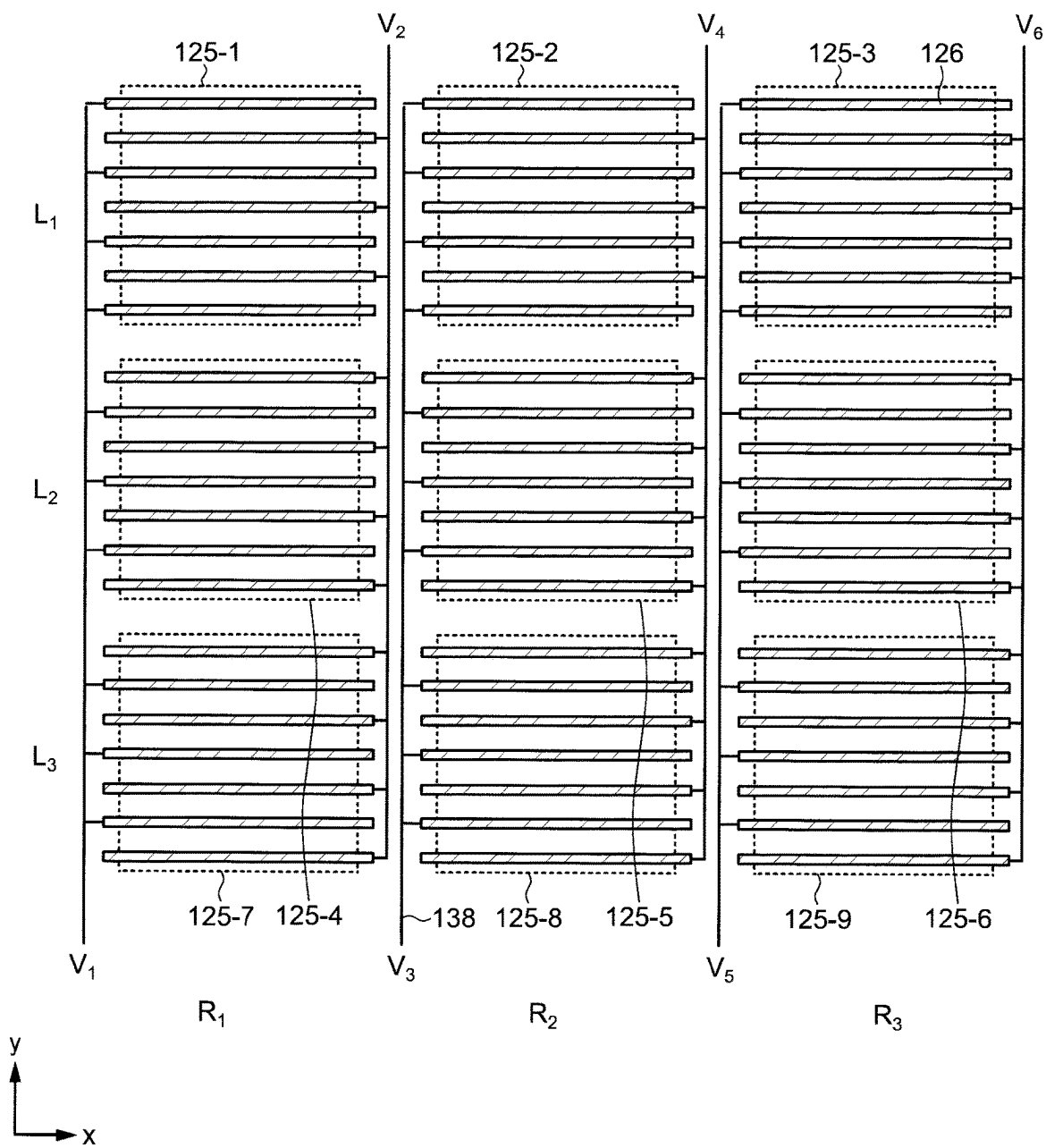


FIG. 41

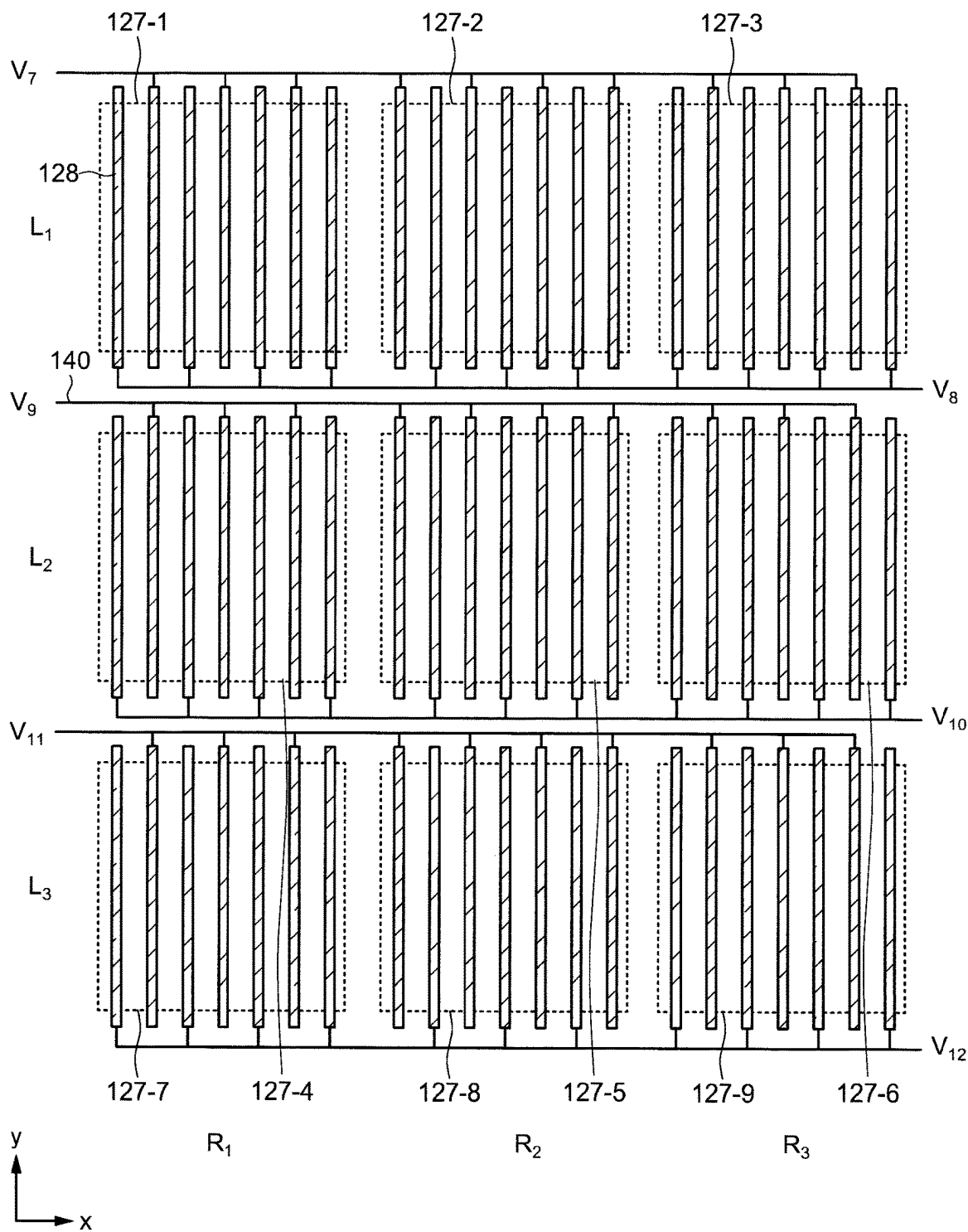


FIG. 42A

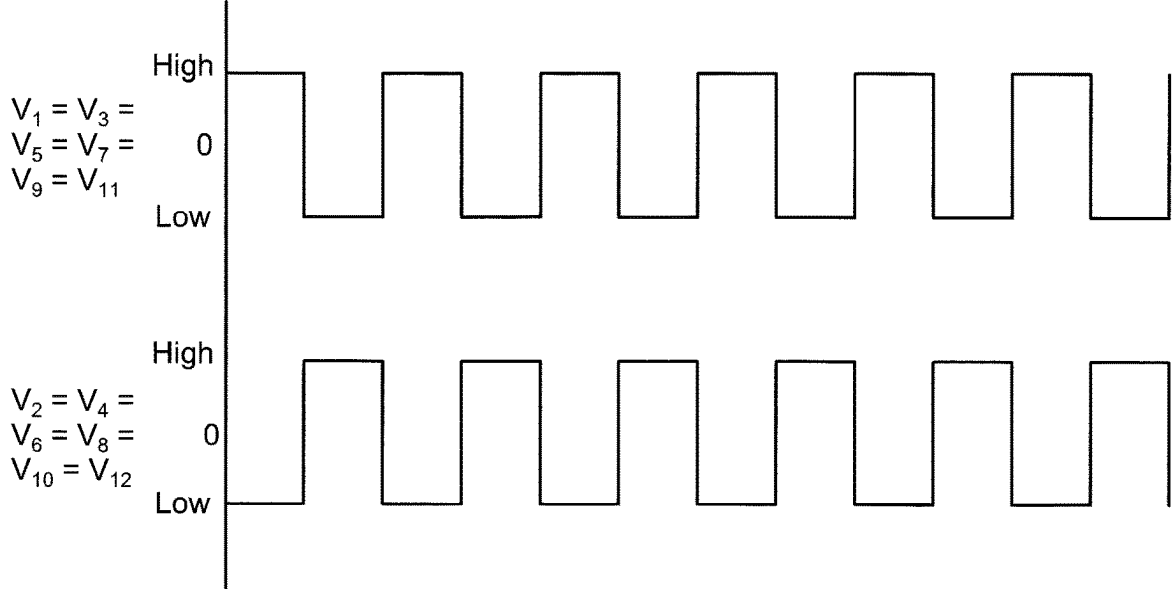
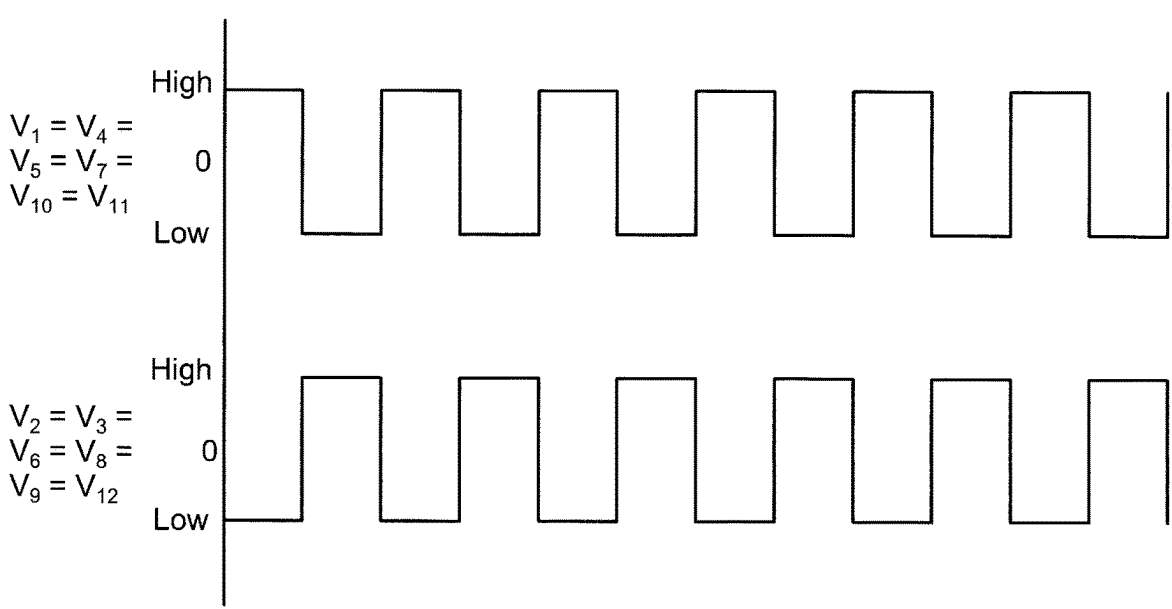


FIG. 42B



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**LIGHTING DEVICE AND OPTICAL
ELEMENT****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a Continuation of International Patent Application No. PCT/JP2022/019928, filed on May 11, 2022, which claims the benefit of priority to Japanese Patent Application No. 2021-086465, filed on May 21, 2021, the entire contents of which are incorporated herein by reference.

FIELD

An embodiment of the present invention relates to a lighting device and a driving method thereof. For example, an embodiment of the present invention relates to a lighting device capable of arbitrarily controlling an illuminated region and a driving method thereof.

BACKGROUND

In recent years, lighting devices have been developed in which an illuminated region and an illumination distance of a light source can be controlled by controlling the light emitted from the light source using a liquid crystal lens. For example, the lighting devices disclosed in Japanese Patent Application Publications No. 2010-230887, 2016-057541, and 2019-169435 include a liquid crystal cell having a liquid crystal layer and electrodes sandwiching the liquid crystal layer as well as a light source overlapping the liquid crystal cell. In these lighting devices, the orientation of liquid crystal molecules in the liquid crystal layer is controlled by the electric field between the electrodes to allow the liquid crystal cell to function as a lens, thereby controlling the light distribution.

SUMMARY

An embodiment of the present invention is a lighting device. The lighting device includes a light source, a first liquid crystal cell over the light source, and a second liquid crystal cell over the first liquid crystal cell. The light source includes a plurality of light-emitting elements arranged in a matrix shape with m rows and n columns. Each of the first liquid crystal cell and the second liquid crystal cell includes: a first substrate; a plurality of first electrode groups located over the first substrate and arranged in a matrix shape with m rows and n columns; a liquid crystal layer over the plurality of first electrode groups; and a second substrate over the liquid crystal layer. In each of the first liquid crystal cell and the second liquid crystal cell, each of the plurality of first electrode groups has a plurality of first electrodes extending in a row direction, and the light-emitting element in a j th row and a k th column overlaps with the first electrode group located in the j th row and the k th column. Longitudinal directions of the plurality of first electrodes of the first liquid crystal cell are parallel to longitudinal directions of the plurality of first electrodes of the second liquid crystal cell. n and m are each a natural number larger than 1, j is a variable selected from natural numbers equal to or larger than 1 and equal to or smaller than n , and k is a variable selected from natural numbers equal to or larger than 1 and equal to or smaller than m .

An embodiment of the present invention is an optical element. The optical element includes a first substrate, a

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plurality of first electrode groups, a liquid crystal layer over the plurality of first electrode groups, and a second substrate over the liquid crystal layer. The plurality of first electrode groups is located over the first substrate and is arranged in a matrix shape with m rows and n columns. Each of the plurality of first electrode groups includes a plurality of first electrodes extending in a row direction. In the plurality of first electrode groups arranged in a k th column, the odd-numbered first electrodes along a column direction are connected to a first wiring, while the even-numbered first electrodes are connected to a second wiring. In the plurality of first electrode groups arranged in a $(k+1)$ th column, the odd-numbered first electrodes along the column direction are connected to a third wiring, while the even-numbered first electrodes are connected to a fourth wiring. n and m are each a natural number larger than 1, and k is a variable selected from a natural number equal to or larger than 1 and smaller than n .

BRIEF DESCRIPTION OF DRAWING

FIG. 1 is a schematic perspective view of a lighting device according to an embodiment of the present invention.

FIG. 2A is a schematic top view of a light source of a lighting device according to an embodiment of the present invention.

FIG. 2B is a schematic cross-sectional view of a light source of a lighting device according to an embodiment of the present invention.

FIG. 3A is a schematic top view of a light source of a lighting device according to an embodiment of the present invention.

FIG. 3B is a schematic perspective view of a portion of a light source of a lighting device according to an embodiment of the present invention.

FIG. 4A is a schematic developed and perspective view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 4B is a schematic cross-sectional view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 5A is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 5B is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 6 is a schematic top view of a portion of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 7 is a schematic top view of a portion of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 8A is a schematic cross-sectional view of a portion of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 8B is a schematic cross-sectional view of a portion of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 9A is a schematic cross-sectional view of a portion of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 9B is a schematic cross-sectional view of a portion of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

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FIG. 35A is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 35B is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 36A is a schematic top view of a light source of a lighting device according to an embodiment of the present invention.

FIG. 36B is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 36C is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 36D is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 36E is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 36F is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 37A is a schematic top view of a light source of a lighting device according to an embodiment of the present invention.

FIG. 37B is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 37C is a schematic plane view of an illuminated region of a lighting device according to an embodiment of the present invention.

FIG. 38A is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 38B is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 39 is a schematic perspective view of a lighting device according to an embodiment of the present invention.

FIG. 40 is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 41 is a schematic top view of a liquid crystal cell of a lighting device according to an embodiment of the present invention.

FIG. 42A is a timing chart of a lighting device according to an embodiment of the present invention.

FIG. 42B is a timing chart of a lighting device according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

Hereinafter, each embodiment of the present invention is explained with reference to the drawings. The invention can be implemented in a variety of different modes within its concept and should not be interpreted only within the disclosure of the embodiments exemplified below.

The drawings may be illustrated so that the width, thickness, shape, and the like are illustrated more schematically compared with those of the actual modes in order to provide a clearer explanation. However, they are only an example, and do not limit the interpretation of the invention. In the specification and the drawings, the same reference number is provided to an element that is the same as that which appears in preceding drawings, and a detailed explanation

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may be omitted as appropriate. When a plurality of the same or similar structures is collectively represented, a reference number is used, while a hyphen and a natural number follow the reference number when the structures are independently represented. In addition, when a portion of a structure is represented, a small letter of the alphabet may be provided after the reference number.

In the specification and the claims, unless specifically stated, when a state is expressed where a structure is arranged “over” another structure, such an expression includes both a case where the substrate is arranged immediately above the “other structure” so as to be in contact with the “other structure” and a case where the structure is arranged over the “other structure” with an additional structure therebetween.

In the specification and the claims, an expression that two structures “intersect” includes not only a state where these two structures orthogonally (90°) intersect but also a state where these two structures intersect at an angle of $90^\circ \pm 10^\circ$.

In the specification and the claims, an expression “a structure is exposed from another structure” means a mode in which a portion of the structure is not covered by the other structure and includes a mode where the portion uncovered by the other structure is further covered by another structure. In addition, the mode expressed by this expression includes a mode where the structure is not in contact with the other structure.

First Embodiment

In this embodiment, an optical element according to an embodiment of the present invention, a lighting device **100** including the optical element, and a driving method thereof are explained.

1. Outline Structure of Lighting Device

A schematic perspective view of the lighting device **100** is shown in FIG. 1. As shown in FIG. 1, the lighting device **100** includes, as fundamental components, a light source **110** and two optical elements disposed over and overlapping the light source **110**. One of the optical elements is a first liquid crystal cell **120-1** over the light source **110**, while the other is a second liquid crystal cell **120-2** disposed over and overlapping the first liquid crystal cell **120-1**. The first liquid crystal cell **120-1** and the second liquid crystal cell **120-2** may be in direct contact with or fixed to each other via an adhesive layer **102**. Although not illustrated, the lighting device **100** may further have one or a plurality of liquid crystal cells **120** over or under the second liquid crystal cell **120-2** or between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**. The total number of liquid crystal cells **120** is not restricted and may be equal to or more than 2 and equal to or less than 10, equal to or more than 2 and equal to or less than 6, equal to or more than 2 and equal to or less than 4, or an odd number.

1-1. Light Source

A schematic top view of the light source **110** is shown in FIG. 2A, and a schematic view of the cross section along the chain line A-A' in FIG. 2A is shown in FIG. 2B. The light source **110** has a reflective plate **112** and a plurality of light-emitting elements **114**. One or a plurality of light-emitting elements **114** is provided in each depressed portion **112a**. The reflective plate **112** has a function of imparting directionality to the light emitted from the light-emitting elements **114** and irradiating the liquid crystal cell **120** with the light. Specifically, the reflective plate **112** has a plurality of depressed portions **112a** arranged in a matrix shape with m rows and n columns. Here, m and n are each a natural

number greater than 1, and m and n may be independently 6, 8, 12, 14, or 16, for example. m and n may be the same as or different from each other. Hereafter, the x-direction in the drawings is defined as a row direction, and the y direction is defined as a column direction. The direction perpendicular to both the x direction and the y direction is defined as a z direction. For example, the x direction and the y direction are parallel to the sides of a first substrate 122 or a second substrate 124 described below.

(1) Reflective Plate

The material structuring the reflective plate 112 may be arbitrarily selected and may be a metal such as aluminum and stainless steel, a polymer such as a polyimide, a polycarbonate, and an acrylic resin, or an inorganic oxide such as glass, for example. However, the reflective plate 112 reflects and converges the light from the light-emitting elements 114 in the depressed portions 112a and directs the light toward the liquid crystal cell 120 as shown by the arrows in FIG. 2B. Therefore, when the reflective plate 112 is composed of a material transmitting visible light such as glass and a polymer, it is preferable to configure the surface of the concave portions 112a with a film having a high reflectance to visible light. Examples of such a film include a film containing a metal such as aluminum, silver, gold, chromium, and stainless steel and a laminate of thin films containing a high refractive-index material such as titanium oxide and tantalum oxide and thin films containing a low refractive-index material such as silicon oxide and magnesium fluoride. The shape of the depressed portions 112a is appropriately adjusted to obtain highly directional light from the light-emitting elements 114 in the depressed portions 112a.

There is no restriction on the shape of the reflective plate 112 in the xy plane and may be determined as appropriate according to the environment in which the lighting device 100 is disposed. For example, as shown in FIG. 1A, the shape of the reflective plate 112 in the xy plane may be a square or may be a circle, an ellipse, or a polygon although not illustrated. The planar shape (hereinafter, simply referred to as a planar shape of the depressed portion 112a) of the depressed portion 112a in an upper surface of the reflective plate 112 (an upper surface closer to the liquid crystal cell 120) is not restricted and may be a circle as shown in FIG. 2A or a polygon exemplified by a rectangle as shown in FIG. 3A.

The use of the reflective plate 112 with the depressed portions 112a having the light-converging function allows the light from the light-emitting element 114 to have high directivity, by which light with parallel rays in the beam (also called collimated light) or light with low diffusion (highly directional light) can be obtained from each depressed portion 112a. Therefore, an illuminated plane 116 with the same shape as or a similar shape to the planar shape of the depressed portion 112a can be formed on the liquid crystal cell 120 (FIG. 3B). In other words, each light-emitting element 114 is capable of selectively irradiating a portion of the first liquid crystal cell 120-1 overlapping the depressed portion 112a and providing the illuminated plane 116 with light reflecting the planar shape of the depressed portion 112a.

(2) Light Emitting Device

Each light-emitting element 114 is an element having a function to emit light when supplied with an electric current, and there are no restrictions on its structure. A typical example is a light-emitting diode (LED). A light emitting diode has, as its fundamental components, an electroluminescence element, in which an inorganic emitter such as

gallium nitride and gallium nitride containing indium is sandwiched by a pair of electrodes, as well as a protective film protecting the electroluminescence element, for example, and is configured to emit visible light by electroluminescence.

A single or a plurality of light-emitting elements 114 may be provided in each depressed portion 112a. The emission color of each light-emitting element 114 may also be arbitrarily selected. For example, one or a plurality of light-emitting elements 114 providing white emission may be provided in each depressed portion 112a. Alternatively, the light source 110 may be configured by disposing a red-emissive light-emitting element 114, a green-emissive light-emitting element 114, and a blue-emissive light-emitting element 114 in each depressed portion 112a so that emission with a variety of colors can be obtained.

There is no restriction on the size of each light-emitting element 114, and a light emitting diode with an occupying area equal to or more than $1.0 \times 10^4 \mu\text{m}^2$ and equal to or less than $1.0 \times 10^6 \mu\text{m}^2$, equal to or more than $4.0 \times 10^4 \mu\text{m}^2$ and equal to or less than $5.0 \times 10^5 \mu\text{m}^2$, or equal to or more than $9.0 \times 10^4 \mu\text{m}^2$ and equal to or less than $2.5 \times 10^5 \mu\text{m}^2$ may be used. As an example, a so-called micro-LED with a size of approximately $320 \mu\text{m} \times 300 \mu\text{m}$ may be used as the light-emitting element 114.

1-2. Liquid Crystal Cell

As described above, at least two liquid crystal cells 120 are arranged over the light source 110 in the lighting device 100. The structures of the liquid crystal cells 120 may be the same as or different from each other. The structure of the liquid crystal cells 120 is explained below.

A schematic developed and perspective view of one liquid crystal cell 120 is shown in FIG. 4A, and a schematic view of a cross section along the chain line B-B' in FIG. 4A is shown in FIG. 4B. As shown in these drawings, the liquid crystal cell 120 has a first substrate 122 and a second substrate 124 opposing the first substrate 122, between which a variety of elements structuring a liquid crystal element (a plurality of first electrodes 126, a plurality of second electrodes 128, a liquid crystal layer 136, a first orientation film 132, a second orientation film 134, and the like) is arranged.

(1) First Substrate and Second Substrate

The first substrate 122 and the second substrate 124 serve as a base material for respectively supporting the plurality of first electrodes 126 and the plurality of second electrodes 128 and also provide a space in which the liquid crystal layer 136 is sealed. The first substrate 122 and the second substrate 124 are preferred to include a material exhibiting high transmittance to the light from the light-emitting elements 114 in order to transmit the light from the light source 110 to realize the lighting function. Therefore, it is preferable to configure the first substrate 122 and the second substrate 124 to include, for example, glass, quartz, or a polymeric material such as a polyimide, a polycarbonate, a polyester, and an acrylic resin.

(2) First Electrode

The plurality of first electrodes 126 is provided over the first substrate 122 so as to be in contact with the first substrate 122 or through an undercoat (not illustrated) which is an optional component (FIG. 4B). The plurality of first electrodes 126 is arranged to be parallel to one side of the first substrate 122. The undercoat may be formed with one or a plurality of films containing a silicon-containing inorganic compound such as silicon nitride and silicon oxide. It is preferred to form the first electrode 126 with a conductive oxide exhibiting high transmittance to visible light such as

indium-tin oxide (ITO) and indium-zinc oxide (IZO) in order to provide a high light-transmitting property to the liquid crystal cells 120.

In the liquid crystal cell 120, one first electrode group 125 is formed by a plurality of first electrodes 126, and the plurality of first electrode groups 125 is further arranged in a matrix shape with m rows and n columns (FIG. 5A). Thus, each first electrode group 125 overlaps at least one light-emitting element 114, and the number of the plurality of first electrode groups 125 and the number of the depressed portions 112a are identical.

A portion of the plurality of first electrodes 126 is shown in the schematic top view of FIG. 6. In this drawing, four first electrode groups 125 arranged in two rows and two columns are demonstrated. The plurality of first electrodes 126 all extend in the column direction or the row direction. That is, all of the longitudinal directions of the plurality of first electrodes 126 are parallel to each other. In the following explanation, a configuration in which the plurality of first electrodes 126 extends in the row direction (x direction) is described for convenience. In each first electrode group 125, the plurality of first electrodes 126 is arranged in a stripe form. Each first electrode group 125 overlaps one of the plurality of depressed portions 112a, and, therefore, overlaps the illuminated plane 116 of the light output from the corresponding depressed portion 112a. In other words, the depressed portion 112a located in a jth row and a kth column, which is arbitrarily selected from the plurality of depressed portions 112a, overlaps the first electrode group 125 located in the jth row and the kth column, and the light from the light-emitting element 114 arranged in this depressed portion 112a is selectively applied on the first electrode group 125 located in the jth row and the kth column. As a result, the illuminated plane 116 located in the jth row and the kth column overlaps the first electrode group 125 located in the jth row and the kth column. Therefore, it can be said that one first electrode group 125 is constituted by the plurality of first electrodes 126 overlapping one illuminated plane 116. Here, j is a variable selected from a natural number equal to or larger than 1 and equal to or smaller than n, and k is a variable selected from a natural number equal to or larger than 1 and equal to or smaller than m.

Note that although the plurality of first electrodes 126 extends in a stripe shape in the row direction (x direction) in this embodiment, each first electrode 126 may also have a structure bent at one or more locations, while extending in the row direction as a whole. In addition, the extending directions of the first electrodes 126 may also have an angle of approximately 1 to 10° with respect to the x direction.

In each column, the plurality of first electrodes 126 selected every other one (e.g., the odd-numbered first electrodes in the column direction) is connected to a wiring 138-1 and electrically conducts with each other. The remaining first electrodes 126 in each column (e.g., the even-numbered first electrodes in the column direction) are also connected to another wiring 138-2 and electrically conducts with each other. Thus, the plurality of first electrodes 126 can be alternately supplied with different voltages in each column. The wirings 138 are arranged so as not to overlap the depressed portions 112a. That is, each wiring 138 extends between adjacent depressed portions 112a and between adjacent illuminated planes 116. Furthermore, two wirings 138-1 and 138-2 respectively connected to the first electrodes 126 of the adjacent columns extend between the adjacent depressed portions 112a and between these adjacent illuminated planes 116. In other words, two wirings

138-1 and 138-2 respectively connected to the first electrodes 126 of the adjacent columns extend between the adjacent columns. The wirings 138 may be composed of a metal such as aluminum, copper, molybdenum, tantalum, and tungsten and may include the same material as the first electrodes 126.

The length of the first electrode 126 (length in the x direction which is the longitudinal direction) is greater than the length of the depressed portion 112a in the x direction in the xy plane, and therefore, each first electrode 126 straddles the depressed portion 112a. The width of the first electrode 126 (the length in the y direction intersecting the x direction) may be selected from a range, for example, equal to or more than 2 μm and equal to or less than 10 μm, and the distance between the first electrodes 126 adjacent in the column direction may also be selected from a range, for example, equal to or more than 2 μm and equal to or less than 10 μm. As a typical example, the width and the pitch of the first electrode 126 in the column direction may be 5 μm and 10 μm, respectively.

In this embodiment, this configuration allows the plurality of first electrodes 126 selected every other one in the column direction (e.g., the odd-numbered first electrodes 126 in the column direction) to be connected to one of the wiring 138-1 and the wiring 138-2 and the remaining first electrodes 126 (e.g., the even-numbered first electrodes 126 in the column direction) to be connected to the other of the wiring 138-1 and the wiring 138-2 in the plurality of first electrode groups 125 arranged in each column. In addition, these wirings 138 are each connected to a driver circuit 130 described below. As a result, the first electrode groups 125 can be independently driven in each column according to the potentials supplied from the driver circuit 130. Of course, it is also possible to simultaneously drive the first electrode groups 125 in a plurality of columns or all of the columns by connecting the wirings 138 to each other in each column or by applying the same potential thereto. The methods of connecting each of these wirings and applying the potential are described below.

(3) Second Electrode

The plurality of second electrodes 128 also has the same configuration as the first electrodes 126, but the extending direction thereof is different. Specifically, the plurality of second electrodes 128 is provided over the second substrate 124 so as to be in contact with the second substrate 124 or through an undercoat (not illustrated) which is an optional component (FIG. 4B). The plurality of second electrodes 128 is also arranged to be parallel to one side of the second substrate 124. The first electrodes 126 and the second electrodes 128 are arranged to be sandwiched between the first substrate 122 and the second substrate 124.

plurality of second electrodes 128, and the plurality of second electrode groups 127 is further arranged in a matrix shape with m rows and n columns (FIG. 5B). Thus, each second electrode group 127 overlaps at least one light-emitting element 114, and the number of the plurality of second electrode groups 127 and the number of the plurality of depressed portions 112a are the same as each other.

A portion of the plurality of second electrodes 128 is shown in a schematic top view of FIG. 7. In this drawing, four second electrode groups 127 arranged in two rows and two columns are demonstrated. The plurality of second electrodes 128 each extends in the column direction (y direction). That is, all of the longitudinal directions of the plurality of second electrodes 128 are parallel to each other and orthogonal to the longitudinal directions of the plurality of first electrodes 126. In each second electrode group 127,

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the plurality of second electrodes **128** is arranged in a stripe shape. Each second electrode group **127** overlaps one of the plurality of depressed portions **112a** and thus overlaps the corresponding illuminated plane **116**. In other words, the depressed portion **112a** in the *j*th row and the *k*th column, which is arbitrarily selected from the plurality of depressed portions **112a**, overlaps the second electrode group **127** located in the *j*th row and the *k*th column, and the light from the light-emitting element **114** located in this depressed portion **112a** is applied to the second electrode group **127** located in the *j*th row and the *k*th column through the first electrodes **126** and the liquid crystal layer **136**. As a result, the illuminated plane **116** located in the *j*th row and the *k*th column overlaps the second electrode group **127** located in the *j*th row and the *k*th column. Therefore, it can be said that one second electrode group **127** is constructed by the plurality of second electrodes **128** overlapping one illuminated plane **116**.

Note that, although the plurality of second electrodes **128** extends in a stripe shape in the column direction (*y* direction) in this embodiment, each second electrode **128** may also have a structure bent at one or a plurality of locations, while extending in the column direction as a whole. Moreover, the extending direction of the second electrode **128** may also have an angle of 1 to 10° with respect to the *y* direction.

In each row, the plurality of second electrodes **128** selected every other one (e.g., the odd-numbered second electrodes in the row direction) is connected to a wiring **140-1** and electrically conducts with each other. The remaining second electrodes **128** in each row (e.g., the even-numbered second electrodes **128** in the row direction) are also connected to another wiring **140-2** and electrically conduct with each other. Thus, in each row, the plurality of second electrodes **128** can be alternately supplied with different voltages. Similar to the wirings **138**, the wirings **140** are also arranged so as not to overlap the depressed portions **112a**. That is, each wiring **140** extends between adjacent depressed portions **112a** and between adjacent illuminated planes **116**. Furthermore, two wirings **140-1** and **140-2** respectively connected to the second electrodes **128** in the adjacent rows extend between the adjacent rows.

The length of the second electrode **128** (length in the *y* direction which is the longitudinal direction) is also larger than the length of the depressed portion **112a** in the *y* direction of the *xy* plane, and therefore, each second electrode **128** also straddles the depressed portion **112a**. The width of the second electrode **128** (length in the *x* direction intersecting the *y* direction) is also selected from a range, for example, equal to or more than 2 μm and equal to or less than 10 μm, and the distance between the second electrodes **128** adjacent in the row direction may also be selected from a range, for example, equal to or more than 2 μm and equal to or less than 10 μm. As a typical example, the width and the pitch of the second electrodes **128** in the *x* direction may be 5 μm and 10 μm, respectively.

In this embodiment, this configuration allows the second electrodes **128** selected every other one in the row direction (e.g., the odd-numbered second electrodes **128** in the row direction) to be connected to one of the wiring **140-1** and the wiring **140-2** and the remaining second electrodes **128** (e.g., the even-numbered second electrodes **128** in the row direction) to be connected to the other of the wiring **140-1** and the

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wiring **140-2** in the plurality of second electrode groups **127** arranged in each row. In addition, these wirings **140** are each connected to the driver circuit **130** described below. With this structure, the second electrode groups **125** can be independently driven in each row according to the potential supplied from the driver circuit **130**. Of course, it is also possible to simultaneously drive the second electrode groups **127** in the plurality of rows or all of the rows by connecting the wirings **140** to each other in each row or by applying the same potential thereto. The methods of connecting each of these wirings and applying the potential are described below.

As described above, the first liquid crystal cell **120-1** has the first electrode groups **125** arranged in a matrix shape with *m* rows and *n* columns on the first substrate **122** side as well as the second electrode groups **127** arranged in a matrix shape with *m* rows and *n* columns on the second substrate **124** side. The first electrode groups **125** can be independently driven column-by-column on the first substrate **122** side, while the second electrode groups **127** can be independently driven row-by-row on the second substrate **123** side. This individual driving method is described later.

The first liquid crystal cell **120-1** and the second liquid crystal cell **120-2** are arranged so that the longitudinal directions of the first electrodes **126** thereof are parallel to each other. In this case, the longitudinal directions of the second electrodes **128** in the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2** are also parallel to each other, and the directions in which the first orientation films **132** orient the liquid crystal molecules (hereinafter, referred to as orientation directions) are also parallel to each other. In this case, a configuration in which the first electrodes **126** thereof overlap each other may also be employed. Conversely, the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2** may be arranged so that the longitudinal directions of the first electrodes **126** thereof are perpendicular to each other. In this case, the longitudinal directions of the second electrodes **128** are also perpendicular to each other, and the orientation directions of the first orientation films **132** are also perpendicular to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**.

(4) First Orientation Film, Second Orientation Film, and Liquid Crystal Layer

The first orientation film **132** is provided over the plurality of first electrodes **126**, while the second orientation film **134** is provided over the plurality of second electrodes **128** (under the second electrodes **128** in FIG. 4B). The first substrate **122** and the second substrate **124** are bonded and fixed with a sealing material **118**. The space formed by the first substrate **122**, the second substrate **124**, and the sealing material **118** is filled with the liquid crystal layer **136**.

The first orientation film **132** and the second orientation film **134** contain a polymer such as a polyimide and a polyester, and their surfaces are subjected to a rubbing treatment. The rubbing treatment is performed so that the orientation direction of the first orientation film **132** is perpendicular to the direction in which the first electrodes **126** extend (see the arrow in FIG. 6) and the orientation direction of the second orientation film **134** is perpendicular to the direction in which the second electrodes **128** extend (see the arrow in FIG. 7). Therefore, the orientation direction of the first orientation film **132** and the orientation direction of the second orientation film **134** are orthogonal. Here, the orientation direction is the direction of the long axis of the liquid crystal molecules when they are oriented under the influence of the orientation film. Instead of the rubbing treatment, the orientation directions of the first orientation

film 132 and the second orientation film 134 may be produced by photo-orientation. The photo-orientation is a rubbing-less orientation treatment using light, and polarized light in the ultraviolet region is applied to orientation films which have not been subjected to the rubbing treatment from a predetermined direction, for example. This causes a photoreaction in the orientation films, thereby introducing anisotropy to their surface to provide the ability to control liquid crystal orientation.

The liquid crystal layer 136 contains liquid crystal molecules. The structure of the liquid crystal molecules is not limited. Thus, the liquid crystal molecules may be nematic liquid crystal, smectic liquid crystal, cholesteric liquid crystal, or chiral smectic liquid crystal.

The thickness d of the liquid crystal layer 136 (see FIG. 4B), i.e., the distance between the first orientation film 132 and the second orientation film 134, is also arbitrarily determined, but is preferred to be greater than the pitch of the first electrodes 126 or the second electrodes 128. For example, the thickness of the liquid crystal layer 136 may be preferably set to be equal to or more than 2 times and equal to or less than 10 times, equal to or more than 2 times and equal to or less than 5 times, or equal to or more than 2 times and equal to or less than 3 times the pitch of the first electrodes 126 or the second electrodes 128. A specific thickness of the liquid crystal layer 136 may be selected from a range, for example, equal to or more than 20 μm and equal to or less than 60 μm or equal to or more than 20 μm and equal to or less than 50 μm . Although not illustrated, spacers may be provided in the liquid crystal layer 136 to maintain this thickness throughout the lighting device 100. Note that, if the thickness of the liquid crystal layer 136 described above is employed in a liquid crystal display device, the high responsiveness required for displaying moving images cannot be obtained, and it is difficult to realize the functions as a liquid crystal display device.

(5) Other Components

The driver circuit 130 is provided over the first substrate 122 to generate signals for illumination and supply them to the first electrodes 126 and the second electrodes 128 (FIG. 4A and FIG. 5A). The driver circuit 130 may be fabricated over the first substrate 122 by appropriately combining a variety of patterned conductive films, semiconductor films, and insulating films or by mounting an IC chip with an integrated circuit formed over a semiconductor substrate over the first substrate 122. Alternatively, the driver circuit 130 may not be provided over the first substrate 122, but an IC chip may be provided as the driver circuit 130 over a connector such as a flexible printed circuit board (FPC) connected to the wirings 138 and 140 extending from the first electrodes 126 and the second electrodes 128.

2. Operation Principle

As described above, the light emitted from the light-emitting element 114 provided in each depressed portion 112a of the reflective plate 112 is selectively applied to one first electrode group 125, passes through the liquid crystal layer 136, and is further applied to one second electrode group 127. In addition, each first electrode group 125 and each second electrode group 127 are respectively provided with the plurality of first electrodes 126 and the plurality of second electrodes 128 arranged in a stripe shape. Therefore, the liquid crystal layer 136 functions as a sort of liquid crystal lens by controlling the voltages applied to the plurality of first electrodes 126 and the plurality of second electrodes 128 respectively included in each first electrode group 125 and each second electrode group 127. As a result, since the spread of the light output from each depressed

portion 112a can be individually controlled, the illuminated region of the light extracted from the light source 110 through the two liquid crystal cells 120 can be controlled in a diverse and arbitrary manner. Hereinafter, the operation principle and driving method of the lighting device 100 are explained. Here, the "illuminated region" refers to as a region where the light is applied onto an object when the lighting device 100 is driven. However, the illuminated region varies depending on the angle between the travelling direction of the light and the surface on the object and the distance between the lighting device 100 and the object. Therefore, the "illuminated region" is defined as a region where the light from the lighting device 100 is applied on a plane perpendicular to the normal line of the main surface of the second substrate 124 of the liquid crystal cell 120.

2-1. Non-Driving State

FIG. 8A and FIG. 8B show schematic views of the cross section of the liquid crystal cell 120 in a non-driving state. FIG. 8A is a schematic view observed from the row direction (x direction), while FIG. 8B is a schematic view observed from the column direction (y direction). In FIG. 8A and FIG. 8B, the liquid crystal molecules are schematically depicted as ellipses.

As described above, the orientation directions of the first orientation film 132 and the second orientation film 134 are respectively orthogonal to the directions in which the plurality of first electrodes 126 and the plurality of second electrodes 128 extend. Therefore, when the liquid crystal cells 120 are not driven, that is, when no voltage is applied to the plurality of first electrodes 126 and the plurality of second electrodes 128, the orientation of the liquid crystal molecules is not affected by the electric field and is determined by the orientation directions. As a result, at the vicinity of the first electrodes 126, the liquid crystal molecules are oriented with their longitudinal axes along the direction (y direction) perpendicular to the direction (x direction) in which the first electrodes 126 extend. On the other hand, at the vicinity of the second electrodes 128, the liquid crystal molecules are oriented with their longitudinal axes along the direction (x direction) perpendicular to the direction (y direction) in which the second electrodes 128 extend. Therefore, the orientation direction of the liquid crystal molecules rotates about the z direction as a center axis when it approaches the second substrate 124 from the first substrate 122 and is twisted by 90°.

2-2. Driving State

In the driving state, a pulsed alternating voltage (alternating square wave) is applied to the plurality of first electrodes 126 of either one or both of the first liquid crystal cell 120-1 and the second liquid crystal cell 120-2 so that the phase is inverted between adjacent first electrodes 126. Similarly, a pulsed alternating voltage (alternating square wave) is applied to the plurality of second electrodes 128 of either one or both of the first liquid crystal cell 120-1 and the second liquid crystal cell 120-2 so that the phase is inverted between adjacent second electrodes 128. In each liquid crystal cell 120, the frequency of the alternating voltages is the same. The alternating voltage may be selected from a range equal to or more than 5 V and equal to or less than 50 V or equal to or more than 5 V and equal to or less than 30 V, for example. The application of the alternating voltage generates an electric field (transverse electric field) between adjacent first electrodes 126 and adjacent second electrodes 128 as respectively shown by the arrows in FIG. 9A and FIG. 9B. Meanwhile, an electric field (vertical electric field) is also generated between the first electrodes 126 and the second electrodes 128. However, as described above, the

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thickness d of the liquid crystal layer **136** is larger than the distance between adjacent first electrodes **126** and between adjacent second electrodes **128**. Therefore, the vertical electric field is significantly smaller than the transverse electric field, and each liquid crystal molecule is oriented according to the transverse electric field.

When the transverse electric field is generated in the liquid crystal layer **136**, the liquid crystal molecules located approximately midway between the adjacent first electrodes **126** maintain their initial orientation state on the first substrate **122** side, because the direction of the transverse electric field is almost parallel to the first substrate **122**. However, since the direction of the electric field tilts in the z direction when it approaches the first electrodes **126**, the liquid crystal molecules also tilt in the z direction and their angle (tilt angle) increases. As a result, the liquid crystal molecules in the liquid crystal layer **136** on the first substrate **122** side are oriented in an upward convex arc (FIG. 9A). The same is applied to the second substrate **124** side. The liquid crystal molecules located approximately midway between the adjacent second electrodes **128** maintain their initial orientation state because the direction of the transverse electric field is approximately parallel to the second substrate **124**. However, since the direction of the electric field tilts in the z direction as it approaches the second electrodes **128**, the liquid crystal molecules also tilt in the z direction and their angle (tilt angle) increases. Accordingly, the liquid crystal molecules in the liquid crystal layer **136** on the second substrate **124** side are oriented in a downward convex arc (FIG. 9B).

Due to the orientation change of the liquid crystal molecules, the light incident on the liquid crystal layer **136** diffuses according to the refractive index distribution of the arc-oriented liquid crystal molecules on the first substrate side **122** and further diffuses according to the refractive index distribution of the arc-oriented liquid crystal molecules on the second substrate **124** side. As a result, the liquid crystal cell **120** functions as a lens diffusing light. This light diffusion mechanism is explained in detail using FIG. 10 and FIG. 11. FIG. 10 is a schematic perspective view showing the orientation of the liquid crystal molecules shown in FIG. 9A and FIG. 9B, and FIG. 11 is a schematic view showing the behavior of the light passing through the two liquid crystal cells **120**. The following explanation uses, as a model, an example in which the extending directions of the first electrodes **126** are parallel to each other and the extending directions of the second electrodes **128** extends are also parallel to each other between the two liquid crystal cells **120**.

As described above, when a pulsed alternating voltage is applied to the plurality of first electrodes **126** so that the phase is inverted between adjacent first electrodes **126** and to the plurality of second electrodes **128** so that the phase is inverted between adjacent second electrodes **128**, the transverse electric fields orthogonal to each other are generated on the first electrode **126** side and the second electrode **128** side as shown in FIG. 10. As a result, the liquid crystal molecules in the liquid crystal layer **138** are oriented convexly upward between the adjacent first electrodes **126** on the first substrate **122** side and convexly downward between the adjacent second electrodes **128** on the second substrate **124** side. Furthermore, the orientation of the liquid crystal molecules is twisted by 90° as it approaches from the first electrode **126** to the second electrode **128**.

As shown in FIG. 11, the light emitted from the light source **110** first enters the first liquid crystal cell **120-1**. This light has a polarization component **150** in the y direction

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(straight arrow in the drawing) and a polarization component **156** in the x direction (circled symbol with a cross in the drawing). Hereafter, the y directional and x -directional polarization components of the light before entering the liquid crystal cell **120** are respectively referred to as S and P components for convenience, and these terms will be used regardless of the change in polarization axis.

Since the liquid crystal molecules are oriented along the y direction on the first electrode **126** side, the liquid crystal layer **136** has a refractive index distribution in the y direction. Therefore, the S component **150** incident on the liquid crystal layer **136** diffuses in the y direction due to the refractive index distribution in the y direction on the first electrode **126** side. When passing through the liquid crystal layer **136**, this light is optically rotated by the twist of the orientation of the liquid crystal molecules, and the polarization axis thereof changes to the x direction. Since the liquid crystal layer **136** has a refractive index distribution in the x direction on the second electrode **128** side, this light is further diffused in the x direction. As a result, when the S component **150** passes through the liquid crystal layer **136** of the first liquid crystal cell **120-1**, this component becomes a S component **152** diffused in the x direction and the y direction.

On the other hand, the refractive index distribution exists in the y direction on the first electrode **126** side. Therefore, the P component **156** incident on the first liquid crystal cell **120-1** is not affected by the refractive index distribution but is optically rotated by the twist of the orientation of the liquid crystal molecules without being diffused so that the polarization axis is changed to the y direction. In addition, since the refractive index distribution on the second electrode **128** side exists in the x direction, the P component **156** whose polarization axis has been changed to the y direction is not affected by the refractive index distribution. As a result, the P component **156** is optically rotated but is not diffused so as to become a P component **158** when passing through the liquid crystal layer **136** of the first liquid crystal cell **120-1**.

Next, the light which has passed through the first liquid crystal cell **120-1** is considered. As described above, the longitudinal directions of the first electrodes **126** are parallel to each other and the longitudinal directions of the second electrodes **128** are also parallel to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2** in this model. Therefore, in the liquid crystal layer **136** of the second liquid crystal cell **120-2**, there is a refractive index distribution in the y direction on the first electrode **126** side and a refractive index distribution in the x direction on the second electrode **128** side.

As described above, the S component **150** becomes the S component **152** diffused in the x direction and the y direction when passing through the first liquid crystal cell **120-1**. This S component **152** does not diffuse on the first electrode **126** side of the second liquid crystal cell **120-2** because its polarization axis is orthogonal to the direction of the refractive index distribution. The S component **152** is optically rotated according to the twist of the orientation of the liquid crystal molecules, and its polarization axis is changed to the y direction when passing through the liquid crystal layer **136**. However, since the refractive index distribution is in the x direction on the second electrode **128** side, this component is not affected by the refractive index distribution. As a result, although the S component **152** is optically rotated by the second liquid crystal cell **120-2**, it does not diffuse to become a S component **154**. In summary, the S component **150** emitted from the light source **110** is optically

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rotated by the first liquid crystal cell **120-1** while being diffused in the x direction and the y direction so as to become the S component **152**, and is then optically rotated by the second liquid crystal cell **120-2** without being diffused to eventually become a S component **154** diffused in the x direction and they direction.

On the other hand, the P component **158** incident on the liquid crystal layer **136** of the second liquid crystal cell **120-2** diffuses in the y direction according to the refractive index distribution in the y direction on the first electrode **126** side. This light is optically rotated by the twist of the orientation of the liquid crystal molecules so that its polarization axis is changed to the x direction when passing through the liquid crystal layer **136**. The light is then diffused in the x direction because the liquid crystal layer **136** has the refractive index distribution in the x direction on the second electrode **128** side. As a result, the P component **158** is optically rotated and is simultaneously diffused in the x direction and the y direction to become a P component **160** when passing through the second liquid crystal cell **120-2**. In summary, the P component **156** emitted from the light source **110** is optically rotated by the first liquid crystal cell **120-1** without being diffused, and is then optically rotated by the second liquid crystal cell **120-2** while being diffused in the x direction and the y direction to eventually become a P component **160** diffused in the x direction and they direction.

Since the degree of orientation of the liquid crystal molecules can be controlled by the voltages applied to the first electrodes **126** and the second electrodes **128**, the degree of light diffusion can also be controlled by the voltages applied to the first electrodes **126** and the second electrodes and **128**. Therefore, in accordance with the mechanism described above, the degree of the diffusion of the light applied onto each first electrode group **125** and each second electrode group **127** can be independently controlled by the voltages applied to the first electrodes **126** and second electrodes **128**.

Note that the light diffusion is controlled by the transverse electric fields generated between adjacent first electrodes **126** and between adjacent second electrodes **128**. Therefore, the light diffusion can be performed as long as a potential difference is provided between adjacent first electrodes **126** and/or between adjacent second electrodes **128** in each liquid crystal cell **120**. Hence, a constant voltage may be applied to the plurality of first electrodes **126** so that the voltage is different between the adjacent first electrodes **126**, or an alternating voltage may be provided to the plurality of first electrodes **126** selected every other one while providing a constant voltage to the remaining first electrodes **126**. The same is applied to the second electrodes **128**.

3. Light Distribution Control

By using the mechanism described above, the illuminated region of the light source **110** can be arbitrarily controlled. This feature is explained below. In the following explanation, it is assumed that, as shown in FIG. **12**, an alternating voltage V_1 is applied to the plurality of first electrodes **126** selected every other one, while an alternating voltage V_2 is applied to the remaining first electrodes **126** in the first liquid crystal cell **120-1**. In addition, it is assumed that an alternating voltage V_3 is applied to the plurality of second electrodes **128** selected every other one, while an alternating voltage V_4 is applied to the remaining second electrodes **128** in the first liquid crystal cell **120-1**. In the second liquid crystal cell **120-2**, it is assumed that an alternating voltage V_5 is applied to the plurality of first electrodes **126** selected every other one, while an alternating voltage V_6 is applied to the remaining first electrodes **126**. In addition, it is assumed

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that an alternating voltage V_7 is applied to the selected second electrodes **128** selected every other one, while an alternating voltage V_8 is applied to the remaining second electrodes **128** in the second liquid crystal cell **120-2**. In this model, the first electrodes **126** are parallel to each other, and the second electrodes **128** are also parallel to each other between the two liquid crystal cells **120**.

3-1. Non-Driving State of Liquid Crystal Cells

When the liquid crystal cells **120** are in a non-driving state, no electric field is generated between the first electrodes **126** and between the second electrodes **128**. Therefore, since there is no refractive index distribution in the liquid crystal layer **136**, the S component **150** and the P component **152** do not experience any diffusion effect, although they are optically rotated by each liquid crystal cell **120**. Thus, when all of the light-emitting elements **114** provided in the plurality of depressed portions **112a** are turned on as shown in FIG. **13A**, for example, the light does not greatly spread when passing through the two liquid crystal cells **120**, and only reflects the diffusion of the light output from each depressed portion **112a**. As a result, in the xy plane, the illuminated region A_0 of the light source **110** is almost similar to the shape of the light source **110** in the xy plane (FIG. **13B**).

3-2. Driving State of Liquid Crystal Cells

(1) Selective Diffusion in y Direction

As an example, a case where the liquid crystal cells **120** are driven according to the timing chart in FIG. **14A** is considered. Here, in each of the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**, an alternating voltage is applied to the plurality of first electrodes **126** so that the phase is inverted between adjacent first electrodes **126**, and a constant voltage or no voltage is applied to the plurality of second electrodes **128**. The constant voltage may be 0 V or an intermediate potential relative to the aforementioned alternating voltage.

When the liquid crystal cells **120** are driven in this manner, the S component **150** of the light from the light source **110** diffuses in the y direction while its polarization axis is changed to the x direction by the liquid crystal layer **136** in the first liquid crystal cell **120-1** as can be understood from FIG. **11**. Since there is no electric field between the second electrodes **128**, this light is emitted from the first liquid crystal cell **120-1** without being diffused on the second electrode **128** side to become the S component **152** diffused in the y direction. This S component **152** is not affected by the refractive index distribution when it enters the second liquid crystal cell **120-2** because the refractive index distribution caused by the transverse electric field generated between the first electrodes **126** of the second liquid crystal cell **120-2** is in the y direction. In addition, since there is no electric field between the second electrodes **128**, this S component **152** is optically rotated without being diffused. In summary, the S component **150** becomes the S component **154** diffused in the y direction when passing through the two liquid crystal cells **120**.

On the other hand, the P component **156** of the light from the light source **110** is not affected by the refractive index distribution because the transverse electric field is in the y direction on the first electrode **126** side of the first liquid crystal cell **120-1**. Furthermore, since there is no electric field between the second electrodes **128**, there is no refractive index distribution on the second electrode **128** side. Therefore, the P component **156** is not diffused but is optically rotated according to the twist of the orientation of the liquid crystal molecules in the liquid crystal layer **136** to become the P component **158**. When this P component **158**

enters the second liquid crystal cell **120-2**, this component is diffused in the y direction by the refractive index distribution in the y direction on the first electrode **126** side, while the polarization axis is changed to the x direction by the liquid crystal layer **136**. This light is emitted from the second liquid crystal cell **120-2** without being diffused on the second electrode **128** side because there is no electric field between the second electrodes **128**. In summary, the P component **156** becomes the P component **160** diffused in the y direction when passing through the two liquid crystal cells **120**.

By the above mechanism, the light from the light source **110** is diffused only in they direction. Therefore, the lighting device **100** provides an illuminated region A_1 diffused in the y direction compared with the illuminated region A_0 formed when the two liquid crystal cells **120** are not driven (FIG. **14B**). Without going into detail, in the case of obtaining an illuminated region selectively diffused in the x direction, the liquid crystal cells **120** are driven so that, in each of the first liquid crystal cell **120-1** and the second liquid crystal cell, an alternating voltage is applied to the plurality of second electrodes **128** to allow the phase to be inverted between adjacent second electrodes **128**, while applying a constant voltage or no voltage to the plurality of first electrodes **126** (FIG. **15A** and FIG. **15B**).

(2) Diffusion in x Direction and y Direction

As another example, a case where the liquid crystal cells **120** are driven according to the timing chart in FIG. **16A** is considered. Here, in the first liquid crystal cell **120-1**, an alternating voltage is applied to the plurality of first electrodes **126** so that the phase is inverted between adjacent first electrodes **126**, while a constant voltage or no voltage is applied to the plurality of second electrodes **128**. On the other hand, in the second liquid crystal cell **120-2**, a constant voltage or no voltage is applied to the plurality of first electrodes **126**, while an alternating voltage is applied to the second electrodes **128** so that the phase is inverted between adjacent second electrodes **128**. The constant voltage may be 0 V or an intermediate potential with respect to the aforementioned alternating voltage.

As can be understood from FIG. **11**, when the liquid crystal cells **120** are operated in this manner, the S component **150** of the light from the light source **110** is diffused in the y direction in the first liquid crystal cell **120-1** due to the refractive index distribution in the y direction on the first electrode **126** side and is optically rotated by the liquid crystal layer **136** to become S component **152**. When the S component **152** enters the second liquid crystal cell **120-2**, the S component **152** is optically rotated by the liquid crystal layer **136** and the polarization axis thereof is changed to the y direction since there is no refractive index distribution on the first electrode **126** side. However, since the refractive index distribution on the second substrate **124** side is in the x direction, this component is not diffused. In summary, the S component **150** becomes the S component **154** diffused only in the y direction when passing through the two liquid crystal cells **120**.

On the other hand, the P component **156** of the light from the light source **110** is not affected by the refractive index distribution because the transverse electric field on the first electrode **126** side of the first liquid crystal cell **120-1** is in the y direction. In addition, since there is no electric field between the second electrodes **128**, there is no refractive index distribution on the second electrode **128** side. Therefore, the P component **156** is not diffused but is optically rotated according to the twist of the orientation of the liquid crystal molecules in the liquid crystal layer **136** to become the P component **158** whose polarization axis is in the y

direction. When this P component **158** enters the second liquid crystal cell **120-2**, this P component **158** is optically rotated while being diffused in the x direction to become the P component **160** because, although there is no refractive index distribution on the first electrode **126** side, there is a refractive index distribution in the x direction on the second electrode **128** side. In summary, the P component **156** becomes the P component **160** diffused only in the x direction when passing through the two liquid crystal cells **120**.

By the above mechanism, the S component **150** and the P component **156** of the light from the light source **110** are selectively diffused in the y direction and the x direction, respectively. Therefore, unlike the illuminated region A_0 formed when the two liquid crystal cells **120** are not driven, the lighting device **100** provides a cross-shaped illuminated region A_1 (FIG. **16B**).

(3) Asymmetric Diffusion in x Direction and y Direction

As another example, a case where the liquid crystal cells **120** are driven according to the timing chart shown in FIG. **17A** is considered. Here, in the two liquid crystal cells **120**, an alternating voltage is supplied to the plurality of first electrodes **126** so that the phase is inverted between adjacent first electrodes **126**, while an alternating voltage is supplied to the plurality of second electrodes **128** so that the phase is inverted between adjacent second electrodes **128**. However, the voltages supplied to the plurality of first electrodes **126** and the plurality of second electrodes **128** are different in magnitude. Here, an example is used for explanation where the former is larger than the latter.

As can be understood from FIG. **11**, when the liquid crystal cells **120** are driven in this manner, the S component **150** of the light from the light source **110** is optically rotated while being diffused in the y direction due to the refractive index distribution in the y direction on the first electrode **126** side and is also diffused in the x direction due to the refractive index distribution in the x direction on the second electrode **128** side in the first liquid crystal cell **120-1**. However, since the voltage applied to the first electrodes **126** is greater than the voltage applied to the second electrodes **128**, this light becomes the S component **152** largely diffused more in the y direction. This S component **152** is not diffused but is optically rotated in the second liquid crystal cell **120-2**. In summary, the S component **150** becomes the S component **154** largely diffused more in the y direction than in the x direction.

On the other hand, the P component **156** of the light from the light source **110** is optically rotated without being diffused in the first liquid crystal cell **120-1** to become the P component **158**. When this P component **158** enters the second liquid crystal cell **120-2**, this component is optically rotated and is simultaneously diffused in the y direction due to the refractive index distribution on the first electrode **126** side and in the x direction due to the refractive index distribution on the second electrode **128** side. However, since the voltage applied to the first electrodes **126** is greater than the voltage applied to the second electrodes **128**, this component becomes the P component **160** largely diffused more in the y direction. In summary, the P component **156** also becomes the P component **160** largely diffused more in they direction than in the x direction.

Due to the above mechanism, the S component **150** and the P component **156** of the light from the light source **110** are each diffused in the y direction largely more than in the x direction. Therefore, the lighting device **100** provides an illuminated region A_1 greatly expanded more in the y direc-

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tion compared to the illuminated region A_0 formed when the two liquid crystal cells **120** are not driven (FIG. **17B**).

As described above, the implementation of the embodiment of the present invention enables the light from the light source **110** to be arbitrarily controlled, leading to the production of illuminated regions with a variety of shapes.

Second Embodiment

In this embodiment, a driving method of the lighting device **100** different from the driving method described in the First Embodiment is explained. An explanation of the structures the same as or similar to those described in the First Embodiment may be omitted.

In the driving method of the lighting device **100** according to the present embodiment, a part of the plurality of light-emitting elements **114** is driven (local dimming). This allows the illuminated region to be changed to more diverse shapes and also reduces power consumption.

As an example, a model shown in FIG. **18A** is considered where only a part of the light-emitting elements **114** is driven while the other light-emitting elements **114** are not turned on. Here, the light-emitting elements **114** in the depressed portions **112a** located in the fourth row (1-4) among the depressed portion **112a** arranged in a matrix of 8 rows and 8 columns are turned on. When the two liquid crystal cells **120** are not driven, the illuminated region A_0 of the light source **110** in the xy plane is approximately similar to the shape of the light source **110** in the xy plane. Therefore, as shown in FIG. **18B**, the illuminated region A_0 is linear.

In this state, the liquid crystal cells **120** are driven to selectively diffuse the light in the x direction. Specifically, the light emitted from the light source **110** is selectively diffused in the x direction, i.e., the row direction, by driving the liquid crystal cells **120** according to the timing chart shown in FIG. **15A**. As a result, it is possible to provide an illuminated region A_1 enlarged in the x direction compared with the illuminated region A_0 (FIG. **18C**). Although not illustrated, the light can also be diffused in the y direction by driving the liquid crystal cells **120** as appropriate, thereby providing illuminated regions with arbitrary shapes using the light from the light source **110**.

Thus, by performing the local dimming in this manner, the illuminated region of the light source **110** can also be controlled, which allows the formation of illuminated regions with more diverse shapes. In addition, since only a part of the light-emitting elements **114** can be selectively driven, power consumption can be reduced.

Third Embodiment

In this embodiment, a driving method of the lighting device **100** different from the driving methods described in the First and Second Embodiments is explained. An explanation of the structures the same as or similar to those described in the First and Second Embodiments may be omitted.

In the driving method of the lighting device **100** described in this embodiment, the local dimming is performed similar to the Second Embodiment. At the same time, the liquid crystal cells **120** are partially driven. That is, a portion of the plurality of first electrode groups **125** and second electrode groups **127** provided in the liquid crystal cells **120** is selectively driven, which enables further reduction of power consumption and more diverse light-distribution control.

As an example, a model is considered in which the light-emitting elements **114** provided in the depressed por-

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tions **112a** located in the fifth row R_5 among the depressed portions **112a** arranged in a matrix shape with 8 rows and 8 columns are turned on as shown in FIG. **19A**. Schematic enlarged views of the plurality of first electrode groups **125** and second electrode groups **127** in the area surrounded by the dotted line in FIG. **19A** are respectively illustrated in FIG. **19B** and FIG. **20**. In FIG. **19B**, the plurality of first electrodes **126** overlapping the illuminated plane **116**, which is provided by the light-emitting element **114** disposed in one depressed portion **112a**, constitutes one first electrode group **125** (see FIG. **6**). Similarly, in FIG. **20**, the plurality of second electrodes **128** overlapping the illuminated plane **116**, which is provided by the light-emitting element **114** disposed in one depressed portion **112a**, constitutes one second electrode group **127** (see FIG. **6**). Among the nine illuminated planes **116-1** to **116-9** depicted in FIG. **19B** and FIG. **20**, the illuminated planes where the light-emitting elements **114** are turned on are three hatched illuminated planes **116-2**, **116-5**, and **116-8**.

In this state, the two liquid crystal cells **120** are driven using the first electrodes **126** and second electrodes **128** overlapping the illuminated plane **116** provided by one of the depressed portions **112a** in which the illuminating light-emitting element **114** is disposed. For example, the illuminated plane **116-5** is selected, and the first electrodes **126** and the second electrodes **128** overlapping this illuminated plane **116-5** are driven according to the timing chart shown in FIG. **15A**. In accordance with the timing chart in FIG. **15A**, in the two liquid crystal cells **120**, the second electrodes **128** located between the wirings **140-3** and **140-4** among the wirings **140-1** to **140-6** are applied with an alternating voltage so that the phases are inverted between adjacent second electrodes **128**, while applying a constant voltage of 0 V to the other second electrodes **128**. The constant voltage of 0 V is also applied to the first electrodes **126** of the two liquid crystal cells **120**.

By driving the liquid crystal cell **120** in this manner, the illuminated region A_0 , which is linearly formed in the y direction when the liquid crystal cells **120** are not driven, is partially extended in the x direction to obtain a cross-shaped illuminated region A_1 as shown in FIG. **21**. Such a deformation of the illuminated region can be achieved by driving only a portion of the second electrodes **128**, which also results in a reduction of power consumption of the liquid crystal cells **120**.

Alternatively, the illuminated region can be changed to a variety of shapes by varying the voltage applied to the second electrodes **128** according to the timing chart in FIG. **15A**. For example, as shown in FIG. **22A**, all of the light-emitting elements **114** arranged in the depressed portions **112a** in the fifth row R_5 are turned on, while the second electrodes **128** overlapping the depressed portions **112a** in the first row L_1 to the third row L_3 are driven according to the timing chart in FIG. **15A**. At this time, the voltage applied to the second electrodes **128** is increased in the order from the first row L_1 to the third row L_3 , by which the linearly formed illuminated region A_0 can be transformed into an arrow-shaped illuminated region A_1 as shown in FIG. **22B**.

By controlling the voltages applied to the first electrode groups **125** and the second electrode groups **127** on a row-by-row or column-by-column basis, the power consumption can be reduced, and the illuminated region can be arbitrarily transformed into a variety of shapes.

Fourth Embodiment

In this embodiment, a driving method of the lighting device **100** different from the driving methods described in

the First to Third Embodiments is explained. An explanation of the structures the same as or similar to those described in the First to Third Embodiments may be omitted.

In the driving method of the lighting device **100** described in this embodiment, when the liquid crystal cells **120** are partially driven, a row (hereinafter, referred to as a driving row) overlapping the depressed portions **112a** from which the target light for diffusion is output, and the rows (hereinafter, referred to as non-driving adjacent rows), which are adjacent to the driving row and overlap the depressed portions **112a** having the light-emitting elements **114a** that are not driven or output the light that is not a target for diffusion, are synchronized. This synchronization is carried out so that there is no potential difference between the first electrodes **126** in the driving row and two wirings **138** arranged between the non-driving rows or between the first electrodes **126** in the driving row and the first electrodes **126** in the non-driving adjacent rows. Alternatively, a column (hereinafter, referred to as a driving column) overlapping the depressed portions **112a** from which the target light for diffusion is output, and the columns (hereinafter, referred to as non-driving adjacent columns), which are adjacent to the driving column and overlap the depressed portions **112a** having the light-emitting elements **114a** that are not driven or output the light that is not a target for diffusion, are synchronized. This synchronization is carried out so that there is no potential difference between the second electrodes **128** in the driving column and two wirings **140** arranged between the non-driving columns or between the second electrodes **128** in the driving column and the second electrodes **128** in the non-driving adjacent columns.

As an example, a case is considered in which the reflective plate **112** has the depressed portions **112a** arranged in 8 rows and 8 columns, and light is emitted from the light-emitting elements **114** in the depressed portions **112a** of the fourth column R_4 and the fifth column R_5 in the fourth row L_4 and the fourth column R_4 and the fifth column R_5 in the fifth row L_5 as shown in FIG. 23A. Here, it is assumed that only the light from the light-emitting elements **114** in the two depressed portions **112a** in the fourth row L_4 is selectively diffused in the x direction. A schematic top view of the second electrodes **128** in the area surrounded by the dotted line in FIG. 23A is shown in FIG. 23B. According to the timing chart shown in FIG. 15A for diffusing the light in the x direction, among the second electrodes **128** of the two liquid crystal cells **120**, the second electrode groups **127** in the fourth row L_4 (the second electrode groups **127** overlapping the illuminated planes **116-1** and **116-2** in FIG. 23B), which is a driving row, is applied with alternating voltages V_5 and V_6 with different phases to each other, while a constant voltage or no voltage is applied to the first electrodes **126**. With this driving method, the light incident on the illuminated planes **116-1** and **116-2** is diffused in the x direction.

However, if the voltages V_7 and V_8 of the wirings **140** connected to the second electrodes **128** of the fifth row L_5 , which is the non-driving adjacent row are kept constant, a potential difference is also generated between the two wirings **140** located between the driving row L_4 and the non-driving adjacent row L_5 (here, wirings **140-2** and **140-3**). Similarly, a potential difference is also generated between the second electrodes **128** in the fourth row L_4 and the second electrodes **128** in the fifth row L_5 . These potential differences cause the liquid crystal molecules in the liquid crystal layer **136** to produce a downwardly convexed refractive index distribution in the y direction between the fourth row L_4 and the fifth row L_5 . As a result, a lens effect is also

generated in the liquid crystal layer **136** between the fourth row L_4 and the fifth row L_5 , and the direction of this refractive index distribution is orthogonal to that of the lens effect of the liquid crystal layer **136** over the fourth row L_4 . Hence, a part of the light applied to the driving row is also diffused in the unintended y direction, which inhibits precise light distribution control.

Hence, in this embodiment, the wirings **140** between the driving row and the non-driving adjacent row are synchronized with each other in each liquid crystal cell **120**. In the example demonstrated in FIG. 23C, the wiring **140-3** connected to the second electrodes **128** in the fifth row L_5 , which is the non-driving adjacent row, and located between the driving row L_4 and the non-driving adjacent row L_5 and the wiring **140-2** connected to the second electrodes **128** in the fourth row L_4 , which is the driving row, and located between the driving row L_4 and the non-driving row L_5 are synchronized with each other. As a result, there is no potential difference between the wirings **140-2** and **140-3** located between the driving row and the non-driving adjacent row and between the second electrodes **128** in the driving row and the second electrodes **128** in the non-driving adjacent row, by which the generation of the refractive index distribution is suppressed in the liquid crystal layer **136** between the driving row and the non-driving adjacent row. However, all of the second electrodes **128** in the non-driving adjacent row L_5 are synchronized with each other because the light is not diffused in the non-driving adjacent row L_5 . Therefore, the other wiring **140-4** connected to the second electrodes **128** in the non-driving adjacent row L_5 are also synchronized with the wiring **140-3**. As a result, the generation of the refractive index distribution is suppressed in the non-driving adjacent rows and also between the wiring **140-2** or the second electrodes **128** in the driving row L_4 and the second electrodes **128** in the non-driving adjacent line L_5 , by which unintended light diffusion can be suppressed and precise light distribution can be performed.

As another example, a case is explained in which light is emitted from the light-emitting elements **114** in the depressed portions **112a** of the fourth row L_4 in the fourth column R_4 and the fourth column R_4 in the fifth row L_5 as shown in FIG. 24A, and this light is diffused in the y direction. The arrangements of the first electrodes **126** and the second electrodes **128** in the region surrounded by the dotted rectangle in FIG. 24A are respectively shown in FIG. 25 and FIG. 26. In FIG. 25 and FIG. 26, the light from the depressed portions **112a** gives illuminated planes **116-2** and **116-5**.

In this case, all of the first electrodes **126** in the third column R_3 , which is the non-driving adjacent column, are synchronized with the first electrodes **126** selected every other one in the fourth column R_4 , which is the driving column, in each liquid crystal cell **120** as shown in FIG. 24B. Similarly, all of the first electrodes **126** in the fifth column R_5 , which is the non-driving adjacent column, are synchronized with the first electrodes **126** selected every other one in the fourth column, which is the driving column. Specifically, an alternating voltage V_3 is applied to the first electrodes **126** selected every other one, while an alternating voltage V_4 inversed in phase with respect to the alternating voltage V_3 is applied to the other first electrodes **126** in the fourth column R_4 . In the third column R_3 , alternating voltages V_1 and V_2 with the same voltage and phase as the alternating voltage V_3 are applied. On the other hand, alternating voltages V_5 and V_6 with the same voltage and phase as the alternating voltage V_4 are applied in the fifth column. Note that, since the light is not diffused in the x

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direction, a constant voltage or no voltage is applied to all of the second electrodes **128** in the liquid crystal cell **120**. As a result, since there is no potential difference between the wirings **138** disposed between the fourth column and the third column and between the wirings **138** disposed between the fourth column and the fifth column, the generation of the refractive index distribution causing the diffusion of the light in the x direction can be suppressed. Accordingly, unintended light diffusion can be prevented, and precise light distribution can be performed.

As another example, a case is explained in which the light is emitted from the light-emitting elements **114** in the depressed portions **112a** of the fourth column R_4 to the sixth column R_6 in the fifth row L_5 as shown in FIG. **24C**, and this light is diffused in the x direction. The arrangements of the first electrodes **126** and the second electrodes **128** in the region surrounded by the dotted rectangle in FIG. **24C** are also respectively shown in FIG. **25** and FIG. **26**, and the light from the depressed portions **112a** forms the illuminated planes **116-4** to **116-6** in FIG. and FIG. **26** in this case.

In this case, all of the second electrodes **128** in the fourth row L_4 , which is the non-driving adjacent row, are synchronized with the second electrodes **128** in the fifth row L_5 , which is the driving row, in each liquid crystal cell **120** as shown in FIG. **24D**. Specifically, an alternating voltage V_9 is applied to the second electrodes **128** selected every other one, while an alternating voltage V_{10} inverted in phase with respect to the alternating voltage V_9 is applied to the other second electrodes **128** in the fifth row. Furthermore, alternating voltages V_7 and V_8 with the same voltage and phase as the alternating voltage V_9 are applied in the fourth row L_4 . Note that, since the light is not diffused in the y direction, a constant voltage or no voltage is applied to all of the first electrodes **126** in each liquid crystal cell **120**. As a result, since there is no potential difference between the wirings **140** disposed between the fourth row L_4 and the fifth row L_5 , the generation of the refractive index distribution causing the diffusion of the light in the y direction can be suppressed. Accordingly, unintended light diffusion can be suppressed, and precise light distribution can be performed.

Fifth Embodiment

In this embodiment, two modified examples of the lighting device **100** described in the First Embodiment are explained. An explanation of the structures the same as or similar to those described in the First to Fourth Embodiments may be omitted.

1. Modified Example 1

1-1. Structure

In the lighting device **100** described in the First Embodiment, the plurality of first electrodes **126** in one column does not conduct with the first electrodes **126** in other columns (e.g., adjacent columns) and is independently controlled from the first electrodes **126** in the other columns (see FIG. **6**). Similarly, the plurality of second electrodes **128** in one row does not conduct with the second electrodes **128** in other rows (e.g., adjacent rows) and is independently controlled from the second electrodes **128** in the other rows (see FIG. **7**).

In contrast, in this modified example, the plurality of first electrodes **126** selected every other one in one column (e.g., fourth column R_4) is electrically connected to and conducts with the plurality of first electrodes **126** selected every other one in the adjacent column (e.g., third column R_3) in at least

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one of the liquid crystal cells **120** as shown in FIG. **27**. Other first electrodes **126** in this one column (i.e., fourth column R_4) may conduct with the plurality of first electrodes **126** selected every other one in the opposite adjacent column (i.e., fifth column R_5).

Furthermore, or alternatively, the plurality of second electrodes **128** selected every other one in one row (e.g., fourth row L_4) is electrically connected to and conducts with the plurality of second electrodes **128** selected every other one in the adjacent row (e.g., fifth row L_5) in at least one of the liquid crystal cells **120** as shown in FIG. **28**. Other second electrodes **128** in this one row (i.e., fourth row L_4) may conduct with the plurality of second electrodes **128** selected every other one in the opposite adjacent row (i.e., third row L_3 which is not illustrated).

These connections enable the reduction in number of the wirings **138** and **140**, which increases the degree of freedom in designing the liquid crystal cell **120**. In addition, since the distances between adjacent rows and columns can be reduced, the lighting device **100** can be miniaturized.

1-2. Driving Method

An example of the driving method of the liquid crystal cell **120** according to the first modified example is explained. Here, a case is considered in which the reflective plate **112** has the depressed portions **112a** arranged in 8 rows and 8 columns as shown in FIG. **29A**, the light is emitted from the light-emitting elements **114** from the depressed portions **112a** of the fourth column R_4 in the fourth row L_4 and the fourth column R_4 in fifth row R_5 , and this light is diffused in the y direction. FIG. **27** schematically shows the arrangement of the first electrodes **126** and the wirings **138** connected thereto in the region surrounded by the dotted line in FIG. **29A**, and the light-emitting elements **114** form the illuminated planes **116-2** and **116-5**. In order to diffuse the light in the y direction, alternating voltages V_2 and V_3 with different phases are applied to the first electrodes **126** of the fourth column R_4 , which is the driving column, among the first electrodes **126** of the two liquid crystal cells **120**, while a constant voltage or no voltage is applied to the second electrodes **128** according to the timing chart of FIG. **14A** (FIG. **29B**). As a result, the light incident on the illuminated planes **116-2** and **116-5** can be diffused in the y direction.

At this time, all of the first electrodes **126** in the third column R_3 , which is one of the non-driving adjacent columns, may be synchronized with the plurality of first electrodes **126** selected every other one in the fourth column R_4 in order to suppress unintended light diffusion. Similarly, all of the first electrodes **126** in the fifth column R_5 , which is the other non-driving adjacent column, may be synchronized with the other plurality of first electrodes **126** selected every other one in the fourth column R_4 . In this modified example 1, since the plurality of first electrodes **126** selected every other one in the fourth column R_4 , which is the driving column, and the plurality of first electrodes **126** selected every other one in the third column R_3 , which is the non-driving adjacent column, conduct with each other, an alternating voltage V_1 with the same voltage and phase as the alternating voltage V_2 may be applied to the remaining first electrodes **126** in the third column R_3 . Similarly, since the remaining first electrodes **126** in the fourth column R_4 and the plurality of first electrodes **126** selected every other one in the fifth column R_5 , which is the non-driving adjacent column, conduct with each other, an alternating voltage V_4 with the same voltage and phase as the alternating voltage V_3 may be applied to the remaining first electrodes **126** in the fifth column R_5 (FIG. **29B**). When the liquid crystal cells **120** are driven in this manner, the contribution of the

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refractive index distribution between the first electrodes **126** in two adjacent columns is significantly decreased. As a result, unintended light diffusion can be suppressed.

As another example, a case is considered in which the reflective plate **112** has the depressed portions **112a** arranged in 8 rows and 8 columns, and light is emitted from the light-emitting elements **114** in the depressed portions **112a** of the third column R_3 to the fifth column R_5 in the fifth row L_5 and is diffused in the x direction as shown in FIG. 29C. FIG. 28 schematically shows the arrangement of the second electrodes **128** and the wirings **140** connected thereto in the region surrounded by the dotted line in FIG. 29C, where the light-emitting elements **114** form the illuminated planes **116-4** to **116-6**. In order to diffuse the light in the x direction, alternating voltages V_6 and V_7 with different phases from each other are applied to the second electrodes **128** in the fifth row L_5 , which is the driving row, among the second electrodes **128** in the two liquid crystal cells **120**, while a constant voltage or no voltage is applied to the first electrodes **126** according to the timing chart shown in FIG. 15A (FIG. 29D). Accordingly, the light on the illuminated planes **116-4** to **116-6** is diffused in the x direction.

At this time, the second electrodes **128** in the fourth row L_4 , which is the non-driving adjacent row, may be synchronized with the plurality of second electrodes **128** selected every other one in the fifth row L_5 in order to suppress unintended diffusion. In this modified example 1, since the plurality of second electrodes **128** selected every other one in the fifth row L_5 , which is the driving row, and the plurality of second electrodes **128** selected every other one in the fourth row L_4 , which is the non-driving adjacent row, conduct with each other, an alternating voltage V_5 with the same voltage and phase as the alternating voltage V_6 may be applied to the remaining second electrodes **128** in fourth row L_4 (FIG. 29D). When the liquid crystal cells **120** are driven in this manner, the contribution of the refractive index distribution between the second electrodes **128** in two adjacent rows is significantly decreased. As a result, unintended light diffusion can be suppressed.

2. Modified Example 2

2-1. Structure

In the lighting device **100** of the present modified example 2, the plurality of first electrodes **126** selected every other one in each column is electrically connected to and conducts with each other via the wiring **138** in at least one of the two liquid crystal cells **120** as shown in FIG. 30. The remaining first electrodes **126** are independently controlled in each row. Further or alternatively, the plurality of second electrodes **128** selected every other one in each row is electrically connected to and conducts with each other via the wiring **140** in at least one of the two liquid crystal cells **120** as shown in FIG. 31. The remaining second electrodes **128** are independently controlled in each row. Similar to the modified example 1, the number of wirings **138** and **140** can be reduced by adopting this connection, thereby improving the degree of freedom in designing the liquid crystal cells **120**. In addition, since the distances between adjacent rows and columns can be reduced, the lighting device **100** can be miniaturized.

2-2. Driving Method

An example of the driving method of the liquid crystal cells **120** in the present modified example 2 is explained below. Here, a case is considered in which the reflective plate **112** has the depressed portions **112a** arranged in 8 rows and 8 columns, and light is emitted from the light-emitting

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elements **114** in the depressed portions **112a** of the fourth column R_4 in the fourth row L_4 and the fifth row L_5 and is diffused in the y direction as shown in FIG. 32A. FIG. 30 schematically shows the arrangement of the first electrodes **126** and the wirings **138** connected thereto in the region surrounded by the dotted line in FIG. 32A, where the light-emitting elements **114** form the illuminated planes **116-2** and **116-5**. A constant voltage V_1 is applied to the plurality of the first electrodes **126** selected every other one in the fourth column R_4 , which is the driving column, among the first electrodes **126** in two liquid crystal cells **120**, while an alternating voltage V_3 is applied to the remaining first electrodes **126** in order to diffuse the light in the y direction. A constant voltage or no voltage is applied to the second electrodes **128** (FIG. 32B). By this driving method, the light incident on the illuminated planes **116-2** and **116-5** is diffused in the y direction.

At this time, the first electrodes **126** selected every other one in the third column R_3 and the fifth column R_5 , which are the non-driven adjacent columns, may be synchronized with the plurality of first electrodes **126** selected every other one in the fourth column R_4 in order to suppress unintended diffusion. In this modified example, since the plurality of first electrodes **126** selected every other one in the non-driving adjacent columns conducts with the plurality of first electrodes **126** selected every other one in the fourth column R_4 , which is the driving column, and applied with a constant voltage, a constant voltage V_2 or V_4 with the same voltage as the constant voltage V_1 may be applied to the remaining first electrodes **126** in each non-driving adjacent column (FIG. 32B). When the liquid crystal cells **120** are driven in this manner, the contribution of the refractive index distribution between the first electrodes **126** in two adjacent columns is significantly reduced. As a result, unintended light diffusion can be suppressed.

As another example, a case is considered in which the reflective plate **112** has the depressed portions **112a** arranged in 8 rows and 8 columns, and light is emitted from the light-emitting elements **114** in the depressed portions **112a** of the third column R_3 to the fifth column R_5 of the fifth row L_5 and is diffused in the x direction as shown in FIG. 32C. FIG. 31 schematically shows the arrangement of the second electrodes **128** and the wirings **140** connected thereto in the region surrounded by the dotted line in FIG. 32C, where the light-emitting elements **114** form the illuminated planes **116-4** to **116-6**. In order to diffuse the light in the x direction, a constant voltage V_5 is applied to the plurality of the second electrodes **128** selected every other one in the fifth column L_5 , which is the driving row, among the second electrodes **128** in the two liquid crystal cells **120**, while an alternating voltage V_7 is applied to the remaining second electrodes **128**. A constant voltage or no voltage is applied to the first electrodes **126** (FIG. 32D). With this driving method, the light incident on the illuminated planes **116-4** to **116-6** is diffused in the x direction.

At this time, the second electrodes **128** in the fourth row L_4 , which is the non-driving adjacent row, may be synchronized with the plurality of second electrodes **128** selected every other one in the fifth row L_5 in order to suppress unintended diffusion. In this modified example, since the plurality of second electrodes **128** selected every other one in the non-driving adjacent rows conducts with the plurality of second electrodes **128** selected every other one in the fifth row L_5 , which is the driving row, and applied with a constant voltage, a constant voltage V_6 with the same voltage as the constant voltage V_5 may be applied to the remaining second electrodes **128** in the non-driving adjacent rows (FIG. 32D).

When the liquid crystal cells **120** are driven in this manner, the contribution of the refractive index distribution between the second electrodes **128** in two adjacent rows is significantly decreased. As a result, unintended light diffusion can be suppressed.

Sixth Embodiment

In this embodiment, modified examples 3 and 4 of the lighting device **100** described in the First Embodiment are explained. An explanation of the structures the same as or similar to those described in the First to Fifth embodiments may be omitted. 1. Modified Example 3

The lighting device **100** according to the modified example 3 differs from the lighting device **100** described in the First Embodiment in that, as shown in FIG. 33A and the schematic view of the cross section along the chain line C-C' thereof (FIG. 33B), a single second electrode **128** overlapping all of the depressed portions **112a** is provided in at least one of the liquid crystal cells **120**. The single second electrode **128** overlaps the plurality or all of the first electrode groups **125**. In this case, the light diffusion occurs only on the first electrode **126** side because the refractive index distribution cannot be formed on the second electrode **128** side of each liquid crystal layer **136**. Therefore, in this modified example 3, it is preferable to configure the lighting device **100** so that the extending directions of the first electrodes **126** are orthogonal to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**. When this modified example 3 is employed, patterning of the second electrode **128** is no longer necessary, by which the manufacturing process can be shortened, and the lighting device **100** can be produced at a lower cost.

2. Modified Example 4

The lighting device **100** according to the modified example 4 differs from the lighting device **100** described in the First Embodiment in that, as shown in FIG. 34A and the schematic view of the cross section along the chain line D-D' thereof (FIG. 34B), no second electrode **128** is provided in at least one of the liquid crystal cells **120**. In this case, the light diffusion also occurs only on the first electrode **126** side since the refractive index distribution cannot be formed on the second electrode **128** side of each liquid crystal layer **136**. Therefore, in this modified example 3, it is preferable to configure the lighting device **100** so that the extending directions of the first electrodes **126** are orthogonal to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**. When this modified example 4 is employed, the fabrication process of the second electrode **128** becomes unnecessary, by which the manufacturing process can be shortened, and the lighting device **100** can be provided at a lower cost. In addition, since the absorption of light by the second electrode **128** is completely eliminated, the power consumption of the lighting device **100** can be reduced.

Seventh Embodiment

In this embodiment, modified examples 5 and 6 of the lighting device **100** described in the First Embodiment are explained. An explanation of the structures the same as or similar to those described in the First to Sixth embodiments may be omitted.

1. Modified Example 5

Schematic plan views of the first substrate **122** and the second substrate **124** of the first liquid crystal cell **120-1** of

the lighting device **100** according to the modified example 5 are shown in FIG. 35A, while schematic plan views of the first substrate **122** and the second substrate **124** of the second liquid crystal cell **120-2** are shown in FIG. 35B. As shown in these drawings, the plurality of first electrodes **126** is arranged so that the longitudinal direction thereof is inclined from at least one side of the first substrate **122** in at least one of the liquid crystal cells **120** of the lighting device **100** according to the modified example 5. The angle (first angle) between the longitudinal direction of the first electrodes **126** and the one side of the first substrate **122** may be arbitrarily set, for example, at equal to or more than 5° and equal to or less than 85° or equal to or more than 30° and equal to or less than 60°, and is typically 45°. Similar to the lighting device **100** described in the First Embodiment, the orientation direction of the first orientation film **132** is perpendicular to the longitudinal direction of the first electrodes **126** (see the white arrow).

Similarly, in each liquid crystal cell **120**, the plurality of second electrodes **128** is arranged so that the longitudinal direction thereof is inclined from at least one side of the second substrate **124**. The angle (second angle) between the longitudinal direction of the second electrodes **128** and the one side of the second substrate **124** may also be set arbitrarily, for example, at equal to or more than 5° and equal to or less than 85° or equal to or more than 30° and equal to or less than 60°, and is typically 45°. Similar to the first orientation film **132**, the orientation direction of the second orientation film **134** is perpendicular to the longitudinal direction of the second electrodes **128** (see the white arrow).

In each liquid crystal cell **120**, the first electrodes **126** and the second electrodes **128** are arranged so that their longitudinal directions are orthogonal to each other. Note that the longitudinal directions of the first electrodes **126** are parallel or perpendicular to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**. Similarly, the longitudinal directions of the second electrodes **128** are parallel or perpendicular to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**. Therefore, when the first substrate **122** and the second substrate **124** are each rectangular, it is preferred to set the first angle and the second angles at 45° to minimize the projected area in the xy plane.

Employment of the aforementioned arrangement allows both the plurality of first electrodes **126** and the plurality of second electrodes **128** to be inclined from the row direction (x direction) and the column direction (y direction) of the depressed portions **112a** in the first electrode group **125** and the second electrode group **127** overlapping each depressed portion **112a**. Thus, it is possible to diffuse the light from the light-emitting elements **114** disposed in the depressed portions **112a** in the direction inclined from the row direction and the column direction. For example, as shown in FIG. 36A, when all of the light-emitting elements **114** disposed in each of the depressed portions **112a** arranged in a matrix of 8 rows and 8 columns emit light, a rectangular illuminated region **A₀** reflecting the matrix shape is obtained when the liquid crystal cells **120** are not driven (FIG. 36B). In contrast, when the liquid crystal cells **120** are driven as appropriate, the light can be diffused in directions inclined from the x direction and the y direction (i.e., in the first direction and/or second direction), resulting in the formation of illuminated regions **A₁** with a variety of shapes as shown in FIG. 36C to FIG. 36F.

Furthermore, it is also possible to apply the local dimming described in the Second Embodiment. For example, as shown in FIG. 37A, when light is output from the depressed

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portions **112a** along the diagonal among the depressed portions **112a** arranged in a matrix of 8 rows and 8 columns, it is possible to obtain an illuminated region **A₁** expanded in a direction inclined from the x direction and the y direction (FIG. 37C) compared with the illuminated region **A₀** obtained when the liquid crystal cells **120** are not driven (FIG. 36B).

2. Modified Example 6

Schematic plan views of the first substrate **122** and the second substrate **124** of the first liquid crystal cell **120-1** of the lighting device **100** of the modified example 6 are shown in FIG. 38A, while schematic plan views of the first substrate **122** and the second substrate **124** of the second liquid crystal cell **120-2** are shown in FIG. 38B. As shown in these drawings, the first substrate **122** is divided into two regions in at least one liquid crystal cell **120** in the lighting device **100** of the modified example 6. The first electrodes **126** are arranged in one region (first region) so that the longitudinal direction thereof is parallel to at least one side of the first substrate **122**, whereas the first electrodes **126** are arranged in the other region (second region) so that the longitudinal direction thereof is inclined from that side similar to the modified example 5.

Similarly, the second substrate **124** is also divided into two regions. The second electrodes **128** are arranged in one region (third region) so that the longitudinal direction thereof is parallel to at least one side of the second substrate **124**, whereas the second electrodes **128** are arranged in the other region (fourth region) so that the longitudinal direction thereof is inclined from that edge, similar to the modified example 5.

In each liquid crystal cell **120**, the first regions and the third region overlap each other, and the second region and the fourth region overlap each other. Furthermore, the longitudinal direction of the first electrodes **126** is orthogonal to the longitudinal direction of the second electrodes **128**. The longitudinal directions of the first electrodes **126** are parallel or perpendicular to each other, and the longitudinal directions of the second electrodes **128** are also parallel or perpendicular to each other between the first liquid crystal cell **120-1** and the second liquid crystal cell **120-2**. Note that, the orientation direction is perpendicular to the longitudinal direction of the first electrodes **126** or the second electrodes **128** in each of the first region to the fourth region (see white arrows).

Employment of such an arrangement allows the light output from the depressed portions **112a** to be diffused in the x direction and they direction in a portion of the lighting device **100** (i.e., in a portion where the first region and the third region overlap each other) and to be diffused in a direction inclined from the x direction and the y direction in the other portion (i.e., in a portion where the second region and the fourth region overlap each other). Therefore, it is possible to form illuminated regions with more diverse shapes.

Furthermore, the liquid crystal cells **120** having the structure described in the First Embodiment may be combined with the pair of liquid crystal cells **120** according to the modified example 5 or the modified example 6 in the lighting device **100**. Adjacent liquid crystal cells **120** may be in direct contact with each other or may be fixed through an adhesive layer **102**. For example, FIG. 39 shows an example in which a third liquid crystal cell **120-3** and a fourth liquid crystal cell **120-4** with the plurality of first electrodes **126** and the plurality of second electrodes **128** respectively

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inclined from the sides of the first substrate **122** and the second substrate **124** are disposed over a first liquid crystal cell **120-1** and a second liquid crystal cell **120-2** with the plurality of first electrodes **126** and the plurality of second electrodes **128** respectively extending parallel to the sides of the first substrate **122** and the second substrate **124**. Since the light can be diffused in a direction inclined from the x direction and the y direction (e.g., inclined by 45°) in addition to the x direction and the y direction by driving the first to fourth liquid crystal cells **120** as appropriate in such a configuration, the illuminated region can be transformed in a variety of ways.

Eighth Embodiment

In the present embodiment, a modified example of the driving method of the lighting device **100** described in the First Embodiment is explained. An explanation of the structures the same as or similar to those described in First to Seventh Embodiments may be omitted.

As an example, the arrangements of the first electrodes **126** and the second electrodes **128** of the lighting device **100** having the first electrode groups **125** and the second electrode groups **127** each arranged in a matrix of three rows and three columns are respectively shown in FIG. 40 and FIG. 41. As described in the First Embodiment, alternating voltages with different phases from each other are respectively applied to the first electrodes **126** selected every other one and the other first electrodes **126** in each column. Similarly, alternating voltages with different phases from each other are respectively applied to the second electrodes **128** selected every other one and the other second electrodes **128** in each row. Since the alternating voltages are applied to the wirings **138** and **140** according to the timing chart shown in FIG. 42A in the driving method described in the First Embodiment, the phases of the two wirings **138** between adjacent columns are opposite, and the phases of the two wirings **140** between adjacent rows are also opposite. Hence, since a potential difference is generated between the wirings **140** arranged between adjacent rows and between the wirings **138** arranged between adjacent columns, a refractive index distribution is also generated between adjacent liquid crystal cells **120**, i.e., between adjacent depressed portions **112a**. Since no illuminated plane **116** is formed between adjacent depressed portions **112a**, such an unintended refractive index distribution does not seriously influence the light distribution control. However, an increase in the aforementioned potential difference may affect the orientation of the liquid crystal layer **136** overlapping the illuminated planes **116**.

Therefore, when generating a larger transverse electric field, for example, it is preferable to apply an alternating voltage with the same voltage and the same phase to two wirings **138** arranged between adjacent columns to avoid the generation of a potential difference therebetween. Similarly, it is preferable to apply an alternating voltage with the same voltage and the same phase to two wirings **140** arranged between adjacent rows to avoid the generation of a potential difference therebetween. Thus, it is preferable to provide the alternating voltages to the wirings **138** and **140** according to the timing chart shown in FIG. 42B.

In this embodiment, light can also be diffused independently in the x direction and the y direction. In this case, the voltages of the first electrodes **126** and the second electrodes **128** may be appropriately adjusted in each liquid crystal cell **120** as described in the First Embodiment. In addition, when only the light-emitting elements **114** of a part of the

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depressed portions **112a** are driven and the first electrodes **126** of a part of the columns or the second electrodes **128** of a part of the rows are driven as in the Fourth Embodiment, the wirings **138** may be applied with alternating voltages with different phases so that the phases of the first electrode **126** selected every other one and the other first electrodes **126** are different from each other in the non-driving adjacent columns. Similarly, alternating voltages with different phases may be applied to the wirings **140** so that the phases of the second electrodes **128** selected every other one and the other second electrodes **128** are different from each other in the non-driving adjacent rows.

The aforementioned modes described as the embodiments of the present invention can be implemented by appropriately combining with each other as long as no contradiction is caused. Furthermore, any mode which is realized by persons ordinarily skilled in the art through the appropriate addition, deletion, or design change of elements or through the addition, deletion, or condition change of a process is included in the scope of the present invention as long as they possess the concept of the present invention.

It is understood that another effect different from that provided by each of the aforementioned embodiments is achieved by the present invention if the effect is obvious from the description in the specification or readily conceived by persons ordinarily skilled in the art.

What is claimed is:

1. A lighting device comprising:

a light source comprising a plurality of light-emitting elements arranged in a matrix shape with m rows and n columns;

a first liquid crystal cell over the light source; and

a second liquid crystal cell over the first liquid crystal cell, wherein each of the first liquid crystal cell and the second liquid crystal cell comprises:

a first substrate;

a plurality of first electrode groups located over the first substrate and arranged in a matrix shape with m rows and n columns;

a liquid crystal layer over the plurality of first electrode groups; and

a second substrate over the liquid crystal layer,

wherein, in each of the first liquid crystal cell and the second liquid crystal cell,

each of the plurality of first electrode groups comprises a plurality of first electrodes extending in a row direction,

the light-emitting elements in a jth row and a kth column overlap with one of the plurality of first electrode groups located in the jth row and the kth column,

wherein longitudinal directions of the plurality of first electrodes of the first liquid crystal cell are parallel to longitudinal directions of the plurality of first electrodes of the second liquid crystal cell, and

wherein n and m are each a natural number larger than 1, j is a variable selected from natural numbers equal to or larger than 1 and equal to or smaller than n, and k is a variable selected from natural numbers equal to or larger than 1 and equal to or smaller than m.

2. The lighting device according to claim 1,

wherein each of the first liquid crystal cell and the second liquid crystal cell further comprises a single second electrode between the liquid crystal layer and the second substrate, and

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wherein, in each of the first liquid crystal cell and the second liquid crystal cell, the second electrode overlaps the plurality of first electrode groups.

3. The lighting device according to claim 1,

wherein each of the first liquid crystal cell and the second liquid crystal cell further comprises a plurality of second electrode groups arranged in a matrix shape with m rows and n columns between the liquid crystal layer and the second substrate, and

wherein, in each of the first liquid crystal cell and the second liquid crystal cell,

each of the plurality of second electrode groups comprises a plurality of second electrodes extending in a column direction, and

the light-emitting element in the jth row and the kth column overlaps one of the plurality of second electrode groups in the jth row and the kth column.

4. The lighting device according to claim 1,

wherein the light source comprises a reflective plate having a plurality of depressed portions arranged in a matrix shape with m rows and n columns,

the plurality of light-emitting elements is arranged in the corresponding plurality of depressed portions, and

the plurality of depressed portions is configured so that light from the light-emitting element in the depressed portion located in the jth row and the kth column is selectively applied to the one of the plurality of first electrode groups located in the jth row and the kth column in the first liquid crystal cell.

5. The lighting device according to claim 1,

wherein, in each of the first liquid crystal cell and the second liquid crystal cell,

the plurality of first electrodes is configured to be applied with a constant voltage or a first alternating voltage,

a phase of the first alternating voltage is inverted between the first electrodes adjacent in the same column when the first alternating voltage is applied, and

the constant voltage is different between the first electrodes adjacent in the same column when the constant voltage is applied.

6. The lighting device according to claim 3,

wherein, in each of the first liquid crystal cell and the second liquid crystal cell,

the plurality of second electrodes is configured to be applied with a constant voltage or a second alternating voltage,

a phase of the second alternating voltage is inverted between the second electrodes adjacent in the same row when the second alternating voltage is applied, and

the constant voltage is different between the second electrodes adjacent in the same row when the constant voltage is applied.

7. The lighting device according to claim 1,

wherein a thickness of the liquid crystal layer is equal to or larger than 20 μm and equal to or smaller than 60 μm in each of the first liquid crystal cell and the second liquid crystal cell.

8. The lighting device according to claim 1,

wherein each of the first liquid crystal cell and the second liquid crystal cell further comprises:

a first orientation film between the plurality of first electrode groups and the liquid crystal layer; and

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- a second orientation film located between the liquid crystal layer and the second substrate and in contact with the liquid crystal layer, and
 wherein, in each of the first liquid crystal cell and the second liquid crystal cell,
 an orientation direction of the first orientation film is perpendicular to the longitudinal directions of the plurality of first electrodes, and
 an orientation direction of the second orientation film is perpendicular to the orientation direction of the first orientation film.
9. The lighting device according to claim 1,
 wherein, in each of the first liquid crystal cell and the second liquid crystal cell, the plurality of first electrodes selected every other one in the kth column electrically conducts with the plurality of first electrodes selected every other one in the column adjacent to the kth column.
10. The lighting device according to claim 3,
 wherein, in each of the first liquid crystal cell and the second liquid crystal cell, the plurality of second electrodes selected every other one in the jth row electrically conducts with the plurality of second electrodes selected every other one in the row adjacent to the jth row.
11. The lighting device according to claim 1,
 wherein, in each of the first liquid crystal cell and the second liquid crystal cell, the plurality of first electrodes selected every other one in each column electrically conducts with each other.
12. The lighting device according to claim 3,
 wherein, in each of the first liquid crystal cell and the second liquid crystal cell, the plurality of second electrodes selected every other one in each row electrically conducts with each other.
13. The lighting device according to claim 1,
 wherein the light source is configured so that the plurality of light-emitting elements is controlled independently from each other.
14. The lighting device according to claim 3,
 wherein, in each of the first liquid crystal cell and the second liquid crystal cell,
 the first substrate has a first region and a second region, the second substrate has a third region and a fourth region respectively overlapping the first region and the second region,
 the plurality of first electrodes is parallel to a side of the first substrate in the first region,
 the plurality of first electrodes is inclined from the side of the first substrate in the second region,
 the plurality of second electrodes is parallel to a side of the second substrate in the third region, and
 the plurality of second electrodes is inclined from the side of the second substrate in the fourth region.
15. The lighting device according to claim 1, further comprising:
 a third liquid crystal cell having the same structure as the first liquid crystal cell and located between the first liquid crystal cell and the second liquid crystal cell; and
 a fourth liquid crystal cell having the same structure as the second liquid crystal cell and located over the second liquid crystal cell,

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- wherein the longitudinal directions of the plurality of first electrodes are parallel between the first liquid crystal cell and the third liquid crystal cell,
 wherein the longitudinal directions of the plurality of first electrodes are parallel between the second liquid crystal cell and the fourth liquid crystal cell, and
 wherein the longitudinal directions of the plurality of first electrodes intersect each other between the first liquid crystal cell and the second liquid crystal cell.
16. An optical element comprising:
 a first substrate;
 a plurality of first electrode groups located over the first substrate and arranged in a matrix shape with m rows and n columns;
 a liquid crystal layer over the plurality of first electrode groups; and
 a second substrate over the liquid crystal layer,
 wherein each of the plurality of first electrode groups comprises a plurality of first electrodes extending in a row direction,
 wherein, in a portion of the plurality of first electrode groups arranged in a kth column, odd-numbered first electrodes in a column direction are connected to a first wiring, while even-numbered first electrodes are connected to a second wiring,
 wherein, in a portion of the plurality of first electrode groups arranged in a (k+1)th column, odd-numbered first electrodes in the column direction are connected to a third wiring, while even-numbered first electrodes are connected to a fourth wiring, and
 wherein n and m are each a natural number larger than 1, and k is a variable selected from natural numbers equal to or larger than 1 and less than n.
17. The optical element according to claim 16, further comprising a driver circuit connected to the first wiring to the fourth wiring and configured to independently supply a potential to the first wiring to the fourth wiring.
18. The optical element according to claim 16, further comprising a plurality of second electrode groups located under the substrate and arranged in a matrix shape with m rows and n columns,
 wherein each of the plurality of second electrode groups comprises a plurality of second electrodes extending in a column direction,
 wherein, in a portion of the plurality of second electrode groups arranged in a jth row, odd-numbered second electrodes in the row direction are connected to a fifth wiring, while even-numbered second electrodes are connected to a sixth wiring,
 wherein, in a portion of the plurality of second electrode groups arranged in a (j+1)th row, odd-numbered second electrodes in the column direction are connected to a seventh wiring, while even-numbered second electrodes are connected to an eighth wiring, and
 wherein j is a variable selected from natural numbers equal to or larger than 1 and less than m.
19. The optical element according to claim 18, further comprising a driver circuit connected to the first wiring to the eighth wiring and configured to independently supply a potential to the first wiring to the eighth wiring.

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